Informatica® Intelligent Data Management Cloud

October 2026

Organization Administration

Informatica Intelligent Data Management Cloud Organization Administration
October 2026

© Copyright Informatica LLC 2021, 2026

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

U.S. GOVERNMENT RIGHTS Programs, software, databases, and related documentation and technical data delivered to U.S. Government customers are "commercial computer software" or "commercial technical data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, duplication, disclosure, modification, and adaptation is subject to the restrictions and license terms set forth in the applicable Government contract, and, to the extent applicable by the terms of the Government contract, the additional rights set forth in FAR 52.227-19, Commercial Computer Software License.

A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html. Other company and product names may be trade names or trademarks of their respective owners.

Portions of this software and/or documentation are subject to copyright held by third parties. Required third party notices and AI assistant acknowledgements are included with the product.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2026-10-16

Table of Contents

Preface 7

Informatica Resources 7

- Informatica Documentation 7
- IDMC web site 7
- IDMC Communities 7
- IDMC Marketplace 7
- Data Integration connector documentation 8
- Informatica Knowledge Base 8
- IDMC Trust Center 8
- IDMC Pulse Dashboard 8
- Informatica Global Customer Support 8

Chapter 1: Introducing Administrator 9

General and security settings 11

Chapter 2: Organizations 13

Setting up an organization 14

- Authentication properties 14
- Fingerprint authentication 16
- Connection Credentials storage location 16
- Enterprise Data Catalog integration properties 17
- Data Integration service properties 18

Organization settings 18

Sub-organizations 20

- Adding a sub-organization 22
- Removing a sub-organization 24
- Disabling or enabling a sub-organization 25
- Switching to a different organization 25
- Denying parent organization access to a sub-organization 26
- Add-on connectors in sub-organizations 26
- Exporting and importing assets in sub-organizations 26
- Sub-organization capability and service support matrix 27

Additional production organizations and sandbox organizations 27

- Creating an additional organization 28

Chapter 3: Licenses 30

License categories 30

License types 30

Sub-organization licenses 31

- Editing sub-organization licenses 32

Synchronizing licenses with the parent organization . . . 32
Configuring the organization type. . . . 32
License expiration. . . . 33

Chapter 4: Metering. . . . 34

Informatica processing unit metrics. . . . 34
IPU scalars. . . . 34
IPU meters. . . . 35
Rate cards. . . . 38
IPU usage for disabled and deleted sub-organizations. . . . 39
Viewing IPU metrics. . . . 40
IPU metrics reports. . . . 43
Setting up IPU consumption alerts. . . . 47
Metering tags. . . . 47
Feature-based license metrics. . . . 49
Viewing license metrics. . . . 49
Viewing usage details. . . . 54
Metering usage reports. . . . 55

Chapter 5: Source control configuration. . . . 57

Source control configuration for sub-organizations. . . . 58
Repository access using OAuth. . . . 58
Working with an on-premises repository. . . . 59
Enabling source control for an organization. . . . 59
Changing the source control repository URL. . . . 61
Disabling source control for an organization. . . . 62
Configuring repository access. . . . 62
Source control best practices. . . . 63

Chapter 6: Customer-managed encryption keys. . . . 65

Creating and enabling a customer-managed key. . . . 67
Frequently asked questions about customer-managed keys (CMK). . . . 67

Chapter 7: Secrets managers. . . . 70

AWS Secrets Manager configuration. . . . 71
IAM role configuration for AWS Secrets Manager. . . . 72
Instance profile configuration for AWS Secrets Manager. . . . 74
Cross-account access configuration for AWS Secrets Manager. . . . 75
AWS Secrets Manager secret names. . . . 78
AWS Secrets Manager connection properties. . . . 78
Azure Key Vault configuration. . . . 79
Azure Key Vault secret names. . . . 80
Azure Key Vault connection properties. . . . 80

HashiCorp Vault configuration. 80
HashiCorp Vault authentication. 81
HashiCorp Vault secret formats. 81
HashiCorp Vault connection properties. 81
Enabling and disabling a secrets manager. 82
Configuring a secrets manager for a sub-organization. 83
Configuring a connection to use the secrets manager. 83
Connection property format. 85

Chapter 8: Permissions. 88

Rules and guidelines for permissions. 89
Configuring permissions. 90

Chapter 9: Schedules. 92

Configuring a blackout period. 93
Repeat frequency. 93
Time zones and schedules. 94
Daylight Savings Time changes and schedules. 94
Configuring a schedule. 95
Exporting schedules. 96
Troubleshooting scheduled tasks. 96

Chapter 10: CLAIRE preferences. 97

CLAIRE generative AI services. 97
CLAIRE recommendations. 98

Chapter 11: Bundle management. 99

Types of bundles. 99
Installing a bundle. 100
Copying a bundle. 101
Upgrading a bundle. 101
Uninstalling a bundle. 102

Chapter 12: Connection Credentials storage location 103

Chapter 13: Service-specific settings. 105

Enterprise Data Catalog integration properties. 105
Data Integration service properties. 106
Application Integration properties. 106

Chapter 14: Custom branding. 107

Logo and favicon guidelines. 107
Configuring custom branding for an organization. 107

Chapter 15: Informatica support access. 109

Removing Informatica support access. 109

Chapter 16: Event monitoring. 111

Chapter 17: Troubleshooting security. 113

Chapter 18: Cipher suites used by IDMC. 114

Index. 115

Preface

Use *Organization Administration* to learn how to set up and maintain your IDMC℠ organization and sub-organizations. Learn how to manage licenses and monitor license usage, configure source control, configure object permissions, create schedules, manage bundles, monitor events, and troubleshoot security issues.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

IDMC Pulse Dashboard

The IDMC Pulse Dashboard provides system status and upgrade information for all PODs and services across IDMC.

Follow the services you use and subscribe to alerts to receive notifications about upgrades, maintenance, production incidents, and upcoming releases. The IDMC Pulse Dashboard displays the production status of all the IDMC products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single service or all IDMC services. Subscribing to all services is the best way to be certain you never miss an update.

For more information, see the Service Status and Communications guide.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introducing Administrator

Administrator provides organization management capabilities across IDMC.

Use Administrator to manage the following aspects of your organization:

Organization and sub-organizations

Manage the general properties for your organization and sub-organizations. Create and manage sub-organizations. Back up and restore your organizations.

For information about organizations and sub-organizations, see Chapter 2, “Organizations” on page 13.

Licenses

View your organization's licenses and manage sub-organization licenses.

For information about licenses, see Chapter 3, “Licenses” on page 30.

Ecosystem and SAML single sign-on

Configure single-sign on settings for Microsoft Azure. Enable single sign-on capability for a SAML third-party identity provider.

For information about Microsoft Azure single sign-on settings and information about enabling and configuring SAML single sign-on, see *User Administration*.

Metering

View metering information such as job limits, usage, and Informatica processing unit (IPU) balances.

For information about metering, see Chapter 4, “Metering” on page 34.

General and security settings

Configure settings for your organization, including source control settings, CLAIRE preferences, user authentication, Turbo mode for Application Integration, connection credentials storage location, Enterprise Data Catalog integration properties, Data Integration Service properties, email notification settings, support access settings. and secret vault toggle.

For more information about general and security settings, see “General and security settings” on page 11.

Users, user groups, and user roles

Create and configure individual user accounts to allow access to your organization. Create groups of users that can perform the same tasks. Create and configure roles to define the privileges for your users and user groups.

For information about users, user groups, and user roles, see *User Administration*.

Permissions

Configure the access rights that users and user groups have for objects such as Secure Agents, Secure Agent groups, connections, and schedules.

For information about permissions and configuring permissions, see Chapter 8, “Permissions” on page 88.

Runtime environments

Download and install Secure Agents. Create and configure Secure Agent groups.

For information about Secure Agents, Secure Agent groups, and downloading and installing a Secure Agent, see *Runtime Environments*.

Serverless runtime environments

Use a runtime environment that Data Integration manages to reduce maintenance overhead.

Note: To use a serverless runtime environment, you must have a private cloud on the AWS cloud platform.

For information about serverless runtime environments, see *Runtime Environments*.

Secure Agent services

Configure settings for the microservices that the Secure Agent uses for data processing such as the Elastic Server, CIH Processor, Data Integration Server, EDC Search Agent, and Process Server.

For information about Secure Agent services and their configuration, see *Secure Agent Services*.

Advanced clusters

Manage the ephemeral clusters that your organization can use to process data integration jobs.

For information about advanced clusters, see *Advanced Clusters*.

Schedules

Create schedules to run tasks or taskflows at specified times or at regular intervals. Define a blackout period in which no scheduled tasks or jobs in your organization can run.

For information about schedules and organization blackout periods, see Chapter 9, “Schedules” on page 92.

Add-on bundles

Install, copy, upgrade, and uninstall sets of related mappings, mapping tasks, and mapplets that Data Integration users can use in data integration projects.

For information about managing add-on bundles, see Chapter 11, “Bundle management” on page 99.

Data services repository

Download out-of-the-box data services that process industry-standard messages and use them to create custom data services to process custom messages.

For information about the data services repository, see *Data Services Repository*.

Event monitoring

Monitor events for the assets, licenses, users, and Secure Agents in your organization through the asset and security logs.

For information about asset and security logs, see Chapter 16, “Event monitoring” on page 111.

File transfer

Configure your organization's file server to securely send and receive files from a business partner's remote server. Configure a connection, and then send the files to your partners using the IDMC REST API.

For information about file servers and file transfer, see *File Transfer*.

Note: Some of the functionality that's mentioned in the help might not be available due to your organization's IDMC license agreement.

General and security settings

Configure general, notification, and security settings on the **Settings** page.

Based on your organization's licenses, you can configure the following settings:

Source control settings

Configure source control for your organization to enable version management for projects, folders, and assets. Set up a connection to the global source control repository for your organization and configure read and write access to the repository. You can also enable project-level source control so that users can link source control repositories to specific projects.

For more information, see Chapter 5, “Source control configuration” on page 57.

CLAIRE preferences

Configure CLAIRE preferences for your organization to enable or disable CLAIRE GPT and CLAIRE recommendations for mappings.

For more information, see Chapter 10, “CLAIRE preferences” on page 97.

Upgrade settings for Secure Agent services

If a service that supports rolling upgrades encounters an error during an upgrade, you can specify whether to continue or stop upgrading the service.

You can also configure a restart schedule for services that need to be restarted after minor upgrades.

For more information, see *Secure Agent Services*.

Authentication

Configure password requirements, multi-factor authentication, idle timeout, and the authentication type.

If your device supports it, you can also configure fingerprint authentication.

For more information, see *Secure Agent Services*.

Turbo mode for Application Integration

If your organization has a usage-based license and was created before April 30, 2025, you can configure Application Integration processes to run in the Turbo mode.

For more information, see “Application Integration properties” on page 106.

Connection Credentials

Specify where to store credentials on the Informatica Cloud or the local Secure Agent.

For more information, see “Connection Credentials storage location ” on page 16.

Enterprise Data Catalog

If your organization uses data catalog discovery in Data Integration, you can configure Enterprise Data Catalog integration properties for your organization and sub-organizations.

For more information, see "Enterprise Data Catalog integration properties" on page 17.

Data Integration Service

Set the time zone and email addresses for job notifications in Data Integration.

For more information, see "Data Integration service properties" on page 18.

Custom branding settings

Configure custom branding settings for your parent organization and apply them to your sub-organizations. You can also configure custom branding settings for each sub-organization based on your requirement. The custom branding settings include logo, color theme, and favicon.

For more information, see Chapter 14, "Custom branding" on page 107.

Notification settings

Configure whether users in your organization receive email notifications. You can set the default behavior for new users and override the setting for all users.

For more information, see "Notifications" in *User Administration*.

Informatica support access

Control whether Informatica employees are allowed to view your organization details.

For more information, see Chapter 15, "Informatica support access" on page 109.

Customer-managed encryption key settings

Use your own master key to encrypt your organization's encryption keys instead of using Informatica's master key.

For more information, see Chapter 6, "Customer-managed encryption keys" on page 65.

Secrets manager configuration settings

Configure your organization to retrieve sensitive connection credentials from an external secrets manager like AWS Secrets Manager or Azure Key Vault.

For more information, see Chapter 7, "Secrets managers" on page 70.

CHAPTER 2

Organizations

An organization is a secure area within the IDMC repository that stores your licenses, user accounts, data integration assets such as mappings and tasks, and information about jobs and security. You might have access to one or more organizations.

By default, the organization that you create when you start your free trial is a production organization.

Based on your licenses, you might also have access to the following organizations:

Sub-organizations

Administrators in the production organization can create one or more sub-organizations. Sub-organizations are child organizations of the production organization. They are automatically linked to the parent organization.

Each sub-organization has its own set of assets, connections, runtime environments, and users. However, the parent organization can share runtime environments and add-on connectors with a sub-organization. Administrators in the parent organization can switch to a sub-organization unless the sub-organization forbids this.

You cannot create any other organizations from a sub-organization.

Additional production organizations and sandbox organizations

Administrators in the production organization can create additional production organizations and sandbox organizations.

These organizations are automatically linked to the production organization for IPU usage, but they are otherwise completely independent organizations. They do not share assets, connections, runtime environments, or users with the production organization. Administrators in the production organization cannot switch into an additional production organization or a sandbox organization.

If the production organization has the license to create sub-organizations, administrators in the additional production organizations and sandbox organizations can create sub-organizations for their organizations.

The administrator of an organization maintains the organization and its sub-organizations. Log in to IDMC as an administrator to set up your organization, create and manage schedules, and monitor activities related to assets and security.

Setting up an organization

When you set up an organization, you configure the organization properties, sub-organizations, additional organizations, licenses, runtime environments, and user accounts.

To set up your company's organization, perform the following steps:

1. Configure organization properties such as the organization name and address, authentication information, and notification email addresses.
2. Verify that your organization has the appropriate licenses.
3. Optionally, create one or more sub-organizations and configure licenses for the sub-organizations.
4. Optionally, create additional production organizations and sandbox organizations.
5. Configure runtime environments and Secure Agents.
6. Set up users, user groups, and roles.

You might also need to download and install non-native connectors for your organization. For example, if users in your organization create tasks that read data from Teradata tables, you need to download and install the add-on connector for Teradata. For more information about downloading and installing add-on connectors, see *Connections*.

Authentication properties

You can configure authentication properties for your organization and sub-organizations. Authentication properties control password restrictions and IP address filtering.

Password restrictions are enforced when users create or change their passwords. If you change the password expiration date from "never" to a number of days, then users with passwords that are older than the number of days will be required to change their passwords the next time that they log in to IDMC.

The following table describes the authentication properties:

Property	Description
Minimum Password Length	Minimum password length required for a valid password. Must be a number between 8 and 25 characters. If the organization uses multi-factor authentication, the minimum password length is 8 characters. Otherwise, the minimum password length is 15 characters.
Minimum Character Mix	Minimum number of character types required for a valid password. Passwords can contain a mix of the following character sets: - Lowercase alphabetic characters - Uppercase alphabetic characters - Numeric characters - Special characters For example, if you set **Minimum Character Mix** to 1, then passwords must contain at least one of the character sets. If you set **Minimum Character Mix** to 2, then passwords must contain at least two of the character sets.
Password Reuse	Controls whether users can reuse passwords.
Password Expires	Determines how often users must reset their passwords.

Property	Description
Enable Multi-Factor Authentication	Enables multi-factor authentication for native human users. By default, multi-factor authentication is enabled when organizations are provisioned. With multi-factor authentication, native human users receive a verification code through email when they log in to the user interface. The email address for each human user must be valid. Categorize users as human or non-human users on the **Users** page. For more information, see *User Administration*.
Session Idle Timeout	Amount of time before a user's session times out due to inactivity. IDMC displays a warning message to the user 60 seconds before the user is logged out. Default is 30 minutes, which is the recommended value.
Authentication Type	Authentication type used after a user logs in. Default is **Session ID**. For JWT authentication, select a duration for tokens to expire. Default is 30 minutes. When you change the authentication type, the new type takes effect at the next login. The change doesn't affect sessions that are in progress. Before you use the JWT authentication type, modify custom scripts to refresh the tokens before they expire. For more information, see the JWT Support Knowledge article. **Note:** Don't use the JWT authentication type if your organization uses REST V2 Connector to call IDMC APIs.
Use Trusted IP Ranges	Enables IP address filtering. IP address filtering uses trusted IP address ranges in addition to account passwords to prevent unauthorized users from accessing your organization. When you enable IP address filtering, a user with a valid login must also have an IP address within the range of trusted IP addresses, or the user can't log in to your organization. This option applies to all Informatica clients that connect to IDMC, including human and non-human users, REST API users, Secure Agents, and Secure Agent services. When you enable this option, you must also enter one or more trusted IP address ranges. Use the following guidelines: - If you create a serverless runtime environment when trusted IP ranges are enabled, you must add the IP addresses of the DMZ NAT gateway to the list of trusted IP addresses. For a list of the DMZ NAT gateway addresses, see *Runtime Environments*. - If you enable trusted IPs and use Cloud Integration Hub, ensure that you've also allowed the outbound IP addresses under Core Services for your POD. For more information, see POD Availability and Networking.
Allowed Trusted IP Ranges	The trusted ranges of IP addresses from which users can log in to access the organization. IDMC supports IP address formats in IP version 4 (IPv4) and version 6 (IPv6). Fields for the trusted IP address range appear when you enable IP address filtering. To enter additional address ranges, click **+**. To ensure seamless communication between resources across both IPv4 and IPv6 networks, enable a dual stack configuration. For example, if your virtual machine uses one network type, IPv4 or IPv6, and the servers that host the connectors use another network type, enable dual stack on your virtual machine to ensure seamless communication regardless of their network type. **Note:** If you enter an invalid IP address range, users cannot access your organization. Contact your network administrator for valid IP address ranges.

Fingerprint authentication

Fingerprint authentication verifies that a Secure Agent starts up on the same machine it was set up on. You can configure settings to customize how your organization handles fingerprint mismatches.

The first time an agent starts up, IDMC uses anonymized device attributes to create a unqiue fingerprint of the Secure Agent machine. Every subsequent time that the agent starts up, IDMC checks the fingerprint and, if there's a mismatch, sends an alert to the notifications tray for administrators.

Note: If you uninstall and reinstall the Secure Agent on the same machine, the fingerprint remains the same.

Configure settings for fingerprint authentication in the Authentication section of the **Settings** page. The following table describes the fingerprint authentication properties:

Property	Description
Fingerprint Enforcement	Prevents the Secure Agent from starting up and logs an error when there's a fingerprint mismatch. IDMC logs the following error in the agentcore.log file: Internal error. Agent <Secure Agent ID> fingerprint is not matching with the previous stored value for request <request ID>. When you disable this property and there's a fingerprint mismatch, the Secure Agent starts up normally. Default is off.
Fingerprint Mismatch Email Notifications	To receive mismatch alerts by email, enter an email address. IDMC checks the email format, but doesn't verify the email address. Be sure to allow emails from the address "admin@informaticacloud.com". IDMC sends an email with the following message: There was a fingerprint mismatch while logging in agent with name <Secure Agent name> for Organization <organization ID>. The agent was last active on <date in UTC>.

Connection Credentials storage location

You can configure where to store the connection properties for your organization and sub-organizations. To specify where to store the connection properties, configure the **Connection Credentials** on the **Settings** page in Administrator.

You can store connection properties in either of the following locations:

Informatica Cloud

When you store connection properties on the cloud, the connection properties are stored in the IDMC repository and are always available. The connections are encrypted by the IDMC key management service.

IDMC backs up connection properties regularly as part of standard backup procedures.

Local Secure Agent

You might store connection properties with a local Secure Agent if you need the connection properties to reside within your firewall. When you enable this option, the properties for all connections that are listed on the **Connections** page are stored with the local agent.

Note: In organizations subject to FedRAMP, you can't store connection properties with a local Secure Agent.

If you choose this option, you can store connection properties with one Secure Agent. Connection properties are stored in the following directory:

```
<Secure Agent installation directory>/apps/Data_Integration_Server/data
```

When you store properties with a local Secure Agent, the Secure Agent must be running so that tasks can run and users can work with connections. Back up connection properties regularly to prevent loss of data. A best practice is to back up connection properties after you change the location or the encryption key for connection properties.

The connections are encrypted by the IDMC key management service. IDMC uses CBC (Cipher Block Chaining) mode 256 AES encryption to store the connections.

If you use an external secrets manager like AWS Secrets Manager or Azure Key Vault to store sensitive connection credentials, you need to set the connection credential storage to **Informatica Cloud**. When you do this, sensitive credentials are retrieved from the secrets manager and other connection properties are stored in the IDMC repository. You can't use a secrets manager if you store connection credentials on a local Secure Agent. For more information about secrets manager configuration, see Chapter 7, “Secrets managers” on page 70.

You can change where you want to store connection properties. When you do this, IDMC moves the connection properties to the appropriate location. For example, your license expires, so you configure the organization to store connections on the cloud. IDMC moves the connection properties from the local Secure Agent to IDMC.

Enterprise Data Catalog integration properties

If your organization uses data catalog discovery in Data Integration, you can configure Enterprise Data Catalog integration properties for your organization and sub-organizations. Configure Enterprise Data Catalog integration properties so that users can use catalog assets in mappings, synchronization tasks, and file ingestion and replication tasks.

The Enterprise Data Catalog integration properties that you configure for the organization apply to the data catalog searches that all users in the organization perform. If your organization includes sub-organizations, you can configure different Enterprise Data Catalog integration properties for the parent organization and for each sub-organization.

The following table describes the Enterprise Data Catalog integration properties:

Property	Description
Catalog URL	URL of the Enterprise Data Catalog Service. Use the following format: `http://<fully qualified host name>:<port>` Do not append `/ldmcatalog` at the end of the URL.
Runtime environment	Name of the Secure Agent group that is used to read data from Enterprise Data Catalog. The agents in the group that you select must be able to communicate with Enterprise Data Catalog. Therefore, the Enterprise Data Catalog host must be in the same network as the agent machines or it must have the appropriate ports open for communication.
User name	Enterprise Data Catalog user account that the Secure Agent uses to access Enterprise Data Catalog. This user account must have privileges to view and search for objects in Enterprise Data Catalog and to perform functions using the Enterprise Data Catalog REST API.
Password	Password for the Enterprise Data Catalog user account.
Show the data catalog	Shows and hides the **Data Catalog** page in Data Integration.

Data Integration service properties

Data Integration service properties are used by Data Integration. Configure these properties to set the time zone and default email addresses for job notifications.

You can set the following Data Integration service properties:

Jobs properties

The following table describes the jobs properties:

Property	Description
Schedule Offset	A small amount of time that is added to schedule start times to help prevent server overload at standard schedule start times. An organization has a single schedule offset that is applied to all schedules. The schedule offset does not affect the start time of manually started tasks or taskflows. You cannot change the schedule offset. Even though it is not displayed in the schedule details, the schedule offset for your organization is added to the time range configured for all schedules. This ensures that scheduled tasks run as often as expected. For example, you configure a schedule to run every hour from 8:00 a.m. to 12:00 p.m., and the schedule offset for your organization is 15 seconds. Your schedule runs at 8:00:15, 9:00:15, 10:00:15, 11:00:15, and 12:00:15.
Time Zone	Time zone used to display job execution time stamps in email notifications.

Default email notifications properties

Configure the default email notifications properties to set the default email addresses to use for job failure, warning, and success messages. Enter one or more valid email addresses. Separate email addresses with a comma (,) or semicolon (;).

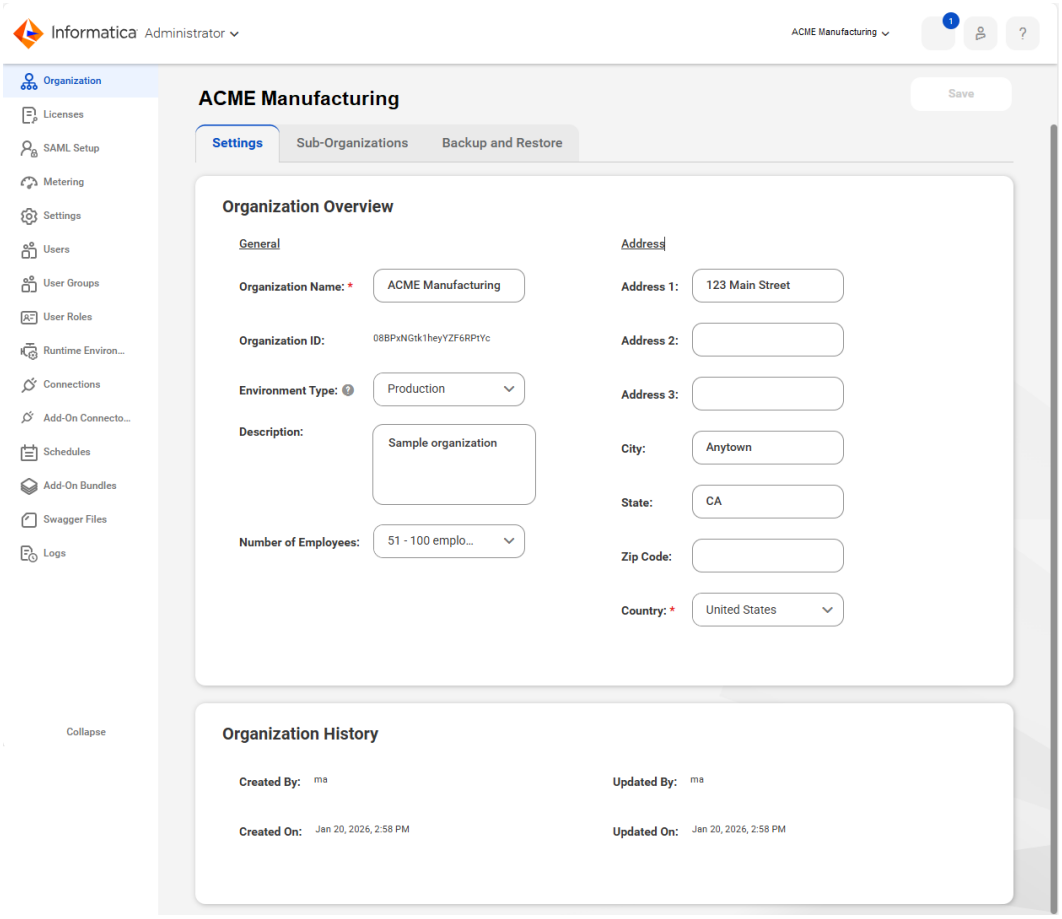
You can also set email notification properties at the task level. When you set email notifications in a task or taskflow, IDMC sends email to the addresses in the task or taskflow instead of the addresses configured for the organization.

Organization settings

You can configure general properties for your organization and sub-organizations and view organization history on the Organization Settings page.

To access the **Organization** page, in Administrator, select **Organization**.

The following image shows the **Organization Settings** page:



The general properties include the following information:

Overview information

The following table describes the overview properties:

Property	Description
Name	Name of the organization. If you change the organization name, the new name appears on the **Organization** menu after you log out and log back in.
ID	ID assigned to your organization when it was created. You cannot change an organization ID.
Parent Organization ID	When you view a sub-organization, this property displays the ID assigned to the parent organization. You cannot change an organization ID.
Organization Status	When you view a sub-organization, this property indicates whether the sub-organization is enabled or disabled.

Property	Description
Environment Type	Environment type for the organization, either Development, Production, QA, or Sandbox. IDMC sets the environment type based on the organization license. In some cases you can edit the value. The environment type is set in the following ways: - When you create your organization by starting your free trial, the environment type is Sandbox. - When you create a production organization or an additional production organization, the environment type is Production. - When you create a sandbox organization, the environment type is initially set to Sandbox. You can set the type to Development, QA, or Sandbox. There is no difference in functionality among the environment types. **Important:** Validate the value periodically to ensure it matches your organization's deployment environment. If the value is incorrect and you are unable to change it, contact Global Customer Support. **Note:** If you are the administrator of a sub-organization but not its parent organization, the user interface allows you to set the environment type to any type, including types that aren't available for your license. To avoid billing issues, don't change the environment type to one that isn't available for your license.
Description	Optional description of the organization.
Number of Employees	Number of employees in the organization.
Deny parent organization access to this sub-organization	When this option is checked, users in the parent organization cannot switch from the parent organization to the sub-organization. Users in the parent organization with the appropriate privileges can make only the following changes to the sub-organization: - Enable and disable the sub-organization - Update the sub-organization licenses - Edit the sub-organization properties such as the organization description and CLAIRE recommendation preferences This option is displayed on the **Organization** page for sub-organizations. This option can be changed when an administrator in the sub-organization logs in to the sub-organization. This option is read-only when a parent organization administrator views the organization properties for the sub-organization. This option is unchecked by default.

Address information

Use the address properties to specify the street address, zip code, state, and country of the organization.

History information

The organization history information displays the date and time that the organization was created, the user who created the organization, the date and time that the organization was last updated, and the user who last updated the organization. IDMC updates the history information when you make changes to the organization.

Sub-organizations

If your organization has the appropriate license, you can create one or more sub-organizations within your organization. Create sub-organizations to represent different business environments within your company.

For example, you might create sub-organizations to represent different clients or different departments in your organization.

You can create a sub-organization from the production organization, from an additional production organization, or from a sandbox organization.

When you create a sub-organization, the organization that you use to create a sub-organization becomes the parent organization. Each sub-organization can have only one parent, and it cannot contain another sub-organization.

Note: Sub-organizations must reside in the same POD (point of deployment) as the parent organization. In a CI/CD (continuous integration/continuous deployment) -oriented approach, all sub-organizations receive feature releases simultaneously, potentially resulting in issues and downtime during the same maintenance window.

Your organization's license controls the number of sub-organizations that you can create. To increase this number, contact Informatica Global Customer Support.

Creating sub-organizations provides the following advantages:

You can manage sub-organization licenses individually or you can automatically synchronize them with the parent organization licenses.

Each sub-organization inherits all feature, connector, and custom licenses from the parent organization except for the license to create sub-organizations and bundle licenses.

Based on your organization's licenses, you can manage sub-organization licenses in the either of the following ways:

- Manage the licenses for your sub-organizations individually. The administrators for the parent organization can disable, enable, and modify the expiration dates for the licenses for each sub-organization. Changes to one sub-organization do not affect other sub-organizations.
- Automatically synchronize sub-organizations licenses with the parent license.

You can manage users and assets separately.

Each sub-organization has its own set of users and assets.

Users whom you create in a sub-organization are unique to the sub-organization. They cannot log in to the parent organization or to other sub-organizations. Only administrators in the parent organization and users in the parent organization that have sub-organization access privileges can access the parent organization and all sub-organizations.

Assets such as mappings and tasks are also unique within an organization. Assets are not shared among sub-organizations or between the parent organization and any sub-organization. If you want to migrate an asset between organizations, export the asset from one organization and import it into a different organization.

You can share runtime environments.

Administrators in the parent organization can share Secure Agent groups with the sub-organizations. When you share Secure Agent groups, users in the sub-organizations can run jobs on the Secure Agents within the group.

Users of a parent organization or sub-organization can use only a Secure Agent that belongs to that organization or sub-organization. Users of a sub-organization cannot use a Secure Agent that belongs to the parent organization.

Note: Share a Secure Agent group when all agents in the group run only the Data Integration Server service. You cannot run other agent services' jobs on a shared Secure Agent group.

For more information about shared Secure Agent groups, see *Runtime Environments*.

You can share resources using bundles.

The Bundle Deployment feature lets you push bundles seamlessly from your parent organization to your sub-organizations. This ensures a smooth and efficient distribution of resources and features across your organizational structure.

For more information about bundles, see Chapter 11, “Bundle management” on page 99.

You can view aggregated IPU consumption metrics.

IPU (Informatica Processing Unit) consumption metrics for each sub-organization are aggregated and rolled up to their respective parent organization. This consolidation gives you a clear overview of how your resources are utilized. The consumption data is consolidated and rolled up within the organizational hierarchy until it reaches the main production organization.

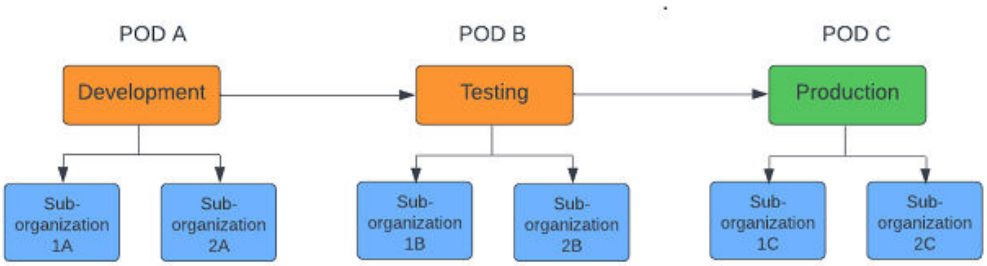
You can switch between organizations without logging in to each one.

Users in the parent organization that have privileges to view sub-organizations can switch between organizations without logging out and logging back in to IDMC.

Sub-organizations example

To observe CI/CD best practices, you want to create separate sub-organizations to represent different areas of your business, such as development, testing, and production. To achieve this, you first create separate parent organizations for development, testing, and production. Each parent organization should ideally be on a different POD, to safeguard against outages if a POD or organization becomes unavailable. Under each parent organization, you create sub-organizations that represent different clients or departments.

The following diagram shows the recommended hierarchy:



Using this hierarchy ensures a step-by-step flow of updates and improvements between environments, beginning with Development, moving to Testing, and concluding in Production. Isolating the environments in this way reduces the chances of unintended changes impacting other workflow stages.

For OEMs (original equipment manufacturers), the sub-organizations can represent individual clients. The structure allows the OEM to maintain control and oversee licensing, while allowing the individual clients to manage their development, testing, and production processes.

For corporations, the sub-organizations can represent different departmental divisions. This structure simplifies administrative monitoring and allows the parent organization to access assets, processes, and other resources set up by the departments.

Adding a sub-organization

To add a sub-organization, you can either create a new sub-organization or link existing organizations.

You can add a sub-organization in either of the following ways:

Create a sub-organization.

Log in to the organization that you want to be the parent organization and create a sub-organization. The new sub-organization is automatically linked to the parent organization.

Link existing organizations.

The organization that you link from becomes the parent organization and the organization that you link to becomes a sub-organization.

Creating a sub-organization

The administrator of a parent organization can create a sub-organization.

To create a sub-organization, you must be a native user with the Admin role or a native user with the "Suborg - create" and "Suborgs - view" privileges.

To create a sub-organization:

1. Log in to the organization that you want to be the parent organization.
2. Open Administrator and select **Organization**.
3. Open the **Sub-Organizations** tab and click **New Sub-Organization**.
4. Enter the properties for the sub-organization and click **Save**.

After you create a sub-organization, verify the licenses, and configure runtime environments, user accounts, and connections so that other people can use it.

Linking organizations

You can create a sub-organization by linking existing organizations. The organization that you link from becomes the parent organization and the organization that you link to becomes a sub-organization.

Before you link an organization, you need the organization ID for the organization that you want to link. You can find this information on the **Organization** page.

Note: If you link a sub-organization that has a license that the parent organization does not have, then the sub-organization loses the license.

You can link an organization if all of the following conditions apply:

- You have a user account with the organization.
- The organization is not the parent of another organization or a sub-organization of another organization.
- You are the administrator of the parent organization, and the parent organization has the license to create sub-organizations.
- The organization that you want to link as a sub-organization does not have the license to create sub-organizations.

You can later unlink the organizations.

To link organizations:

1. Log in to the organization that you want to be the parent organization.
2. Open Administrator and select **Organization**.
3. Open the **Sub-Organizations** tab and click **Link Sub-Organization**.
4. In the **Link Sub-Organization** dialog box, enter the following information:
    - The organization ID for the organization you want to link.

- The user name and password of an administrator in the organization that you want to set up as a sub-organization.

5. Click **Link Sub-Organization** to link the organization.

   The organization displays on the **Sub-Organizations** page.

Removing a sub-organization

To remove a sub-organization, you can unlink organizations or delete the sub-organization.

You can remove a sub-organization in either of the following ways:

Unlink an existing sub-organization from its parent organization.

An unlinked sub-organization becomes a stand-alone organization. You can link it to a different parent organization or re-link it to the original parent organization. If you obtain the license to create sub-organizations for the unlinked organization, you can make it the parent organization for a different sub-organization.

Delete the sub-organization.

When you delete a sub-organization, you delete all of the assets and data associated with the sub-organization. If you have a usage-based license, after deletion, the sub-organization continues to consume IPUs in the current billing period. For more information, see "IPU usage for disabled and deleted sub-organizations" on page 39.

Unlinking a sub-organization

You can unlink a sub-organization from your parent organization. After you unlink an organization, update the unlinked organization with the required licenses unless you plan to link it to a different parent organization.

You can unlink a sub-organization if all of the following conditions apply:

- You have an administrator account with the sub-organization you want to unlink.
- You are an administrator of the parent organization, and the parent organization has the license to create sub-organizations.
- No asset in the sub-organization that you want to unlink uses a shared Secure Agent group as the runtime environment. If any asset in the sub-organization uses a shared Secure Agent group as the runtime environment, update the asset to use a different runtime environment before you unlink the sub-organization.

  Note: This condition does not apply to Application Ingestion and Replication and Database Ingestion and Replication because Application Ingestion and Replication and Database Ingestion and Replication do not support shared runtime environments.

To unlink a sub-organization:

1. Log in to the parent organization.
2. Open Administrator and select **Organization**.
3. Open the **Sub-Organizations** tab.
4. Expand the Actions menu for the sub-organization that you want to unlink and select **Unlink**.
5. In the **Unlink** dialog box, enter the user name and password of a user in the sub-organization with the Admin role.
6. Click **Unlink**.

   The sub-organization is no longer linked to the parent organization.

Deleting a sub-organization

You can delete a sub-organization. When you delete a sub-organization, you delete all of the associated data permanently. If you have a usage-based license, the sub-organization's metering information is retained in the parent organization.

You can delete a sub-organization if you are the administrator of the parent organization.

1. Log in to the parent organization.
2. Open Administrator and select **Organization**.
3. Open the **Sub-Organizations** tab.
4. Expand the Actions menu for the sub-organization that you want to unlink and select **Delete**.

   Note: If you have a usage-based license, the sub-organization continues to consume IPUs until the end of billing period in which the deletion was completed. For more information, see "IPU usage for disabled and deleted sub-organizations" on page 39.

Disabling or enabling a sub-organization

If you are the administrator for a parent organization, you can disable or enable a sub-organization.

When you create a sub-organization, the sub-organization is enabled by default. You might want to disable a sub-organization if you have a separate license agreement with the sub-organization and the license agreement expires. You can re-enable the sub-organization after you disable it.

You can disable or enable a sub-organization even if the sub-organization administrator blocks parent organization access to the sub-organization.

You can perform the following actions:

Disable a sub-organization

When you disable a sub-organization, the organization exists, but sub-organization users cannot log in to the sub-organization or access it through the REST API. Scheduled jobs in the sub-organization do not run.

If you have a usage-based license and you disable a sub-organization, it continues to consume IPUs. For more information, see "IPU usage for disabled and deleted sub-organizations" on page 39.

Enable a sub-organization

When you enable a sub-organization, sub-organization users can log in to the sub-organization and access assets and perform tasks based on their user roles. Users with the appropriate privileges can access the sub-organization through the REST API. Scheduled jobs resume according to their schedules.

Disable or enable a sub-organization on the **Sub-Organizations** tab of the **Organizations** page. In the **Actions** menu for the sub-organization, select **Disable** or **Enable**.

Switching to a different organization

If you are an administrator in a parent organization or a user in a parent organization that has privileges to view sub-organizations, you can switch among organizations. You do not have to log out and log back in to IDMC.

Note: If you switch from a parent organization to a sub-organization, you can't perform the following operations in the sub-organization:

- Create or import data transfer tasks

- Create or import dynamic mapping tasks
- Create, validate, publish, and run taskflows
- Export and import taskflows

To switch to a different organization:

▶ From the **Organization** menu in the upper right corner, select the organization that you want to view.

Denying parent organization access to a sub-organization

If you are the administrator for a sub-organization, you can deny parent organization access to the sub-organization.

When you deny access to the sub-organization, users in the parent organization cannot switch from the parent organization to the sub-organization. Users in the parent organization with the appropriate privileges can make only the following changes to the sub-organization:

- Enable and disable the sub-organization
- Update the sub-organization licenses
- Edit the sub-organization properties such as the organization description and CLAIRE recommendation preferences

To deny parent organization access to the sub-organization, log in to the sub-organization as an administrator. On the **Organization** page, enable the **Deny parent organization access to this sub-organization** option.

Add-on connectors in sub-organizations

To use an add-on connector in a sub-organization, you must install the connector in the parent organization. You cannot install add-on connectors in a sub-organization.

Sub-organizations inherit all connector licenses from the parent organization. If a sub-organization should not use a specific connector, disable the connector license for the sub-organization. For more information about editing and disabling licenses for a sub-organization, see "Editing sub-organization licenses" on page 32.

Exporting and importing assets in sub-organizations

Export and import assets in a sub-organization in the following ways:

- Log in to the sub-organization and export or import assets from within the sub-organization.
- Parent organization administrators can log in to the parent organization, switch to the sub-organization, and import or export Data Integration assets.

  Note: This condition does not apply to taskflows and Application Integration assets.

Sub-organization capability and service support matrix

The capability of sub-organizations vary based on the Informatica service.

The following table summarizes the capabilities for each service:

Service	Licensing capabilities	Metering capabilities	Bundles	Shared Secure Agents	Asset Access
Cloud Data Integration (CDI)	The parent organization has a license to create sub-organizations.	The parent organization must use feature-based licenses.	Available	Supported	Not supported
Cloud Application Integration (CAI)	The parent organization has a license to create sub-organizations.	The parent organization must use feature-based licenses.	Available	Not supported	Not supported
Cloud Data Governance & Catalog (CDGC)	The parent organization has a license to create sub-organizations.	The parent organization must use feature-based licenses.	Not available	Not supported	Not supported
Master Data Management (MDM)	The parent organization has a license to create sub-organizations.	The parent organization must use feature-based licenses.	Not available	Not supported	Supported
Cloud Data Quality (CDQ)	The parent organization has a license to create sub-organizations.	The parent organization must use feature-based licenses.	Available	Supported	Not supported

Additional production organizations and sandbox organizations

If your organization has the appropriate license, you can create additional production organizations and sandbox organizations from the production organization. Create these organizations when you need separate organizations for different assets and users, but you want to manage all IPU usage from the production organization. Additional production organizations and sandbox organizations are automatically linked to the production organization.

You can create the following types of organizations:

Additional production organization

This is a separate production organization. It functions in the same way as the production organization, except that you cannot create additional production organizations or sandbox organizations from an additional production organization.

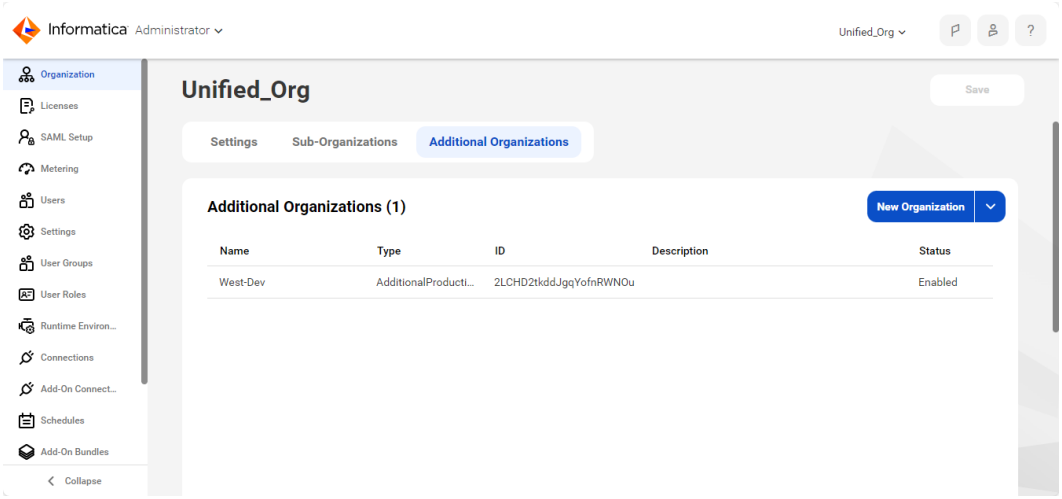
Sandbox organization

This is an organization that is typically used for asset development and testing. There is no difference in functionality between a sandbox organization and an additional production organization.

Additional production organizations and sandbox organizations are completely independent of the production organization and each other. They do not share assets, connections, runtime environments, or users. However, you can export and import assets between them.

The additional production organizations and sandbox organizations that you create inherit all licenses and editions from the production organization. If the production organization has the license to create sub-organizations, administrators in the additional production organizations and sandbox organizations can create sub-organizations for their organizations.

Additional production organizations and sandbox organizations appear on the **Additional Organizations** tab of the **Organization** page in the production organization. The **Additional Organizations** tab is shown in the following image:



When you create an additional production organization or a sandbox organization, you must enter the organization name and a user name for the organization administrator. You can log in to the new organization using this user account and add other users and assets.

After you create an additional production organization or a sandbox organization, you cannot disable or delete it, and you cannot unlink it from the production organization. If you need to do this, contact Informatica Global Customer Support.

Creating an additional organization

Create an additional production organization or a sandbox organization on the **Additional Organizations** tab of the **Organization** page.

Note: Creating additional production organizations or sandbox organizations can incur additional IPU charges for your organization.

To create an additional organization, you must be a native user and have the Admin role with "AdditionalOrg creation" and "AdditionalOrg view" privileges.

To create an additional organization:

1. Log in to the production organization.
2. Open Administrator and select **Organization**.
3. Open the **Additional Organizations** tab.
4. Click **New Organization** and select **Additional Production Organization** or **Sandbox Organization**.

5. On the **New Organization** page, enter a user name for the administrator of the new organization and the organization name.

   The user name must be unique. It must be a valid email address, or it can contain only alphanumeric characters, hyphens, underscores, periods, and apostrophes.
6. Optionally, enter a description, the number of employees, and the address information.
7. Click **Save**.

   The new organization appears on the **Additional Organizations** tab, and the status is Enabled. IDMC sends a welcome email for the new organization administrator to your email address.
8. Log out of the production organization.
9. Click the Confirm Account link in the welcome email, and follow the prompts to activate the account for the new organization administrator.

   When you finish activating the account, IDMC logs you in to the new organization.

After you log in to the new organization, you can add other users and create runtime environments, connections, and assets.

CHAPTER 3

Licenses

Licenses determine the IDMC subscription level for the organization and provide access to IDMC features, connectors, and bundles.

As an administrator, you can review the licenses that are set up for your organization, verify license expiration dates, and manage sub-organization licenses.

You can also review metrics for licensed job limits and usage. For information about reviewing metrics, see Chapter 4, “Metering” on page 34.

License categories

Licenses are categorized as edition licenses, connector licenses, and custom licenses.

The following license categories are available:

Edition licenses

Edition licenses can be feature-based or usage-based. Feature-based edition licenses provide access to IDMC features such as Data Integration mapping tasks, business services components, and fine-grained security. Usage-based edition licenses, such as the license for the Intelligent Cloud Data Management feature, provide access to IDMC using a pre-paid consumption model.

Connector licenses

Connector licenses provide connectivity to entities such as Amazon Redshift, Microsoft SQL Server, and Oracle.

Custom licenses

Custom licenses are licenses that are not part of an edition. They provide access to features, packages, or bundles. If your organization uses a custom license that provides access to a feature that is also included in an edition license, the terms of the custom license override the terms of the edition license.

License types

When you create an organization, IDMC assigns the organization a license type for each licensed edition.

IDMC uses the following types of licenses:

Trial

You can use the edition free of charge for a 30-day period. At the end of the trial period, you can subscribe to the edition. A trial subscription might provide limited access to the features, connectors, and packages that are associated with the license.

Subscription

You can use the licensed edition for the duration of the contract period. Near the end of the contract period, IDMC indicates that the contract is about to expire. Renew the contract to continue to use the edition.

Free subscription

You can use the synchronization task free of charge. A free subscription might provide limited access to the features of the synchronization task.

Sub-organization licenses

A sub-organization has licenses that are maintained by the parent organization. If a sub-organization requires a license that does not belong to the parent organization, contact Informatica Global Customer Support to obtain the license for the parent organization.

When you create a sub-organization, each sub-organization inherits licenses from the parent organization as custom licenses. The sub-organization inherits all licenses except for the following licenses:

- The license to create sub-organizations
- Bundle licenses. To use a bundle in the sub-organization, a user in the sub-organization must install the bundle.

If your organization has the Intelligent Cloud Data Management license, the sub-organization also has this license. The parent organization and the sub-organization share the balance of Informatica processing units (IPUs).

You can manage sub-organization licenses in the following ways:

Manage the licenses individually

When you manage licenses individually, administrators for the parent organization can disable, enable, and shorten the expiration dates for the inherited licenses. They manage the licenses separately for each sub-organization. Sub-organization administrators can view licenses but cannot change them.

This is the default option.

Automatically synchronize sub-organization licenses with the parent organization

If you have the appropriate license, you can automatically synchronize sub-organization licenses with the parent organization. When this license is enabled, each time a license is changed in the parent organization, all sub-organizations inherit the license change.

You might want to enable license synchronization when your organization has many sub-organizations and the sub-organizations have the same licenses.

If license synchronization is not enabled for your organization, then you must manage sub-organization licenses individually.

Note: If you link a sub-organization that has a license that the parent organization does not have, the sub-organization loses the license.

Editing sub-organization licenses

You can edit sub-organization licenses if you are an administrator in the parent organization and if license synchronization between the parent organization and sub-organizations is not enabled. You can edit sub-organization licenses from within the parent organization or from within the sub-organization.

1. Log in to the parent organization.
2. To edit licenses from within the parent organization:
   a. Open Administrator and select **Organization**.
   b. Click **Sub-Organizations**.
   c. Select the sub-organization for which you want to edit licenses.
   d. Click **Licenses**.
3. To edit licenses from within the sub-organization:
   a. From the **Organization** menu in the upper right corner, select the sub-organization for which you want to edit licenses.
   b. Open Administrator and select **Licenses**.
4. Select licenses to enable features, and clear licenses to disable features.
5. Optionally, modify expiration dates.

   All licenses must have expiration dates. You cannot extend a license past its original expiration date.
6. Click **Save**.

Synchronizing licenses with the parent organization

You can automatically synchronize sub-organization licenses with the parent organization. Each time a license is changed in the parent organization, all sub-organizations inherit the license change.

To enable license synchronization, contact Informatica Global Customer Support and request the license for this feature. When the license is enabled for the parent organization, license synchronization with the sub-organizations happens automatically. The parent organization administrator does not have to take any action to synchronize the licenses.

Note: When the license for this feature is enabled, you cannot edit sub-organization licenses individually.

When the license for this feature is enabled and you disable a sub-organization, the sub-organization loses its license settings. When you re-enable the sub-organization, the sub-organization inherits all license settings from the parent organization.

License synchronization between a parent organization and sub-organizations does not affect the license meter counts in the sub-organizations.

Configuring the organization type

When you view the **Licenses** page, you can view details about the organization licenses and other details, including the organization type. The organization type specifies whether the organization is a trial, production, or sandbox organization. If you are a parent organization administrator, you must configure the organization type for each sub-organization.

The organization type is editable when a parent organization administrator views the **Licenses** page for a sub-organization. It is not editable by sub-organization users or in a parent organization.

If you are a parent organization administrator, you can configure the organization type for the sub-organization from within the parent organization or from within the sub-organization. You can change the organization type after you configure it.

1. Log in to the parent organization.
2. To edit the organization type from within the parent organization:
   a. Open Administrator and select **Organization**.
   b. Click **Sub-Organizations**.
   c. Select the sub-organization for which you want to configure the type.
   d. Click **Licenses**.
   e. In the **Type** list in the Sub-Organization Details area, select the organization type.
3. To edit the organization type from within the sub-organization:
   a. From the **Organization** menu in the upper right corner, select the sub-organization for which you want to configure the type.
   b. Open Administrator and select **Licenses**.
   c. In the **Type** list in the Sub-Organization Details area, select the organization type.
4. Click **Save**.

License expiration

When a license expires, you cannot access the features, connectors, or packages that are associated with the license. Scheduled jobs that are associated with the license are also disabled. If all licenses for the organization expire, you cannot log in to IDMC.

You can review the expiration date for licenses on the **Licenses** page in Administrator. To extend a license, contact Informatica Global Customer Support. After you extend a license, you can access the associated features, connectors, and packages, and the scheduled jobs resume processing.

CHAPTER 4

Metering

You can view metering information for your organization and sub-organizations. View metering information on the **Metering** page.

The information on the **Metering** page depends on your organization's license agreement:

- If you have a usage-based license, the **Metering** page displays a dashboard that includes the number of Informatica processing units (IPUs) that you have purchased, have consumed, and have remaining in the current billing cycle. You can also view meters that show the number of IPUs that you have used for each service.
- If you have feature-based licenses, the **Metering** page displays the amounts of computing resources that your organization uses and has remaining. The page also displays the usage limits set through your organization's licenses. The page doesn't mention IPUs since IPUs do not apply to feature-based licenses. The **Metering** page displays information as a dashboard or as a table, based on your licenses.

Informatica processing unit metrics

If your organization has the Intelligent Cloud Data Management feature, your license is based on IDMC usage.

An Informatica processing unit (IPU) is a unit of credit used to pre-pay for using IDMC scalars such as Compute Units and Events Processed. You can monitor your IPU balance and usage on the **Metering** page.

Users with the Admin role receive notification through email when the organization crosses the 25, 50, 75, 95, and 100% IPU consumption thresholds.

IPU scalars

IPUs are based on scalar values. Appropriate scalars are used for each service.

For example, Data Integration usage is measured by Compute Units, and Cloud Integration Hub usage is measured by Events Processed.

The following table lists the primary scalars and their unit of measure:

Scalar value	Unit of measure	Description
API Calls	API call	Number of custom API calls.
Compute Units	Hour	Processing capacity used or consumed.

Scalar value	Unit of measure	Description
Connection Definitions	Connection	Number of connections. If your organization uses a secrets manager to store connection credentials, this number represents the number of connections that are configured to retrieve credentials from the secrets manager in a single day.
Daily Assets Stored	Asset	Number of assets that are stored by each respective service in a single day.
Data Volume	Gigabyte	Volume of data transferred, transformed, or incorporated.
Events Processed	Event	Inbound and outbound instances of data accessing an intermediate storage layer.
Objects Processed	Object	Number of objects processed during assessment and conversion of PowerCenter assets to Cloud Data Integration assets.
Organizations	Organization	Number of additional production organizations, sub-organizations, and sandbox organizations. If you delete a sub-organization, this number includes the deleted sub-organization until the next billing period starts.
Queries	Query	Number of queries submitted.
Rows Processed	Row	Number of rows processed from underlying database logs.

IPU meters

IPU meters are the services and features included with Intelligent Cloud Data Management.

The following table lists the IPU meters with the applicable scalars and metrics:

Meter	Scalar value	Metric
Address Verification - Batch - ROW (Excluding Ireland and Great Britain)	Rows Processed	Per thousand rows
Address Verification - Geo Coding	Rows Processed	Per thousand rows
Address Verification Batch - IRL/GBR (For Ireland and Great Britain)	Rows Processed	Per thousand rows
Address Verification - Real Time -ROW (Excluding Ireland and Great Britain)	Rows Processed	Per thousand rows
Address Verification - Real Time - IRL/GBR (For Ireland and Great Britain)	Rows Processed	Per thousand rows
Advanced Data Integration	Compute Units	Per hour
Advanced Data Integration with Advanced Serverless	Compute Units	Per hour

Meter	Scalar value	Metric
Advanced Data Quality	Compute Units	Per hour
Advanced Data Quality with Advanced Serverless	Compute Units	Per hour
API Center	API Calls	Per million API calls
API Management	API Calls	Per million API calls
Application Ingestion and Replication - Change Data Capture	Rows Processed	Per million rows
Application Ingestion and Replication	Data Volume	Per gigabyte
Application Integration (Turbo mode)	API Calls	Per thousand API calls
Application Integration (Classic mode) **Note:** Compute units are used for IPUs subscribed to before February 7, 2025.	Compute Units	Per hour
Application Integration with Advanced Serverless (Turbo mode)	API Calls	Per thousand API calls
Application Integration with Advanced Serverless (Classic mode) **Note:** Compute units are used for IPUs subscribed to before February 7, 2025.	Compute Units	Per hour
B2B Gateway **Note:** Use this IPU meter to view the metering information for B2B Gateway prior to the October 2025 release.	Compute Units	Per hour
B2B & Industry Solutions **Note:** Use this IPU meter to view the metering information for both B2B Gateway and Industry Solutions from the October 2025 release and later.	Compute Units	Per hour
CLAIRE - GPT	Queries	Per thousand queries
Customer Managed Keys **Note:** Within an organization hierarchy, for each organization with active customer managed keys, an additional charge is applied for a percent of all product IPU usage. For example, if the rate card says 0.2, then an additional charge of 20% is applied.	Org (shown as "Qualified Usage" in the IPU dashboard and reports)	Percent for total IPUs
Data Access Management - Policy Pushdown	Daily Assets Stored	Per hundred assets
Data Access Management - Queries	API Calls	Per thousand API calls
Data Access Management - Rows Processed	Rows Processed	Per million rows

Meter	Scalar value	Metric
Data Governance and Catalog - Catalog Records	Daily Assets Stored	Per hundred thousand assets
Data Governance and Catalog - Governance Records	Daily Assets Stored	Per thousand assets
Data Governance and Catalog - Metadata Record Consumption	API Calls	Per thousand API calls
Data Governance and Catalog - Scanner Execution	Compute Units	Per hour
Data Governance and Catalog - Scanner Execution with Serverless	Compute Units	Per hour
Data Integration	Compute Units	Per hour
Data Integration - Change Data Capture	Rows Processed	Per million rows
Data Integration with Advanced Serverless	Compute Units	Per hour
Data Marketplace Records	Daily Assets Stored	Per hundred daily assets
Data Masking	Compute Units	Per hour
Data Quality	Compute Units	Per hour
Data Quality with Advanced Serverless	Compute Units	Per hour
Data Quality - Real Time	Rows Processed	Per million rows
Data Validation	Compute Units	Per hour
Database Ingestion and Replication	Data Volume	Per gigabyte
Database Ingestion and Replication - Change Data Capture	Rows Processed	Per million rows
File Ingestion and Replication	Data Volume	Per gigabyte
Industry Solutions **Note:** Use this IPU meter to view the metering information for Industry Solutions prior to the October 2025 release. To view metering information for the October 2025 release and later, see the B2B & Industry Solutions meter.	Compute Units	Per hour
INFACore	Compute Units	Per hour
Integration Hub	Events Processed	Per thousand events
Secrets Manager	Connection Definitions	Daily connection definitions
SQL ELT	Rows Processed	Per million rows
Streaming Ingestion and Replication	Data Volume	Per gigabyte

Meter	Scalar value	Metric
Additional Production Organizations	Organizations	Per organization
Associated Sub Organizations IPU Usage **Note:** This is a non-billable, informational meter that aggregates IPU usage for all sub-organizations of sandbox and additional production organizations, plus 6 IPUs per sub-organization.	Organizations	Per organization
Pre-release Organizations	Organizations	Per organization
Sandbox Organizations	Organizations	Per organization
Sub Organizations	Organizations	Per organization

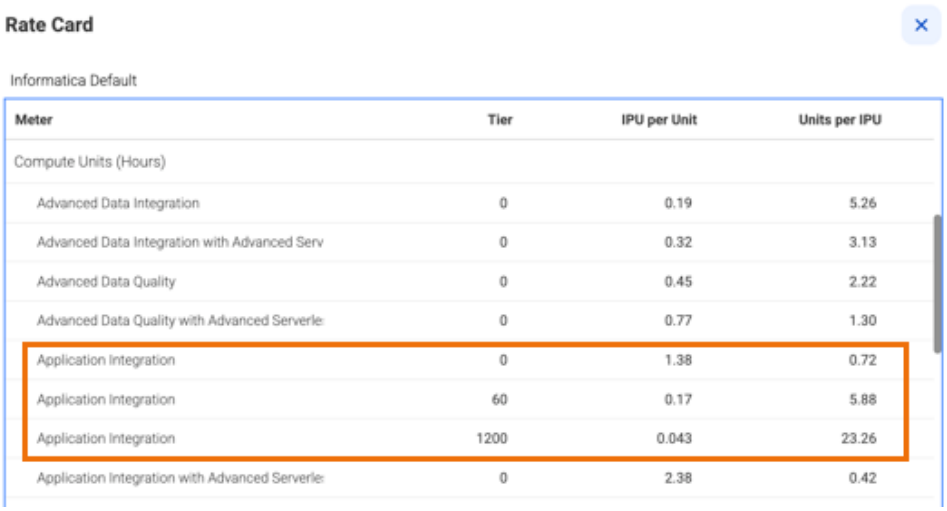
Rate cards

The rate card lists the current IPU rate for each meter and tier.

Note: The sample rate cards in this section might not reflect the current IPU rates for your organization.

Meters with tiered rates

Meters with tiers display the IPU rate per tier. The following image shows a sample rate card with an IPU rate per tier for the Application Integration meter:



In this example, the Application Integration meter uses compute units (hours) as the scalar. The IPU rate for the Application Integration meter is calculated in the following way:

- The first 60 compute hours cost 1.38 IPUs per hour.
- 61 to 1,200 compute hours cost 0.17 IPUs per hour.
- More than 1,200 compute hours cost 0.043 IPUs per hour.

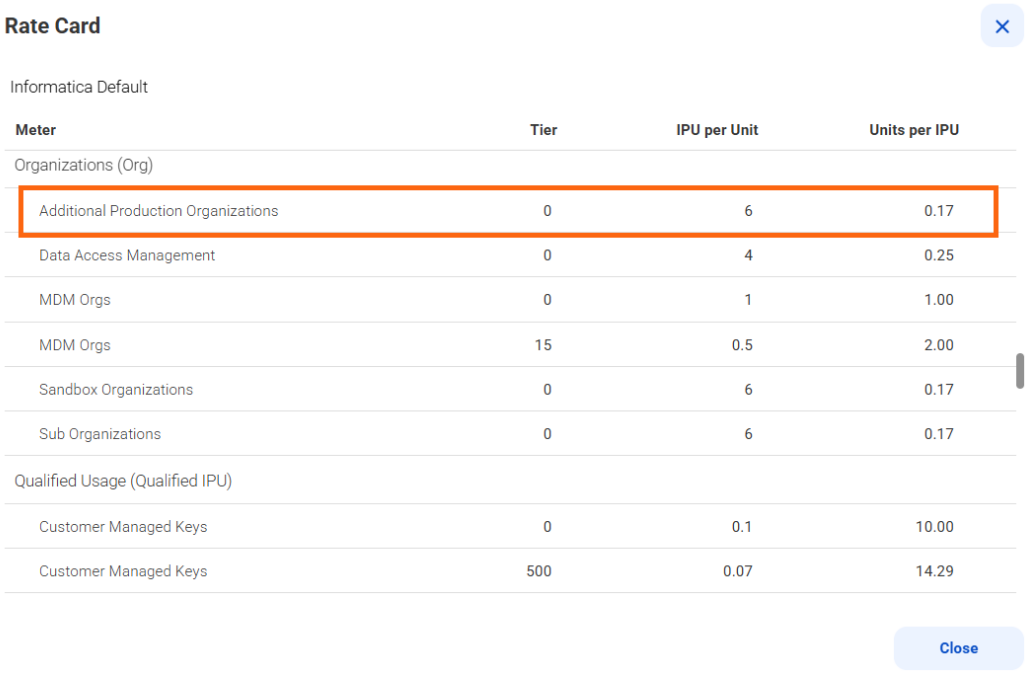
For example, if you use 5,000 compute hours in a billing period, you can calculate the total IPUs using the following equation:

```
(60 * 1.38) + (1140 * 0.17) + (3,800 + 0.043) = 440 IPUs
```

When the next billing period begins, the compute unit count resets to zero.

Meters with flat rates

Meters without tiers display a flat IPU rate. The following image shows a sample rate card with an IPU rate for the Additional Production Organizations meter:



Rate Card

Informatica Default

Meter	Tier	IPU per Unit	Units per IPU
Organizations (Org)			
Additional Production Organizations	0	6	0.17
Data Access Management	0	4	0.25
MDM Orgs	0	1	1.00
MDM Orgs	15	0.5	2.00
Sandbox Organizations	0	6	0.17
Sub Organizations	0	6	0.17
Qualified Usage (Qualified IPU)			
Customer Managed Keys	0	0.1	10.00
Customer Managed Keys	500	0.07	14.29

Close

Because the Additional Production Organizations meter doesn't have tiers, each additional production organization has a flat rate of 6 IPUs per billing period.

IPU usage for disabled and deleted sub-organizations

Sub-organizations are counted in the Organizations scalar value for the Sub-organizations meter. Sub-organizations also consume IPUs based on other meters and scalar values.

Note the following rules regarding IPU consumption for disabled and deleted sub-organizations:

Disabled sub-organizations

When you disable a sub-organization, the sub-organization continues to consume IPUs for meters that use the following scalars:

- Records stored
- Organizations
- Qualified Usage

Deleted sub-organizations

After you delete a sub-organization, that sub-organization consumes six IPUs in the current billing period and doesn't consume IPUs in subsequent billing periods.

After a sub-organization is deleted, the parent organization's metering page continues to display the sub-organization's metering data in historical views.

Viewing IPU metrics

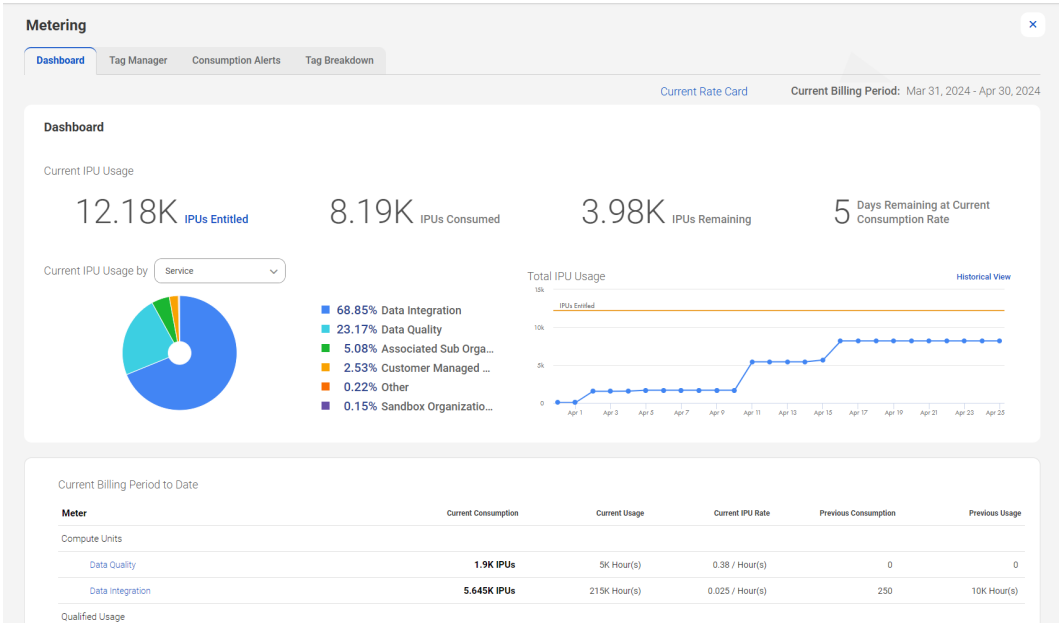
View IPU metrics on the **Metering** page. The **Metering** page displays IPU information in a dashboard view.

The dashboard shows IPU usage for the organization that you're logged in to and linked organizations such as sub-organizations, additional production organizations, and sandbox organizations.

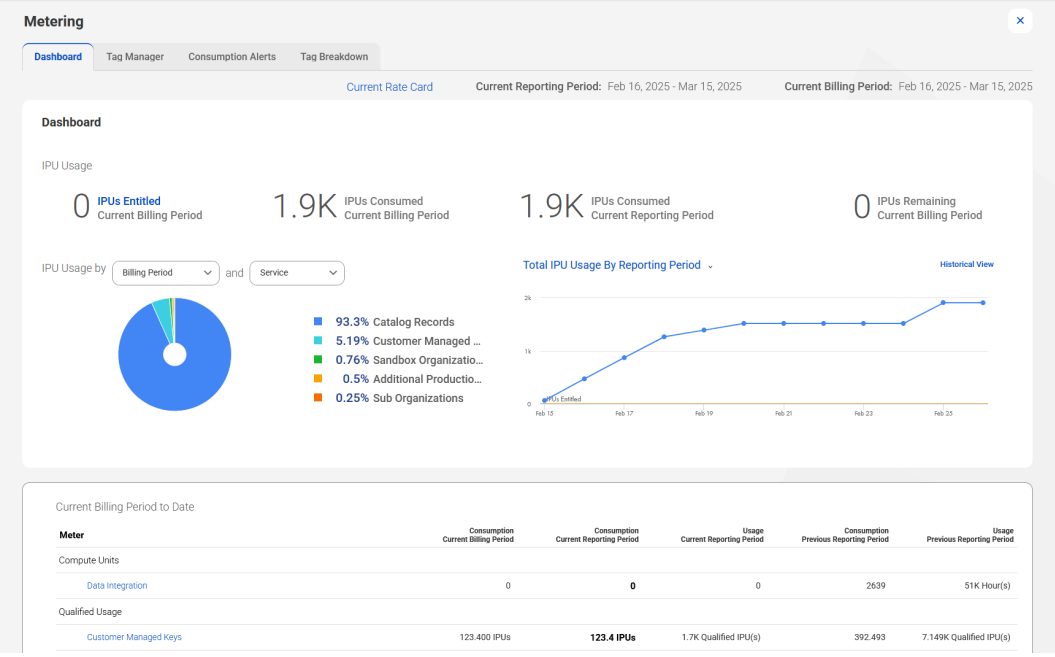
Views on the **Metering** page display metrics based on billing periods and, if applicable, reporting periods. Billing and reporting periods are different than your contract length. Note that billing and reporting periods are based on Coordinated Universal Time (UTC), so the current billing period and current reporting period might appear as a past or future date compared to your time zone.

You can download reports from the **Metering** page. For more information about the data you can download, see "IPU metrics reports" on page 43.

The following image shows the metering dashboard for an account that's set up for monthly billing periods:



The following image shows the metering dashboard for an account that's set up for annual billing periods:



Note: The **Metering** page displays IPU usage metrics that are rounded to three decimal places. The detailed download reports store exact metrics.

IPU billing periods

IPU billing periods can be on a monthly or annual basis. The **Metering** page for monthly billing is slightly different from the page for annual billing.

Monthly billing basis

If your billing period is on a monthly basis, you can view IPU information by billing period. The Metering 1.0 model uses monthly billing.

The metering dashboard for monthly billing includes the following panels:

Current IPU Usage

The **Current IPU Usage** panel includes the following information:

- The total number of IPUs entitled, consumed, and remaining in the current billing period.
- An estimate of the number of days remaining before the IPU balance is depleted based on current usage.
- The IPU usage for the billing period. You can view usage by meter, project, or folder.
  If you view the usage by project or folder, you have the option to see historical consumption by asset location.
- A graph that shows the total IPU usage per day.

To view historical information on IPU consumption, click **Historical View** and select the number of billing periods to include.

Current Billing Period to Date

The **Current Billing Period to Date** panel includes the following information for most meters:

- The IPU rate.

- The IPUs consumed in the current billing period and the previous billing period.
- Usage based on scalars such as compute units, data volume, and rows processed for the current billing period and previous billing period.

To view the details for a meter, click on the meter name. The details page shows the current usage by IPU and scalar, an historical usage graph that shows IPU usage for the selected number of billing periods, and detailed usage information by date.

Annual billing basis

If your billing period is on an annual basis, you can view IPU information by billing period or by reporting period, which is monthly. The Flex IPU metering model uses annual billing.

The metering dashboard for annual billing includes the following panels:

IPU Usage

The **IPU Usage** panel includes the following information:

- The total number of IPUs entitled, consumed, and remaining in the current billing period.
- The total number of IPUs consumed in the current reporting period. The reporting period is the current month.
- The IPU usage for the billing period or reporting period. You can view usage by meter, project, or folder.
  If you view the usage by project or folder, you have the option to see historical consumption by asset location.
- A graph that shows the total IPU usage per day for the billing period or reporting period.

To view historical information on IPU consumption, click **Historical View** and select the number of reporting periods to include.

Current Billing Period to Date

The **Current Billing Period to Date** panel includes the following information for most meters:

- The IPUs consumed in the current billing period.
- The IPUs consumed in the current reporting period and the previous reporting period.
- Usage based on scalars such as compute units, data volume, and rows processed for the current reporting period and previous reporting period.

To view the details for each meter, click the meter name. The details page shows the current usage by IPU and scalar, an historical usage graph that shows IPU and scalar usage for the selected number of reporting periods, and detailed IPU usage information by reporting period.

IPU metrics for multiple organizations

You can view IPU usage information for sub-organizations, sandbox organizations, and additional production organizations.

You can view the following information for organization meters on the metering dashboard in the **Current Billing Period to Date** panel:

Meter	Description
Sandbox Organizations	Displays total IPU usage and consumption by scalar for the sandbox organizations directly linked to the organization that you're logged in to. Includes a link to the details page, which shows details for each sandbox organization.
Sub Organizations	Displays total IPU usage and consumption by scalar for the sub-organizations directly linked to the organization that you're logged in to. Includes a link to the details page, which shows details for each sub-organization.
Additional Production Organizations	Displays total IPU usage and consumption by scalar for the additional production organizations directly linked to the organization that you're logged in to. Includes a link to the details page, which shows details for each additional production organization.
Associated Sub Organizations IPU Usage	When logged in to the production organization, displays total IPU usage for all sub-organizations of sandbox or additional production organizations, along with the rate of 6 IPUs per sub-organization.

For example, a production organization has two sandbox organizations and one additional production organization. The sandbox organizations and the additional production organization each have a sub-organization.

When you log in to the production organization, you can view the following metering information:

- IPU usage, consumption by scalar, and detailed metering data for the production organization.
- Total IPU usage and consumption by scalar for the two sandbox organizations.
- IPU usage and consumption by scalar for the additional production organization.
- Total IPU usage for the sandbox organizations' and additional production organization's sub-organizations.

When you log in to one of the sandbox organizations, you can view the following metering information:

- IPU usage, consumption by scalar, and detailed metering data for the sandbox organization.
- IPU usage and consumption by scalar for the sandbox organization's sub-organization.

Note: A lag of up to 10 minutes might occur before a linked organization's first IPU usage of the day is reflected on the production organization's **Metering** page.

IPU metrics reports

You can download a report that includes a summary of IPU usage, IPU usage by asset location, or a report that includes details for a particular meter.

You can download the following types of reports:

IPU usage history report

The IPU usage summary report includes a summary of IPU usage for the production organization and its sub-organizations, additional production organizations, and sandbox organizations. The report includes data for the selected billing or reporting periods. The report includes the following data by month for each meter:

- Scalar utilization
- IPU consumption
- Organization ID and name

Download the IPU summary report from the **IPU Usage History** page.

IPU metadata breakdown report

The IPU metadata breakdown report includes IPU consumption in the selected billing or reporting periods, broken down either by project and folder or by metering tag. This report includes the following usage:

- Product meter usage for production organizations and their linked organizations.

  Note: This usage might not have project attributes depending on the meter. For more information about which meters have project attributes, see *REST API Reference*.

- The recurring IPU cost to directly link organizations.

  Note: This usage isn't associated with project attributes.

Download the IPU metadata breakdown report from the **Tag Breakdown** tab or the **IPU Usage by Asset Metadata** page.

Table view report

The table view report includes the information that's currently shown in the table on an IPU meter's details page. The report includes only information from the organization that you're in when you request the data. Download a table view report from the IPU meter's details page.

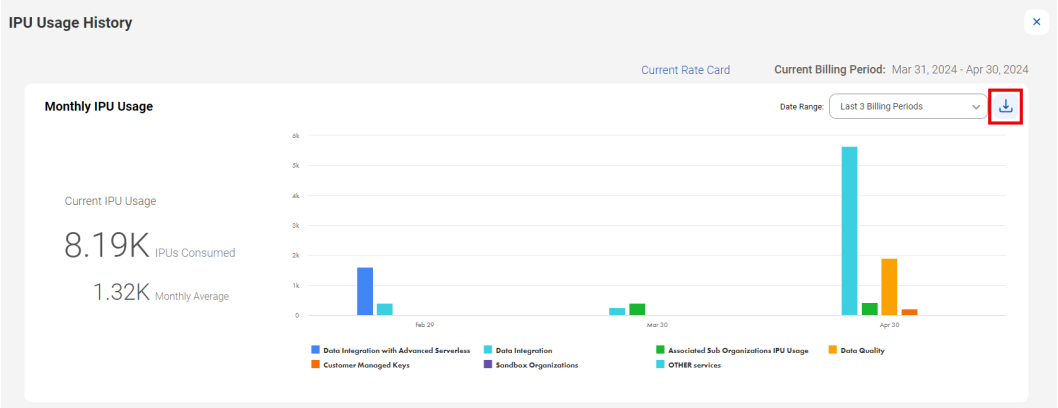
Detailed service report

For some meters, you can download a report that includes service details for a specific date range. The report includes only information from the organization that you're in when you request the data. Download a detailed service report from the IPU meter's details page.

Asset breakdown report

The asset breakdown report includes the information that's shown on an IPU meter's asset breakdown tab. The report includes only information from the organization that you're in when you request the data. Download an asset breakdown report from the asset breakdown tab of the IPU meter's details page.

To download a report, click the **Download** icon. The following image shows the **Download** icon on the **IPU Usage History** page:



Downloading an IPU usage history report

Download a summary report from the **IPU Usage History** page.

1. On the **Metering** page, click **Historical View**.
2. Select the billing periods that you want to include in the report, up to the last 13 billing periods.
3. Click the **Download** icon.
4. Click **Export**.

Downloading an IPU metadata breakdown report

Download an IPU usage report broken down by metadata, such as asset location or metering tags.

1. From the **Metering** page, navigate to either the **Tag Breakdown** tab or the **IPU Usage by Asset Metadata** page.
2. Select whether you want to view IPU usage by project and folder or by a metering tag.
3. Select the billing periods that you want to include in the report, up to the last 13 billing periods.
4. Click the **Download** icon.

Note: Not all meters include project and folder information. If you delete an asset before IDMC processes the metering data, the metering data for the asset also doesn't include the project and folder. The user interface and the downloaded report display assets without project and folder information in a single row. The project value for this row is blank.

Downloading an IPU meter report

Download a report for a specific meter on the meter's details page. You can download a report that's based on the overview table or the asset breakdown table. For some meters, you can download a detailed service report.

1. On the **Metering** page, in the **Current Billing Period to Date** area, click the meter name.
2. Navigate to the **Overview** tab or the **Asset Breakdown** tab, based on the type of report you want to download.
3. Select the billing periods that you want to include in the report.
4. Click the **Download** icon.

5. If the option to download a detailed service report is available, the **Metering Usage Report** dialog box appears when you click the **Download** icon. Perform one of the following tasks:
   - Select the table view report, and then click **Export**.
   - Select the detailed service report, and then select the date range for the report and click **Export**.

Note: Not all meters include asset information, such as the asset name, asset type, project, and folder. If you delete an asset before IDMC processes the metering data, the metering data for the asset also doesn't include this information. On the **Asset Breakdown** tab, the asset name, asset type, project, and folder are blank when the information isn't available. In the downloaded report, the asset information is *null* if it isn't available.

Data Integration detailed reports

Data Integration detailed reports provide metering usage details for jobs that your organization ran within a specified time period.

Detailed reports for Data Integration can include the following fields:

Field	Description
Task ID	Unique ID to identify the task.
Task Name	Name of the job.
Task Object Name	Name of the object used in the job. Applicable to replication tasks and synchronization tasks.
Task Type	Type of task, for example mapping task or replication task.
Task Run ID	Run ID of the job.
Project Name	Project that contains the task.
Folder Name	Folder that contains the task. If the task is located directly under the project, this field is blank.
Org ID	Unique identifier for the organization.
Environment ID	Federated ID of the runtime environment.
Environment	Name of the runtime environment.
Cores Used	Number of cores used by the job.
Assigned SCU:	Requested serverless compute units. Applicable to serverless reports.
Start Time	Time when the job was started. Uses Coordinated Universal Time (UTC).
End Time	Time when the job was completed. Uses Coordinated Universal Time (UTC).
Status	Status of the job.

Field	Description
Metered Value	Processed rows for change data capture and SQL ELT reports. Consumed serverless compute units for serverless reports. Consumed compute hours for all other reports.
Audit Time	Time when the task reported for metering.
OBM Task Time(s)	Actual process time. Applicable to Salesforce OBMSG.

Setting up IPU consumption alerts

You can configure alerts within IDMC or through email to notify you when your organization reaches a configured number of IPUs.

IPU consumption alerts help you track IPU usage by notifying your organization administrator when the organization consumes a specific number of IPUs. Users with the *Metering.admin* privilege can configure thresholds on the **Organizations** tab or the **Tags** tab.

When you're logged into a sub-organization, you'll only have access to view and configure that sub-organization's IPU alerts and thresholds in the **Tags** tab.

1. On the **Consumption Alerts** tab of the **Metering** page, select **Enable Organization Consumption Alerts** to configure consumption alert metrics.
2. Choose how you want to set the threshold for alerts.
   - **IPU absolute value** to configure metrics by whole numbers.
   - **IPU percentage value** to configure metrics by percentages.

   If you decide to switch from one metric to the other, the previous consumption threshold values are removed and you'll need to enter new values. If you configure alerts using an absolute value, the number can't be higher than the number of IPUs you're entitled.
3. In the **Consumption Threshold Value** field, enter the number or percentage of consumed IPUs for which you want to receive alerts. These values are maintained for your reference even if you clear the **Enable Consumption Alerts** check box to turn off alerts.
4. You can configure alerts on the project or folder level in the **Tags** tab. The **Tags** tab displays the project and folders of the organization you're currently logged in to.

After you set up alerts, organization administrators receive alerts within the service and through email. You'll also continue to receive the standard alerts when your organization consumes 25, 50, 75, 95, and 100 percent of its IPUs in addition to your own configurations.
If you consume more IPUs than the values you configured in the **IPU absolute value** or **IPU percentage value** fields, the system doesn't send any more alerts until the next billing period.

Metering tags

Metering tags are asset properties that can help you organize and monitor your cloud resources. Use metering tags to track IPU usage across various departments, projects, or initiatives.

After assigning metering tags to assets, you can view your IPU usage by metering tag on the **Tag Breakdown** tab on the Metering page. When you create a metering tag in an organization, the metering tags apply to all organizations in the hierarchy.

You can use metering tags in the following services:

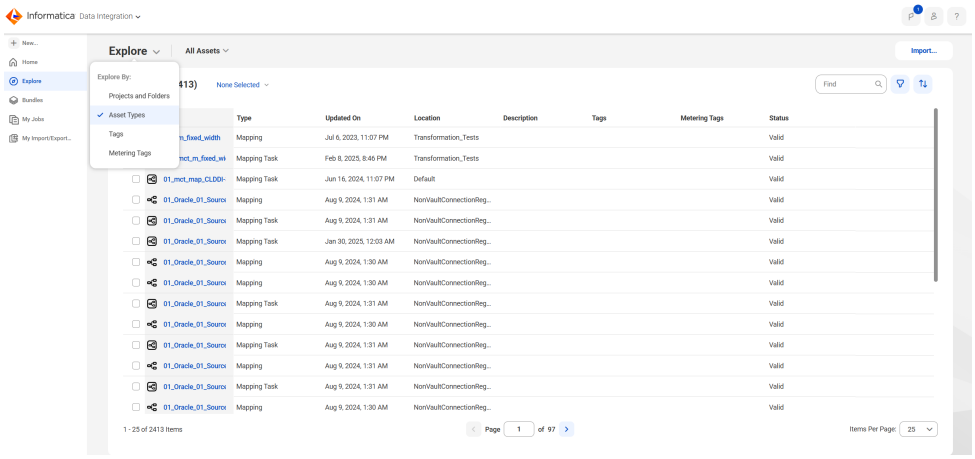
- Application Integration
- Data Ingestion and Replication
- Data Integration
- Data Quality
- Data Profiling

Setting up metering tags

Create metering tags and associate them with assets to organize and track IPU usage across various departments, projects, or initiatves.

To set up metering tags, you must create the metering tag and then associate it with assets.

1. To create a metering tag, perform the following steps:
    a. On the **Metering** page in **Administrator**, click the **Tag Manager** tab.
    b. In the **Metering Tags** section, click the **Add** button to create a tag.
    c. Enter the tag name and tag value in the **Tag Name** and **Tag Values** fields. You can create multiple tag values for each tag name.
    d. Alternatively, use **Allow Ad Hoc Tag Value** to allow people in your organization to create new tag values for this tag name.
    e. Use **Allow Multi-Value Tagging** to allow assets to be tagged with multiple tag values from the same tag name.
    f. To save these metering tag configurations, click **Save**. You can edit tags after you save them.
2. To associate a metering tag with an asset, perform the following steps:
    a. On the **Explore** page in your service, go to **Explore By** > **Asset Types**. The following image shows where to access Asset Types on the Data Integration Explore page:



    b. In the **Actions** menu for an asset, click **Properties**.

c. Use the `Metering Tags` prompt to associate the asset with a metering tag.

The **Tag Breakdown** tab in Administrator displays the IPU usage breakdown of the assets that you associated with a specific metering tag.

Note: When you update the tags that you've assigned to a department, project, or initiative, you see the changes reflect the next day. If you remove the tags and clear your cache, you won't see the tags.

Feature-based license metrics

If your organization has feature-based licenses, the **Metering** page displays information based on your licenses.

The **Metering** page displays the following views depending on the licensed features:

Dashboard view

If you have the appropriate licenses, the **Metering** page displays information in a dashboard view.

The Metrics Summary This Month area shows the total computing resources remaining for the month and each meter type. You can also display a table of all meters that apply to your organization, depending on your organization's editions.

The detail area for each meter type displays metering information for the month, including resource amounts remaining, average resource usage per day, and days of usage remaining. To see more detail for a meter type, you can display a detail chart that shows usage by organization or sub-organization, date range, and runtime environment.

License Metrics view

If you do not have the licenses required to see the dashboard view, the **Metering** page displays license metrics in a table. The table lists summary metrics for all meters that apply to your organization, which is determined by your organization's editions.

You can also navigate to the License Metrics view for all meters from the summary area in the dashboard view.

Note: Feature-based licensing is different from IPU-based licensing. IPUs are not used for feature-based licenses. If your organization uses IPUs, see "Informatica processing unit metrics" on page 34.

Viewing license metrics

You can view a table of all meters used in your organization. For some meters, you can download a report that shows the metering usage. View the table and download the report from the License Metrics view of the **Metering** page.

To open the License Metrics view from the dashboard view, click **All Meters** in the Metrics Summary This Month area. If you do not have the licenses required to see the dashboard view, the License Metrics view is displayed when you open the **Metering** page.

The meters that appear in the License Metrics view are determined by the editions that your organization has. Your organization might also be assigned custom meters. Metering information might not be available for every edition.

If your organization has multiple editions or uses custom meters, a meter might be displayed multiple times with different limits. In this case, the least restrictive limit applies. For example, if one edition has a limit of

500 synchronization jobs per day and another edition has a limit of 100 synchronization jobs per day, the 500 job per day limit applies. The **In Effect** column indicates which limit applies.

The License Metrics view displays the following information for each meter:

Property	Description
Edition	Name of the edition that is associated with the meter.
Service	Service to which the meter applies.
Metering	Name of the meter. For example, the number of synchronization jobs per day, the number of rows processed by mapping jobs per month, or the total number of replication jobs. For Mass Ingestion, this column shows the ingestion type: File Ingestion and Replication, Application Ingestion and Replication, Database Ingestion and Replication, Application Ingestion and Replication - Change Data Capture, Database Ingestion and Replication - Change Data Capture, and Streaming Ingestion and Replication.
Limit	Numeric limit such as the maximum number of jobs or processed rows. The limit applies to the parent organization and to each sub-organization. For example, if the limit is 100 jobs per day, users in the parent organization can run 100 jobs per day, and users in each sub-organization can also run 100 jobs per day. If this field displays -1, there is no limit.
Used	The actual number units consumed, such as the number of jobs run or compute hours used, in the organization or sub-organization during the metering period.
Percent Used	The percentage of units consumed in the organization or sub-organization during the metering period.
In Effect	Indicates whether the meter is in effect for the organization or sub-organization.

Meter definitions

The meters that appear in the License Metrics view of the **Metering** page are determined by the editions that your organization has.

The following table describes the meters that might be in effect based on your editions:

Meter	Definition
Advanced Data Integration	Compute units measured in hours for mappings in advanced mode.
Advanced Data Integration with Advanced Serverless	Serverless compute units measured in hours for mappings in advanced mode.
Advanced Data Quality	Compute units measured in hours for data quality assets in mappings in advanced mode.
Advanced Data Quality with Advanced Serverless	Serverless compute units measured in hours for data quality assets in mappings in advanced mode.
CDI Rows Processed	Number of rows processed by mappings outside of advanced mode.
CDI-E Compute Hours	Deprecated. This meter has been replaced with the Advanced Data Integration meter.

Meter	Definition
Daily/monthly incoming API request maximum	Number of API access requests per day or per month.
Data Governance and Catalog - Catalog Records	Number of technical assets stored in the catalog.
Data Governance and Catalog - Governance Records	Number of governance assets stored in the catalog.
Data Governance and Catalog - Metadata Record Consumption	Number of public API calls made to consume catalog records.
Data Governance and Catalog - Catalog Source Execution	Compute units, measured in hours, for a catalog source sync job execution on a secure agent.
Data Governance and Catalog - Catalog Source Execution with Serverless	Compute units, measured in hours, for a catalog source sync job execution on an Informatica Managed Infrastructure.
Data Integration with Advanced Serverless	Serverless compute units measured in hours for mappings outside of advanced mode.
Data Marketplace Records	Number of data collections created and orders placed over a 30-day period.
Data Quality	Compute units measured in hours for data quality assets in mappings outside of advanced mode.
Data Quality with Advanced Serverless	Serverless compute units measured in hours for data quality assets in mappings outside of advanced mode.
Mass Ingestion meters	For Mass Ingestion Applications, Mass Ingestion Databases, Mass Ingestion Files, and Mass Ingestion Streaming, the number of gigabytes (GBs) ingested by application ingestion and replication, database ingestion and replication, file ingestion and replication, or streaming ingestion and replication jobs per month. For Mass Ingestion Applications - Change Data Capture and Mass Ingestion Databases - Change Data Capture, the number of rows ingested. For MDM Data Ingestion and Replication, the number of gigabytes (GBs) ingested by streaming ingestion and replication tasks run by MDM users who publish real-time events from MDM SaaS.
Taskflow Runs	Number of taskflow runs per day or per month.
Number of PowerCenter jobs per day/ month	Number of PowerCenter jobs per day or per month.
Number of mapping jobs per day/month	Number of mapping jobs per day or per month.*
Number of masking jobs per day/month	Number of masking jobs per day or per month.
Number of replication jobs per day/month	Number of replication jobs per day or per month.
Number of rows processed by PowerCenter jobs per day/month	Number of rows processed by PowerCenter jobs per day or per month.

Meter	Definition
Number of rows processed by mapping jobs per day/month	Number of rows processed by mapping jobs per day or per month.*
Number of rows processed by masking jobs per day/month	Number of rows processed by masking jobs per day or per month.
Number of rows processed by replication jobs per day/month	Number of rows processed by replication jobs per day or per month.
Number of rows processed by synchronization jobs per day/month	Number of rows processed by synchronization jobs per day or per month.
Number of state sync jobs per day	Number of state synchronization jobs per day. State synchronization jobs include the fetchState and loadState jobs that you run through the REST API.
Number of sub-organizations	Number of sub-organizations. If you delete any sub-organizations, this meter includes the deleted sub-organizations.
Number of synchronization jobs per day/month	Number of synchronization jobs per day or per month.
Number of user-management create requests	Number of user management creation requests. User management creation requests include requests to create users, user groups, and custom roles.
Serverless CDI Compute Hours	Deprecated. Serverless compute units measured in hours for mappings outside of advanced mode.
Serverless CDI-E Compute Hours	Deprecated. Serverless compute units measured in hours for mappings in advanced mode.
Total Serverless units used	Deprecated. Total number of serverless compute units used to run tasks.
Total compute hours used by elastic cluster nodes	Deprecated. This meter has been replaced with the Advanced Data Integration meter.
Total number of PowerCenter jobs	Total number of PowerCenter jobs.
Total number of agents	Total number of Secure Agents, including agents that are stopped. Does not include the Informatica Cloud Hosted Agent.
Total number of connections	Total number of connections.
Total number of folders	Total number of folders.
Total number of mapping jobs	Total number of mapping jobs.*
Total number of masking jobs	Total number of masking jobs.
Total number of projects	Total number of projects.
Total number of replication jobs	Total number of replication jobs.

Meter	Definition
Total number of rows processed by PowerCenter jobs	Total number of rows processed by PowerCenter jobs.
Total number of rows processed by mapping jobs	Total number of rows processed by mapping jobs.
Total number of rows processed by masking jobs	Total number of rows processed by masking jobs.
Total number of rows processed by replication jobs	Total number of rows processed by replication jobs.
Total number of rows processed by synchronization jobs	Total number of rows processed by synchronization jobs.
Total number of synchronization jobs	Total number of synchronization jobs.

Guidelines for Data Quality and Data Profiling assets

If you run a mapping outside of advanced mode in which a transformation reads a data quality asset, Informatica records transactions for the mapping in the Data Quality meter. If you run a mapping in advanced mode in which a transformation reads a data quality asset, Informatica records transactions for the mapping in the Data Quality Elastic meter.

If you run a profile on the Data Integration Server, Informatica records transactions for the profile in the Data Quality meter. If you run a profile on an advanced cluster, Informatica records transactions for the profile in the Data Quality Elastic meter.

You can preview the output data on a transformation in a mapping. If you preview the data on a transformation that reads a data quality asset, Informatica calculates the cost in the following ways:

- If the mapping that contains the transformation runs outside of advanced mode, it consumes Data Quality compute hours.
- If the mapping that contains the transformation runs in advanced mode, it consumes Data Quality Elastic compute hours.

Metering serverless compute units

When you view the total serverless compute units used, the meter is based on the number of serverless compute units that your organization uses to run tasks.

When the serverless runtime environment runs a task, the environment creates a virtual machine with resources based on the number of compute units that the task requests.

The minimum task duration is two minutes. If the task completes in less than two minutes, the serverless runtime environment consumes two minutes of compute units. After two minutes, compute units are consumed by the second.

If you cancel the job, the number of consumed compute units is the greater of the following values:

- The time that the job was running before it was canceled
- Two minutes

Guidelines for advanced clusters

To run a mapping in advanced mode, the serverless runtime environment creates an advanced cluster that contains one worker node with resources based on the number of serverless compute units that the task requests. If you run another task, the environment reuses idle worker nodes or adds worker nodes to the cluster to reserve additional resources.

Metering begins when the task starts running and ends when the task is complete. Metering doesn't include the time to compile the job or the time to start the cluster.

Metering doesn't take effect if the job fails before the cluster has been created, such as when the job fails to compile, the cluster fails to start, or you cancel the job before the cluster starts.

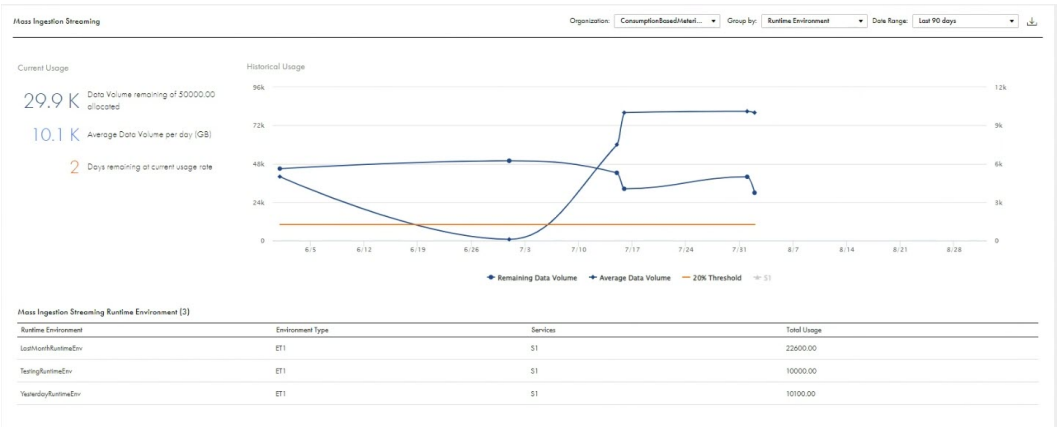
Viewing usage details

You can view detailed metering statistics. To view the detailed statistics, in the dashboard view of the **Metering** page, click **Detail Chart** for the relevant graph.

Detailed metering statistics are available for the following meters:

- Application Integration
- Application Integration with Advanced Serverless
- Catalog Records
- CDI-E Compute Hours
- Serverless CDI-E Compute Hours
- Serverless CDI Compute Hours
- Data Quality
- Data Quality Elastic
- Data Quality with Advanced Serverless
- Data Quality Elastic with Advanced Serverless
- Governance Records
- Mass Ingestion Application
- Mass Ingestion Applications - Change Data Capture
- Mass Ingestion Database
- Mass Ingestion Database - Change Data Capture
- Mass Ingestion Files
- Mass Ingestion Streaming
- MDM Data Ingestion and Replication
- PC2CDI Modernization Service Assessment
- PC2CDI Modernization Service Conversion

The following image shows an example of the details page that shows the total usage for each runtime environment in the last 90 days:



You can customize the view in the following ways:

- If your organization has sub-organizations, you can view usage details for the parent organization or for a sub-organization. You cannot view usage details for a deleted sub-organization.
- You can change the grouping. For example, you can group by runtime environment to see total usage for each runtime environment or you can view the full table to see usage by date.
- You can change the date range that defines the reporting period, such the current month, last month, last 90 days, or last 6 or 13 months.

Metering usage reports

If your organization uses Application Integration, the Data Ingestion and Replication service, advanced clusters, Cloud Data Integration for PowerCenter (CDI-PC) to modernize PowerCenter assets, or a serverless runtime environment, you can download a metering usage report.

A metering usage report contains the following information:

- For Application Integration , the report contains execution details of processes, guides, and OData APIs. For an Application Integration advanced serverless runtime environment, the report contains execution details of processes and guides.
  To view and download the report for processes, subprocesses, guides, and OData APIs that run in the Turbo mode, click the **Application Integration** and the **Application Integration with Advanced Serverless** links in the **API Calls** section of the metering dashboard.

  To view and download the report for processes, guides, and OData APIs that run in the standard mode, click the **Application Integration** and the **Application Integration with Advanced Serverless** links in the **Compute Units** section of the metering dashboard.
- For Application Ingestion and Replication and Database Ingestion and Replication, the report content depends on the load type of the ingestion and replication job. For jobs that perform initial loads, the report contains the volume of data ingested in GBs. For jobs that perform incremental loads, the report contains the number of records or rows processed. For jobs that perform combined initial and incremental loads, both reports are available.
- For File Ingestion and Replication, the report contains details about the volume of data ingested by file ingestion and replication tasks.
- For Streaming Ingestion and Replication, the report contains details about the volume of data ingested by streaming ingestion and replication jobs.

- For advanced clusters, the report contains details about the compute hours for cluster nodes.
- For Cloud Data Integration for PowerCenter (CDI-PC), the report contains details such as the date and time when assessment or conversion jobs were run, organization ID, job details, and the number of objects processed during assessment or conversion.
- For a serverless runtime environment, the report contains details about the number of serverless compute units that were requested and consumed to run tasks.

The metering usage report includes IPU usage from 12:00 a.m. UTC on the start date through 11:59 p.m. UTC on the end date.

If your organization does not use Application Ingestion and Replication, Database Ingestion and Replication, File Ingestion and Replication, Streaming Ingestion and Replication, advanced clusters, or a serverless runtime environment, no metering usage reports are available.

Downloading a metering usage report from Detail Chart

Download a metering usage report from the Detail Chart view of the **Metering** page.

1. In Administrator, select **Metering**.
2. Click **Detail Chart** for the metering usage report you want to download.
3. Select the organization and the date range that you want to view.

   The report data adjusts accordingly.
4. Click the download icon.
5. Click **Export**.

CHAPTER 5

Source control configuration

You can configure source control for your organization to enable version management for projects, folders, and assets. When you configure source control, you can store versions of objects in a cloud-hosted or on-premises Git repository. Configure source control on the **Settings** page.

To use source control with IDMC, you must have the appropriate licenses.

The following table lists the source control repositories that you can use:

Repository	Self-hosted	Cloud-hosted
Atlassian Bitbucket	Supported	Supported for Data Integration assets only
GitHub	Supported	Supported
GitLab	Supported	Supported
Microsoft Azure DevOps	-	Supported

Note: You can't use cloud-hosted Bitbucket or cloud-hosted GitLab if the organization is subject to FedRAMP.

When you configure source control for an organization, users can apply source control to objects. Objects are not checked in automatically. Users can apply source control to individual assets or to all assets in a project or folder. For more information about applying source control to projects, folders, and assets, see the help system for the appropriate IDMC service.

When you configure source control, you configure the connection to a global source control repository for the organization. You can configure source control for your organization in the following ways:

Configure read/write access to the global source control repository.

When you configure read/write access, users in your organization can check in and check out objects, pull versions of objects, and revert objects to a previous version. Users must check out source-controlled objects to change them. Users check out objects exclusively, so one user cannot change an object that is checked out by another user. Users can change objects that are not source-controlled without checking them out.

You might want to configure read/write access for an organization in which you develop projects and assets.

Configure read-only access to the global source control repository.

When you configure read-only access, users in your organization can pull versions of source-controlled objects from the repository. However, users cannot check out or check in objects. Users can make changes to projects, folders, and assets in the organization without checking them out.

You might want to configure read-only access for a test or production organization so that users can test or run the latest versions of assets.

You can change the repository access type. However, to change from read/write to read-only, you must first ensure that no objects are checked out. IDMC doesn't allow you to change the repository access type from read/write to read-only if any objects are checked out.

Warning: When you configure read-only access, users can overwrite source-controlled objects. For example, user John pulls the most recent version of a source-controlled mapping and changes it. If another user pulls any version of the mapping later, John's changes are lost. Configure object permissions and user privileges carefully to prevent users from accidentally overwriting source-controlled assets in your organization.

Enable project-level repositories

You can enable users to specify a branch in the global repository or a different repository for each project. Using different repository branches for your projects can enable parallel development and collaboration across teams in the organization.

You can change the repository URL. To do this, you must first unlink all source-controlled assets. IDMC doesn't allow you to change the repository URL if any assets are source-controlled.

If you want to disable source control after you configure it, unlink all objects from source control and then disable source control for the organization.

You can unlink an object that's checked out by another user if you have the admin role or a user role with the Force Undo Checkout privilege.

Source control configuration for sub-organizations

Configure source control for a sub-organization on the **Settings** page in the sub-organization. As a best practice, each sub-organization should use its own source control repository. Additionally, the source control repository for a sub-organization should be different than the source control repository for the parent organization.

Maintain different source control repositories so that users in one organization do not accidentally overwrite or change assets in another organization.

If you want the parent organization administrator to be able to perform source control operations in the sub-organization, configure the Git user account for the parent organization administrator to have access to the sub-organization's source control repository.

Repository access using OAuth

If your source control repository is cloud-hosted, you can configure an organization to use OAuth authentication instead of personal access tokens to provide access to the repository. Configure OAuth authentication on the **Settings** page.

If you use a GitHub repository, you must have a GitHub access application installed on your repository that allows IDMC to perform source control operations on the organization's GitHub repository. If you don't have this application installed on your repository, you can install it from the **Settings** page.

GitHub repositories with more than 500 assets in a single project or folder require that you accept additional permissions to prevent check-in failures. For more information, see the Checking in large repositories in source control Knowledge article.

Note: Microsoft plans to drop support of Azure DevOps OAuth. If the organization uses Azure DevOps OAuth to access the source control repository, you can change to using Microsoft Entra ID OAuth on the **Settings** page. This change is permanent. You can't revert to Azure DevOps OAuth after you change to Microsoft Entra ID OAuth.

After you change the organization's repository access to Microsoft Entra ID OAuth, each user needs to re-authenticate to avoid interruption in source control access. To re-authenticate, click the User icon on the toolbar and select **Settings**. In the **Source Code Control Credentials** section, click **Drop Authentication** and then reconfigure OAuth authentication.

Working with an on-premises repository

If your source control repository is on-premises, the Secure Agent creates a local copy of the repository on the Secure Agent machine. The Secure Agent performs source control operations in the local repository and then pushes them to the remote repository.

When you use an on-premises repository, ensure that the Secure Agent machine has enough space for the local copy of the repository and for all subsequent version control operations.

The Secure Agent creates the local repository the first time that a user performs a source control operation, such as checking in an asset. When it creates the local repository, it copies the branch that stores IDMC assets. It does not copy other branches. Each time a user performs a source control operation, the agent gets information about the changes from the remote repository to support the operation.

By default, the Secure Agent creates the local repository in the following directory on the Secure Agent machine:

```
<Secure Agent installation directory>/apps/GitRepoConnectApp/data/git_repository/<client URL>/<organization ID>/<branch>/<remote repository name>
```

You can change the local repository directory by editing the **git_local_repository_path** property for the GitRepoConnectApp service on the Secure Agent details page. For more information about changing the value of this property, see *Secure Agent Services*.

Enabling source control for an organization

To enable source control for an organization, configure the type of access and connection to the global source control repository on the **Settings** page. The settings that you configure vary based on the repository type.

1. On the **Settings** page in Administrator, click **Edit** in the Source Control area.
2. Enable the **Enable Source Control** option.
3. Optionally, enable the **Enable Project Level Source Control** option.

   When this option is enabled, users can specify a branch in the global repository or a different repository to use at the project level.

4. Configure the type of access to the source control repository:
   - To configure read/write access, enable the **Allow Push to Git Repository** option.
   - To configure read-only access, disable the **Allow Push to Git Repository** option.
5. To configure access to a cloud-hosted repository, enter the following information:

Option	Description
Platform	Platform type. Select **Cloud**.
Repository Type	Version control system that you use for the organization. You can use one of the following cloud-hosted systems: • GitHub • Microsoft Azure DevOps • Atlassian Bitbucket (for Data Integration) • GitLab
Global Git Repository URL	Repository URL. For example: `https://github.com/MyGitUser/MyRepositoryName.git` The repository URL must use the HTTPS protocol. **Tip:** You can find the repository URL by selecting the clone option in the repository.
Global Git Branch Name	Name of the branch that stores the IDMC objects. The branch that you specify must already exist in the repository. If you do not enter a branch name, IDMC sets the branch name to "master" or "main," based on the name of the default branch in the remote repository.
Allow OAuth access to Git	Enable this option to use OAuth to access the repository. If you use a GitHub repository, a Git access application that authorizes IDMC access must be installed on the organization's repository. To install the application, click **Install Git Access App at GitHub**.

6. To configure access to an on-premises repository, enter the following information:

Option	Description
Platform	Platform type. Select **On-Premise**.
Global Git Repository URL	Repository URL. For example: `https://gitlab.example.com/MyGitUser/MyRepositoryName.git` The repository URL must use the HTTPS protocol. **Tip:** You can find the repository URL by selecting the clone option in the repository.
Global Git Branch Name	Name of the branch that stores the IDMC objects. The branch that you specify must already exist in the repository. If you do not enter a branch name, IDMC sets the branch name to "master" or "main," based on the name of the default branch in the remote repository.
Runtime Environment	Runtime environment used to connect to the Git repository. The repository must be accessible by all agents in the runtime environment that you select.

You cannot configure OAuth access to an on-premises repository.

7. Click **Save**.

   IDMC prompts you for your source control credentials to verify the repository connection. IDMC does not store this information.

   If the connection is valid and you configure read/write access to the repository, IDMC writes a small readme file to the repository to verify that it can push objects to the repository.

After you enable source control, all users that use source control must enter their source control credentials in their user settings. Users cannot see source control columns on the **Explore** page or perform source control actions until they enter their source control credentials. To enter source control credentials, click the **User** icon in the top right corner of the IDMC window, and select **Settings**.

Changing the source control repository URL

To change the source control repository URL, you must first unlink all objects in the organization and then enter the new repository URL on the **Settings** page in Administrator. After you change the URL, all users that use source control must update their source control credentials in the user settings.

1. In each IDMC service that uses the repository, unlink all objects from source control.
2. In Administrator, open the **Settings** page and click **Edit** in the Source Control area.
3. Verify that the **Enable Source Control** option is enabled.
4. Configure the type of access to the source control repository:
   - To configure read/write access, enable the **Allow Push to Git Repository** option.
   - To configure read-only access, disable the **Allow Push to Git Repository** option.
5. Verify the platform and name of the branch that stores the IDMC objects. For a cloud-hosted repository, verify the OAuth access setting, as well.
6. Enter the new repository URL, for example:

   ```
   https://github.com/MyGitUser/MyRepositoryName.git
   ```

   Tip: You can find the repository URL in the following ways based on the repository type:

Repository	How to find URL
Atlassian Bitbucket (self- hosted and cloud-hosted)	Open the repository and select **Clone**.
GitHub (cloud-hosted)	Open the repository and select **Clone or download** > **Clone with HTTPS**.
GitHub Enterprise (self-hosted)	Open the repository and select **Code** > **Clone with HTTPS**.
GitLab (self-managed and cloud-hosted)	Open the repository and select **Code** > **Clone with HTTPS**.
Microsoft Azure DevOps (cloud-hosted)	Open the repository and select **Clone**.

   The repository URL must use the HTTPS protocol.

7. Click **Save**.

   IDMC prompts you for your source control credentials to verify the repository connection. IDMC does not store this information.

   If the connection is valid and you configure read/write access to the repository, IDMC writes a small readme file to the repository to verify that it can push objects to the repository.

After you change the source control repository URL, all users that use source control must update their source control credentials in the user settings. To update source control credentials, click the **User** icon in the top right corner of the IDMC window, and select **Settings**.

Disabling source control for an organization

You can disable source control for an organization. Disabling source control breaks the link between your organization and the source control repository. It does not delete objects in the source control repository.

Before you can disable source control for a read-write organization, all assets must be unlinked.

1. In each IDMC service that uses the repository, unlink all objects from source control.
2. In Administrator, disable source control:
   a. In Administrator, open the **Settings** page.
   b. Click **Edit** in the Source Control area.
   c. Disable the **Enable Source Control** option.
   d. Click **Save**.
3. Optionally, have users in the organization delete their source control credentials in their user settings:
   a. In the top right corner of the IDMC window, click the **User** icon and select **Settings**.
   b. Clear the source control credentials.
   c. Click **Save**.

Configuring repository access

To work with source-controlled objects, specify your repository credentials in IDMC.

Your credentials can include a personal access token or API token, depending on the repository service that you use. If your organization's repository is configured for OAuth access, you can enable OAuth access instead of providing a personal access token or app password.

Personal access tokens and API tokens must be configured to enable full control of private repositories. For information about generating tokens, see the GitHub, GitLab, Azure DevOps Git, or Bitbucket help.

Perform the following steps to configure access to the repository:

1. Click the **User** icon on the toolbar and then select **Settings**.

   Note: Depending on which service you are in, the user settings page might appear different and show different names for the sections.

2. In the **Source Control Access** section, perform one of the following tasks:
   - Enter your repository credentials.
   - Enable OAuth access to the repository.
     If you haven't already authorized access, a Git access app appears for GitHub and Azure DevOps Git repositories. Select to authorize access for IDMC.
3. Click **Save**.

Source control best practices

To ensure that your organization configures and uses source control effectively, use the following guidelines as best practices.

Setup guidelines

Adhere to the following guidelines when you set up source control for your organization:

- Use different organizations for development, testing, staging, and production.

  When you maintain different organizations, you maintain isolation across environments so that changes in one environment do not affect other environments. For example, changes to assets in the testing environment are not accidentally deployed in the production environment.
- Configure development organizations with read/write access to the source control repository, and configure non-development organizations with read-only access to the source control repository.

  This ensures that only users in a development organization can make changes to assets. It also prevents users in a non-development environment from accidentally pushing changes to the source control repository.
- Ensure that only one development organization pushes to a particular source control repository.

  Maintaining separate repositories for development organizations ensures that users in one organization do not accidentally change or overwrite assets in a different organization. Maintaining separate repositories also avoids conflicting asset ID references from different organizations.
- When you enable source control for the organization, select an empty repository.

  Ensure that the repository does not contain a folder named "Explore" because IDMC stores assets under the Explore folder in the Git repository.
- Do not share source control credentials among multiple IDMC users.

  Separate credentials maintain security and make it easier to track which user made a particular change. Additionally, each user gets their own rate limit in GitHub.

Development guidelines

Adhere to the following guidelines as you develop and work with assets:

Guidelines for managing dependencies

Use the following guidelines to manage assets with dependencies:

- Create connections and runtime environments before you pull assets from the repository.

  When required connections and runtime environments exist in the target organization, you can run tasks immediately after you pull them from the repository.

- Ensure that reusable assets such as mappings and components are present in the repository before you use them.

  IDMC does not allow you to save an asset such as a mapping task when the dependent mapping does not exist in the organization.

Guidelines for checking in and checking out assets

Use the following guidelines when you check in and check out assets:

- When you rename or move an asset, check out the asset's first-level dependent assets and include them in the same check-in.

  For example, if you want to rename a mapping that a mapping task uses, and the mapping task is used in a taskflow, check out the mapping and the mapping task. You don't need to check out the taskflow. After you rename the mapping, check in the mapping and the mapping task in one check-in action.

- Enter comments when you check in assets.

  When you check in assets, you might enter a release tag name in the **Summary** field and enter more descriptive comments in the **Description** field. When you do this, the **Git Summary** field in IDMC shows the release tag that is associated with the asset.

- When you check in multiple assets at one time, limit the number of assets to 800 or fewer.

  Checking in more than 800 assets at one time can degrade performance between IDMC and the Git repository service.

CHAPTER 6

Customer-managed encryption keys

You can use your own master key to encrypt your organization's encryption keys.

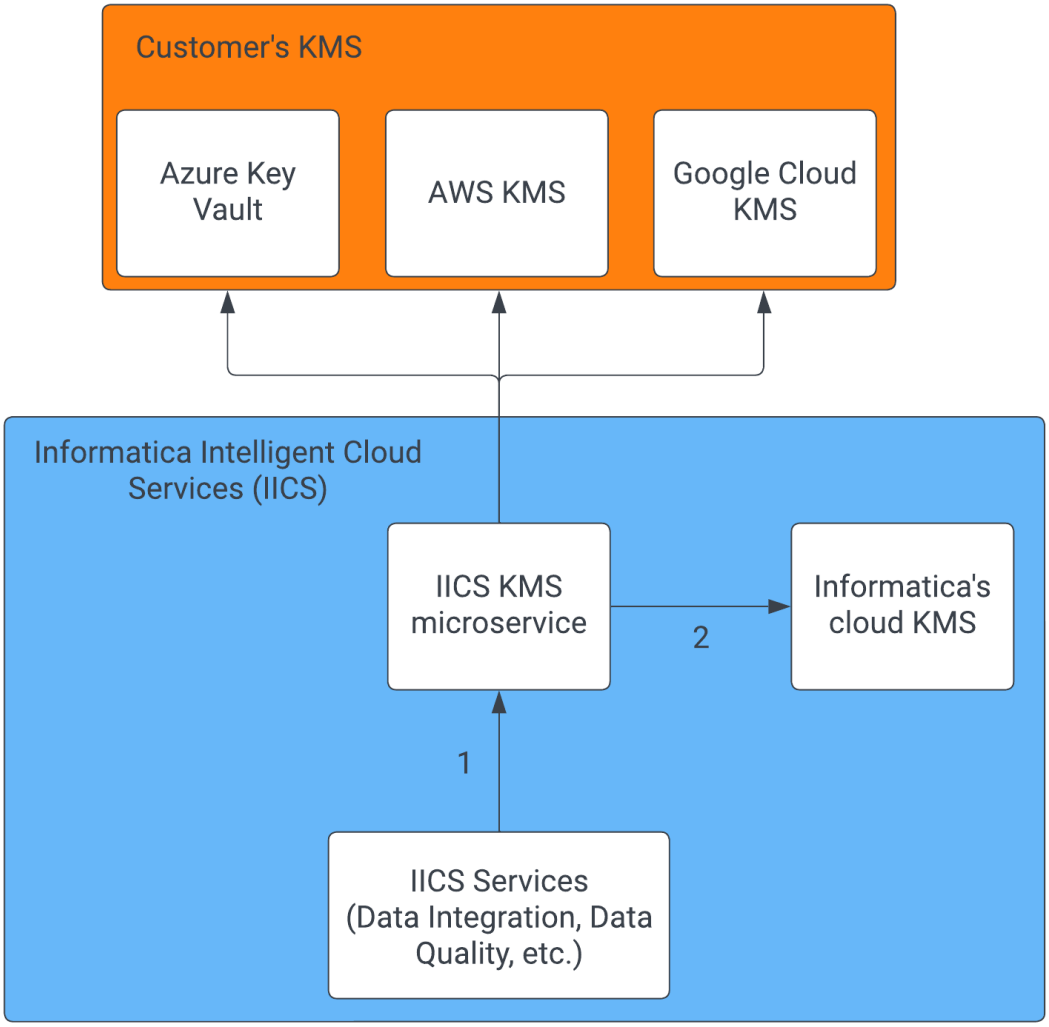
By default, IDMC protects your organization's sensitive data in the cloud using organization-specific encryption keys that are generated and stored in the IDMC key management service (KMS). To prevent malicious access, the keys are encrypted using a master key that is stored in the cloud provider's KMS. The master key is provisioned in Informatica's KMS account and varies by POD.

If you prefer, you can create a customer-managed key (CMK). When you create a CMK, you control access to it. However, you'll need to grant IDMC access to the CMK so that it can encrypt and decrypt your organization's sensitive data.

Creating a CMK offers the following benefits:

- You can restrict and control any access to your data.
- You can restrict the decryption of your data in the event of a data breach.
- You create and hold the key material in your KMS. The key is never exposed to your cloud service provider.
- You maintain full control of the key throughout its lifecycle. You can revoke access or delete the key at any time.

The following image shows how IDMC interfaces with your CMK:



1. IDMC interfaces with the IDMC KMS agnostically.
2. Non-customer managed keys go to Informatica's cloud KMS.

You can create and enable a CMK when you use the following cloud providers' key management services:

- Amazon Web Services
- Microsoft Azure
- Google Cloud

Note: When you create a CMK, your KMS and IDMC POD must use the same cloud provider. For example, if your IDMC POD is USW1 on AWS, then you must store your CMK on AWS KMS. You can't store it on Google Cloud KMS or Azure Key Vault.

Creating and enabling a customer-managed key

To create and enable a customer-managed key, provision the key in your KMS and then enable customer-managed keys in Administrator.

Note: The steps you perform to create and enable a CMK vary based on your cloud provider. For specific instructions, see the following H2L articles:

- Enable Customer Managed Keys for your Organization on Amazon Web Services
- Enable Customer Managed Keys for your Organization on Microsoft Azure
- Enable Customer Managed Keys for your Organization on Google Cloud

In general, you perform the following steps:

1. In your cloud KMS, provision the key and enable cross-account access with IDMC.
2. In Administrator, open the **Security** tab on the **Settings** page, enable the **Enable Customer Managed Keys** option, and enter the key properties.

   Note: To perform this step, you must log in to IDMC with a user account that has both the Admin and Key Admin roles.

   You can test the key after you configure the key properties. It can take up to 24 hours for the key to become active.

After you create and enable a CMK, you can disable it at any time by clearing the **Enable Customer Managed Keys** option on the **Security** tab. When you do this, you'll go back to using Informatica's master key.

Frequently asked questions about customer-managed keys (CMK)

I can't see the **Security** tab on the **Settings** page even though my organization has the appropriate license. Why not?

Log in to IDMC with a user account that has both the Admin and Key Admin roles. If you don't have both roles, you can't see the **Security** tab.

For more information about user roles, see *User Administration*.

When I clicked **Test Managed Key** in on the **Settings** page, the test failed. What should I do?

If you get an error when testing the key, perform the following checks:

- In Administrator, verify that the key settings on the **Settings** page match the settings for the CMK in your cloud KMS.
- In your cloud KMS, verify that the status of the CMK is active.
- In your cloud KMS, verify that the permissions on the CMK allow Informatica cryptographic access to the key.

If you continue to encounter errors, contact Informatica Global Customer Support.

What happens if the CMK is rotated in my KMS?

You can rotate the key in your cloud KMS manually or on a schedule. Rotating a key creates a new version of the key. The old version of the key remains in your cloud KMS and is used for decryption only.

IDMC detects key rotation on Azure Key Vault and Google Cloud KMS. When the CMK is rotated, IDMC decrypts your organization's keys using the old CMK and then encrypts them using the new CMK.

IDMC cannot detect key rotation on AWS KMS. If you use AWS KMS, you'll need to disable customer-managed keys in IDMC and reenable it. To do this, perform the following steps:

1. On the **Settings** page in Administrator, click the **Security** tab and note the **Key ARN** and **Role ARN**.
2. Disable the **Enable Customer Managed Keys** option.
3. Enable the **Enable Customer Managed Keys** option, reenter the key ARN and role ARN, and click the save icon.

What if I need to update the CMK in my KMS?

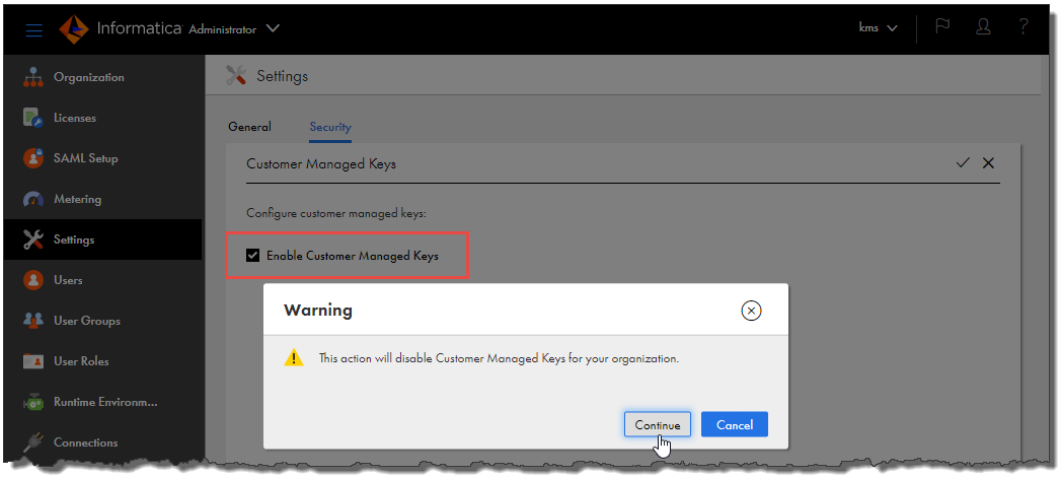
If you need to update the CMK, first provision a new CMK in your cloud KMS. Then update the key details on the **Settings** page in Administrator.

Note: Be sure to keep the old version of the CMK in your cloud KMS active until you update the key details in Administrator.

You can delete the old version of the CMK in your cloud KMS after you update the key details on the **Settings** page.

What if I want Informatica to manage key encryption?

If you want Informatica to manage key encryption, you can disable the **Enable Customer Managed Keys** option on the **Settings** page in Administrator:



When you do this, be sure to keep the current version of the CMK in your cloud KMS active. If the CMK is not active, disabling customer-managed keys fails.

When you disable this option, your organization's encryption keys are once again encrypted using encryption keys that are managed by Informatica. It can take up to 10 minutes for the Informatica encryption keys to become active.

You can disable or delete the CMK in your cloud KMS after you disable the **Enable Customer Managed Keys** option in Administrator.

What if I want to temporarily revoke Informatica's access to the CMK?

If you want to temporarily revoke Informatica's access to the CMK, you can disable the key in your cloud KMS.

When you disable the CMK, IDMC can no longer unencrypt your organization's encrypted data, and any jobs that use the data will fail until you reactivate the CMK in your cloud KMS.

How do I replace the CMK if I suspect it has been compromised?

If you want to replace the CMK, you can delete the key in your cloud KMS and create a new one.

Warning: Deleting the CMK in your cloud KMS results in permanent loss to any encrypted data in IDMC and causes the jobs that use the data to fail.

If you need to replace the CMK, perform the following steps so that you don't lose access to the encrypted data and jobs don't fail:

1. In Administrator, open the **Settings** page, click the **Security** tab, and disable the **Enable Customer Managed Keys** option.
2. In your cloud KMS, delete the CMK.
3. In your cloud KMS, create a new CMK.
4. On the **Settings** page in Administrator, re-enable the **Enable Customer Managed Keys** option and enter the details for the new CMK.

Can I delete the CMK if I don't want Informatica to access any of my encrypted data?

Warning: Deleting the CMK in your cloud KMS results in permanent loss to any encrypted data in IDMC and causes the jobs that use the data to fail.

If you're sure that you want Informatica to forgo all access to your encrypted data in IDMC, you can delete the CMK in your cloud KMS.

What happens if I revoke my CMK?

It is critical that you disable the CMK option from Administrator before revoking your CMK. Otherwise, your organization will be disabled and users can't log in. At this point, it's no longer possible to disable the CMK option because you can't access Administrator.

Requests authenticated using existing JWT tokens will continue to be honored until the tokens expire. Some jobs or processes that were already running at the time of revocation might continue to execute.

To disable the CMK option, clear the check box for **Enable Customer Managed Keys**, as described in "Creating and enabling a customer-managed key" on page 67.

To restore your organization to normal operation, replace your revoked key with a new valid key.

What is the service level agreement after revoking a CMK?

If a CMK is revoked without first being disabled from Administrator, IDMC will invalidate all active sessions within 10 minutes. Attempts to log in to the organization will fail until the CMK is reactivated in the customer's cloud KMS account.

What is the impact of a POD's disaster recovery on a CMK?

Once the disaster recovery process is complete, organizations with customer-managed keys resume normal operation and will honor the CMK status, whether active or disabled.

CHAPTER 7

Secrets managers

You can configure your organization to retrieve sensitive connection credentials from an external secrets manager instead of storing the credentials in the IDMC repository. A secrets manager is also called a secret vault or a key vault.

Using a secrets manager offers the following benefits:

- You retain complete control of your sensitive connection credentials like passwords, OAuth tokens, and API shared secrets.
- You can manage secrets across multiple environments instead of on a per-application basis.
- You can rotate secrets on your schedule without affecting your connections, mappings, or tasks in IDMC.

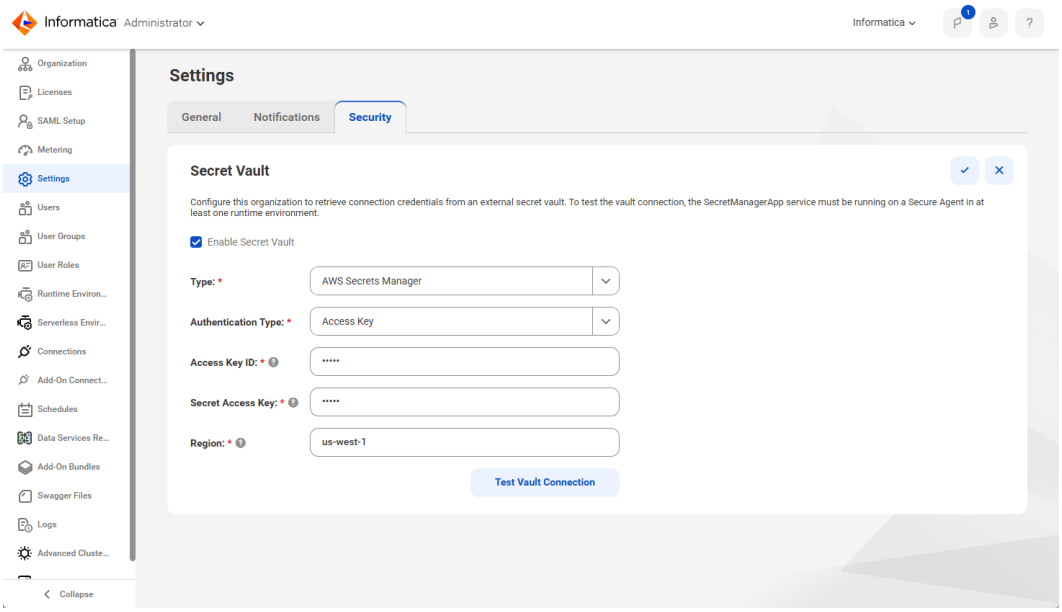
When you enable your organization to use a secrets manager, your Secure Agents can dynamically access sensitive connection credentials from the secrets manager. You can configure one secrets manager for each organization or sub-organization.

You can use one of the following secrets managers:

- AWS Secrets Manager
- Azure Key Vault
- HashiCorp Vault (HCP cloud-hosted)

If you use AWS Secrets Manager, the Secure Agent can access it using role-based, instance profile, or access key authentication.

Configure your organization or sub-organization to use a secrets manager on the **Security** tab of the **Settings** page, as shown in the following image:



To configure your organization to use a secrets manager, you must have the Admin role or the SMS Manage Connection and SMS View Connection feature privileges as well as sufficient privileges to access the Administrator service. The organization must also be configured to store connection credentials on the cloud.

Note: You can't use a secrets manager if your organization uses serverless runtime environments or stores connection credentials on a local Secure Agent.

After you configure your organization to use a secrets manager, you can configure your connections to retrieve credentials from the secrets manager.

AWS Secrets Manager configuration

AWS Secrets Manager is a fully managed AWS service that helps you securely store, manage, and retrieve sensitive information like database credentials, API keys, and other secrets needed by your applications. You can configure your organization to retrieve sensitive connection credentials from AWS Secrets Manager instead of directly entering the credentials in the connection properties.

The Secure Agent can access Secrets Manager using one of the following authentication methods:

- Role-based authentication. If you use role-based authentication, you need to configure the IAM role that the Secure Agent uses.
- Instance profile authentication. If you use instance profile authentication, you need to configure and access an instance profile to the AWS resource that hosts the Secure Agent.
- Access key authentication

If the AWS account that hosts your secrets differs from the account that hosts the Secure Agent, you'll also need to set up both accounts for cross-account access.

To configure your organization to retrieve secrets from AWS Secrets Manager, enable secret vault in Administrator, select AWS Secrets Manager as the secrets manager, and configure the connection properties. Then, you can configure connections to retrieve sensitive credentials from Secrets Manager.

IAM role configuration for AWS Secrets Manager

If you access AWS Secrets Manager using role-based authentication, you need to ensure that the IAM role that the Secure Agent uses to access secrets has the appropriate policies and permissions. You must also attach the role to your EC2 instance.

To configure the IAM role, first define a policy with the appropriate permissions, assign the policy to the role, and then update the role trust policy.

Step 1. Create an IAM policy and assign appropriate permissions.

Depending on your organization's policies, configure one of the following policies for your IAM user:

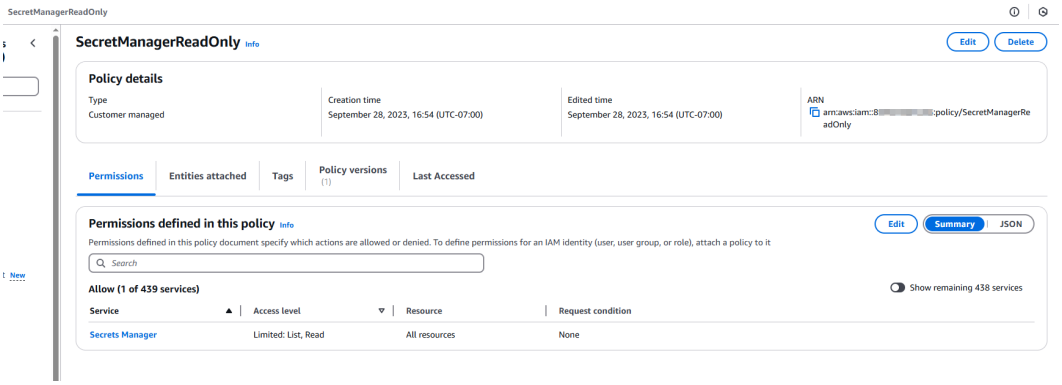
Policy #1

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "secretsmanager:GetSecretValue",
                "secretsmanager:ListSecrets"
            ],
            "Resource": "*"
        }
    ]
}
```

Policy #2

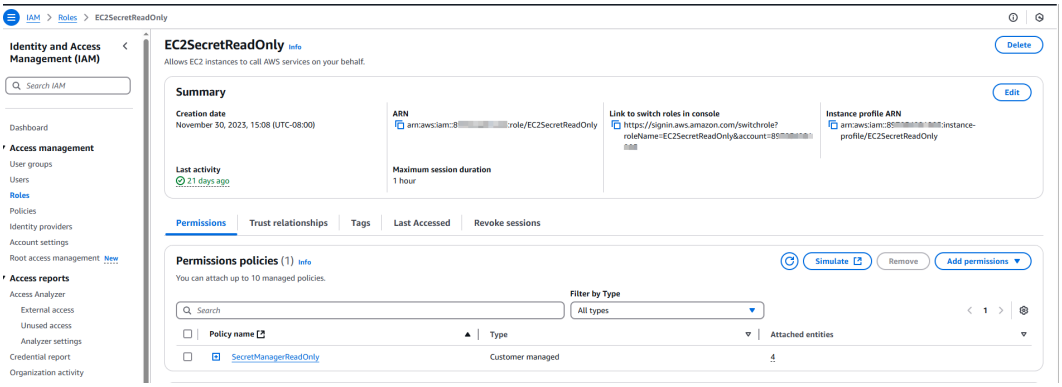
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AwsSecretManagerPolicy",
      "Effect": "Allow",
      "Action": [
        "secretsmanager:GetResourcePolicy",
        "secretsmanager:GetSecretValue",
        "secretsmanager:DescribeSecret",
        "secretsmanager:ListSecretVersionIds"
      ],
      "Resource": [
        "arn:aws:secretsmanager:us-east-1:006102214893:secret:AWS_Secrets_Demo-
voNcel"
      ]
    },
    {
      "Effect": "Allow",
      "Action": "secretsmanager:ListSecrets",
      "Resource": "*"
    }
  ]
}
```

The policy that you select must be able to list and read secrets from Secrets Manager. The following image shows the minimum policies required:



Step 2. Assign the policy to the IAM role.

Assign the policy you created to the role that the Secure Agent uses to access secrets, as shown in the following image:



Step 3. Update the IAM role trust policy.

After you assign the policy, update the IAM role trust policy to define which AWS resources can access the role. To do this, either allow any EC2 VM instance to access the role or allow the EC2 instance's role to assume the role that has permission to read secrets.

If the IAM role is the same role as the EC2 instance's role, you can have the role assume itself.

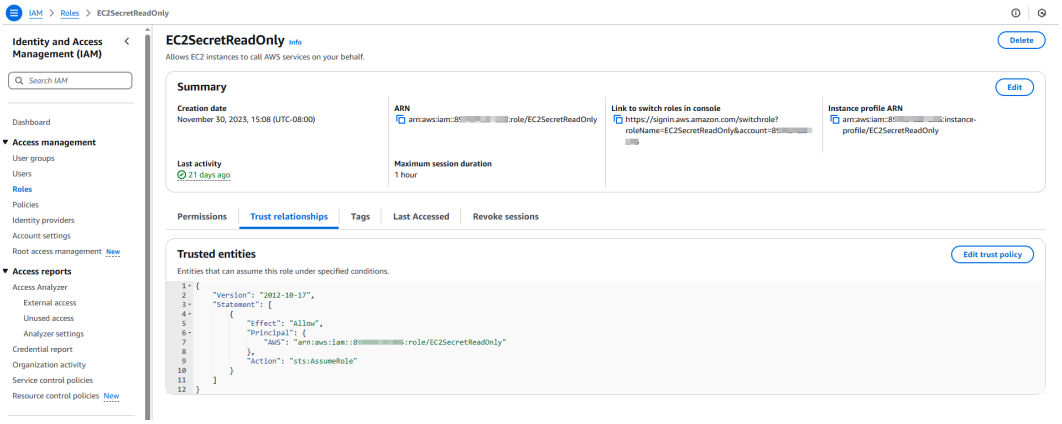
To allow any EC2 VM instance to access the role, configure the following trust policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "ec2.amazonaws.com"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```

To allow the EC2 instance's role to assume the role that has permission to read secrets, configure the following trust policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::<account ID>:role/<EC2 instance's role>"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```

The following image shows the trust policy:



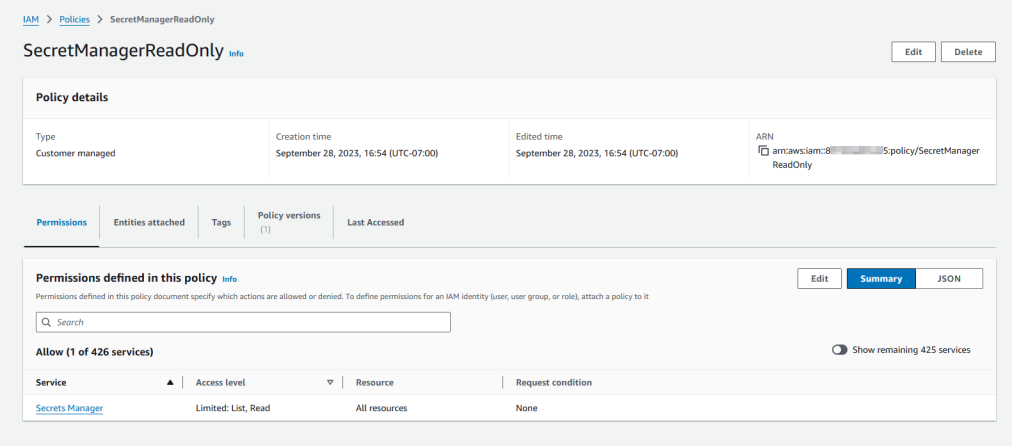
For more information about assigning polices to IAM roles and attaching IAM roles to EC2 instances, see the AWS documentation.

Instance profile configuration for AWS Secrets Manager

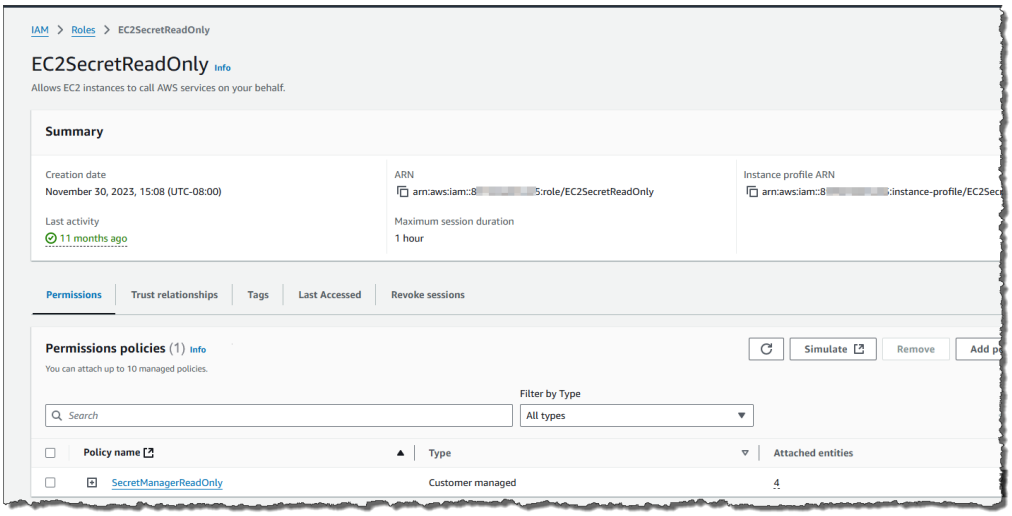
If you access AWS Secrets Manager using instance profile authentication, you need to create an IAM policy that can read secrets, create a role that has this policy, and then attach an instance profile to the AWS resource that hosts the Secure Agent.

1. Log in to the AWS Management Console.
2. Navigate to the IAM console.
3. Under **Access management**, select **Policies**.
4. Click **Create policy**.

5. Create an IAM policy with the **All list actions** and **All read actions** access levels as shown in the following image:



6. Under **Access management**, select **Roles**.
7. Click **Create role**.
8. On the **Select trusted entity** page, select **AWS service** as the trusted entity type and **EC2** as the use case, and then click **Next**.
9. On the **Add permissions** page, select the policy you created, and then click **Next**.
10. Enter a role name, verify the permissions policy, and create the role, as shown in the following image:



11. Navigate to the EC2 dashboard and select the EC2 instance that hosts the Secure Agent.
12. Click **Advanced**.
13. Under **IAM instance profile**, select the IAM role you created, and launch the EC2 instance.

Cross-account access configuration for AWS Secrets Manager

If the AWS account that hosts your secrets differs from the account that hosts the Secure Agent, you'll need to set up both accounts for cross-account access.

When you configure a connection to use a secrets manager, you choose the runtime environment for the connnection. If the runtime environment contains a Secure Agent that is hosted within an AWS account and

this account differs from the account that hosts the secrets, you need to configure cross-account access. Configure cross-account access so that the Secure Agent can access the secrets.

Note: To configure cross-account access, the resource that hosts the secrets needs to be in the same region as the region you choose when you enable your organization to use a secrets manager. For more information about enabling your organization to use a secrets manager, see "Enabling and disabling a secrets manager" on page 82.

Step 1. Set up the account that hosts the secrets.

To set up the account that hosts the secrets, you need to create a customer-managed KMS key, encrypt the secret using the key, and attach a resource policy to the secret. You can't use the AWS managed key for cross-account access.

Perform the following steps:

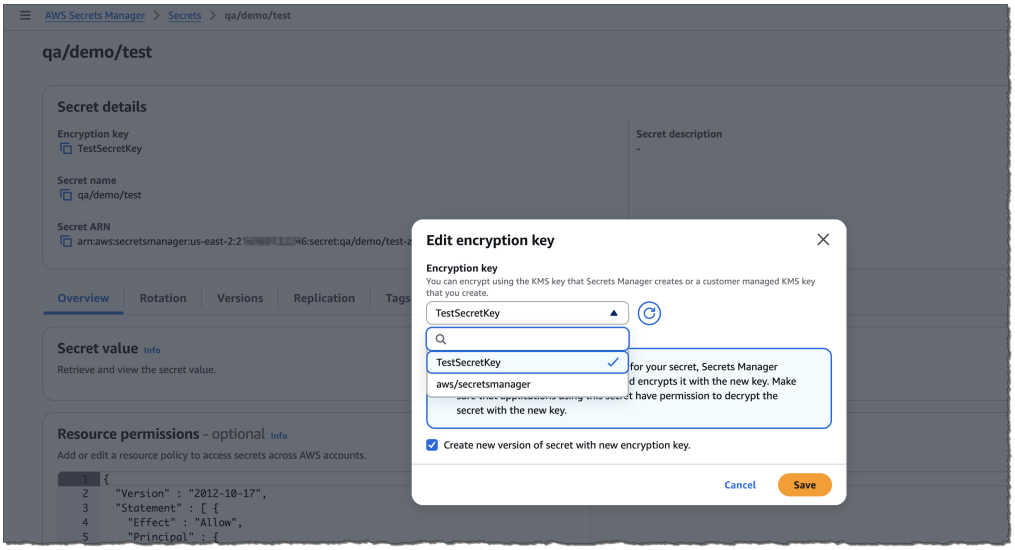
1. Create the KMS key by performing the following steps:
   a. Log in to the AWS Management Console, and search for "Key Management Service" or "KMS."
   b. Navigate to **Customer Managed Keys**.
   c. Create a new customer-managed key that has the following properties:

Property	Value
Key type	Symmetric
Key usage	Encrypt and Decrypt
Key administrators	Select the IAM users or roles that will manage the key.
Key users	Select the IAM users or roles that will use the key to encrypt and decrypt secrets.

   d. Configure the following key policy to allow access to the IAM role in the account that hosts the Secure Agent:

```
{
  "Effect": "Allow",
  "Principal": {
    "AWS": "arn:aws:iam::<Secure Agent account>:role/EC2SecretReadOnly"
  },
  "Action": [
    "kms:Decrypt",
    "kms:DescribeKey"
  ],
  "Resource": "*"
}
```

2. Encrypt the secret in this account using the customer-managed key as shown in the following image, and click **Save**:



3. Attach the following resource policy to the secret:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::<Secure Agent account>:role/<application_role>"
      },
      "Action": "secretsmanager:GetSecretValue",
      "Resource": "*"
    }
  ]
}
```

Step 2. Set up the account that hosts the Secure Agent.

To set up the account that hosts the Secure Agent, attach an identity policy to the role that the agent uses, and then verify that the account that hosts the agent can fetch the KMS key from the account that hosts the secrets.

Attach the following identity policy to the the role that the agent uses to access secrets:

```
{
  "Version" : "2012-10-17",
  "Statement" : [
    {
      "Effect": "Allow",
      "Action": "secretsmanager:GetSecretValue",
      "Resource": "<secret ARN of secrets account>"
    },
    {
      "Effect": "Allow",
      "Action": "kms:Decrypt",
      "Resource": "<KMS key ARN of secrets account>"
    }
  ]
}
```

To verify that the account that hosts the agent can retrieve the KMS key, run the following command from the EC2 instance in the account that hosts the agent:

```
aws secretsmanager get-secret-value --secret-id '<secret ARN>'
```

For more information about configuring cross-account access, see "Access AWS Secrets Manager secrets from a different account" in the AWS documentation.

AWS Secrets Manager secret names

AWS Secrets Manager enforces restrictions on secret names.

In AWS Secrets Manager, secret names can contain only alphanumeric characters and the following special characters:

`/ _ + = . @ - "`

For more information about secret names, see the AWS Secrets Manager documentation.

AWS Secrets Manager connection properties

If you select AWS Secrets Manager as your secrets manager, configure connection properties such as the authentication type and region. The connection properties vary based on the authentication type.

Role-based authentication

Configure the following properties when you access Secrets Manager using role-based authentication:

Property	Description
Type	Secrets manager type. Choose **AWS Secrets Manager**.
Authentication Type	Authentication type that the Secure Agent should use to access Secrets Manager. For role-based authentication, choose **Role Based Access**.
IAM Role	Amazon Resource Name (ARN) of the IAM role that the Secure Agent should use to access secrets. Typically, the format is: `arn:aws:iam::<account>:role/<role name with path>` **Note:** The IAM role must be configured as described in "IAM role configuration for AWS Secrets Manager" on page 72. The AWS resource that hosts the Secure Agent needs to have access to this role.
External ID	External ID required to assume the IAM role.
Region	Region code for the region where your Secrets Manager secrets are hosted, for example, `us-east-2`. Don't enter a full region name like `US East (Ohio)`.
STS Endpoint	STS endpoint URL if you are using a regional or manually configured endpoint. For example, if your service endpoint is US West (N. California), enter the following value: `https://secretsmanager.us-west-1.amazonaws.com` If not specified, the global endpoint `https://sts.amazonaws.com` is used.
STS Endpoint Region	Region in which your service endpoint is located, for example, `us-west-1`. Enter a value for this property if your STS endpoint region differs from your Secrets Manager region. If not specified, the STS endpoint region is assumed to be the same as the Secrets Manager region.

Instance profile authentication

Configure the following properties when you access Secrets Manager using instance profile authentication:

Property	Description
Type	Secrets manager type. Choose **AWS Secrets Manager**.
Authentication Type	Authentication type that the Secure Agent should use to access Secrets Manager. For instance profile authentication, choose **Instance Profile**.
Region	Region code for the region where your Secrets Manager secrets are hosted, for example, `us-east-2`. Don't enter a full region name like `US East (Ohio)`.

Access key authentication

Configure the following properties when you access Secrets Manager using an access key:

Property	Description
Type	Secrets manager type. Choose **AWS Secrets Manager**.
Authentication Type	Authentication type that the Secure Agent should use to access Secrets Manager. For access key authentication, choose **Access Key**.
Access Key ID	AWS access key ID that the Secure Agent should use to access secrets, for example, `AKIAIOSFODNN7EXAMPLE`. The access key ID must be associated with an IAM role that is assigned an access policy with the GetSecretValue and ListSecrets permissions. You need to enter both the access key ID and the secret access key.
Secret Access Key	AWS secret access key that the Secure Agent should use to access secrets, for example `wJalrXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY`. You need to enter both the access key ID and the secret access key.
Region	Region code for the region where your Secrets Manager secrets are hosted, for example, `us-east-2`. Don't enter a full region name like `US East (Ohio)`.

For more information about AWS Secrets Manager properties, see the AWS documentation.

Azure Key Vault configuration

Azure Key Vault is a cloud service provided by Microsoft Azure designed to safeguard cryptographic keys, secrets, and certificates used by cloud applications and services. It helps organizations securely store and control access to tokens, passwords, API keys, encryption keys, and other sensitive information. You can

configure your organization to retrieve sensitive connection credentials from Azure Key Vault instead of directly entering the credentials in the connection properties.

To configure your organization to retrieve secrets from Azure Key Vault, enable secret vault in Administrator, select Azure Key Vault as the secrets manager, and configure the connection properties. Then, you can configure connections to retrieve sensitive credentials from Azure Key Vault.

Azure Key Vault secret names

In Azure Key Vault, secret names can only contain alphanumeric characters and dashes.

For more information about secret names and formats, see the Azure Key Vault documentation.

Azure Key Vault connection properties

If you select Azure Key Vault as your secrets manager, configure connection properties such as the client ID, client secret, and tenant ID.

Configure the following properties:

Property	Description
Type	Secrets manager type. Choose **Azure Key Vault**.
Client ID	Application (client) ID that the Secure Agent should use to connect to your key vault. The client ID is the unique application (client) ID assigned to your app by Azure AD when it was registered. **Tip:** You can find your application (client) ID in your Azure subscription in **Azure Active Directory** > **Enterprise applications** > **Application (client) ID**. The application (client) that you specify must have the Get and List permissions for secrets.
Client Secret	Secret string that the Secure Agent uses to prove its identity when requesting access to the key vault.
Tenant ID	Azure Active Directory (tenant) ID that should be used for authenticating requests to the key vault.
Vault URI	URI of the key vault that stores the connection credentials.
Authority Host	URL of the authority host endpoint. If not specified, the global endpoint `https://login.microsoftonline.com` is used.

For more information about Azure Key Vault properties, see the Azure documentation.

HashiCorp Vault configuration

HashiCorp Vault is a tool designed for managing secrets and protecting sensitive data. It provides a secure way to store, manage, and control access to tokens, passwords, certificates, encryption keys, and other secrets used by your applications. You can configure your organization to retrieve sensitive connection credentials from HashiCorp Vault instead of directly entering the credentials in the connection properties.

To configure your organization to retrieve secrets from Vault, enable secret vault in Administrator, select HashiCorp Vault as the secrets manager, and configure the connection properties. Then, you can configure connections to retrieve sensitive credentials from Vault.

HashiCorp Vault authentication

IDMC uses AppRole authentication to authenticate with HashiCorp Vault. Upon successful authentication, Vault issues a client token to IDMC. The token contains the policies that are attached to the AppRole.

The token can be either of the following types:

- Batch token. Batch tokens have a fixed, short-term time to live and are not renewable. They are not recommended for long-running jobs.
- Service token. Service tokens are suitable for long-running jobs and can be renewed to extend their lifespan. Vault issues service tokens by default.

Because service tokens have a longer lifespan and can be renewed, you must configure the HashiCorp Vault AppRole to issue service tokens to IDMC. If Vault issues a batch token and the token expires, the Secure Agent can't connect to Vault to retrieve secrets.

For more information about client tokens, see the HashiCorp Vault documentation.

HashiCorp Vault secret formats

In HashiCorp Vault, the format of secrets varies based on the secrets engine version.

In Vault, secrets must be in either of the following formats based on the secrets engine version:

Secrets Engine Version	Format
Secrets engine v1	`secret/<secret_path>:<key>`
Secrets engine v2	`secret/data/<secret_path>:<key>`

Note: Because a colon is used to separate the secret path from the key, IDMC can't process keys that have a colon in the path.

For more information about secret names and formats, see the HashiCorp Vault documentation.

HashiCorp Vault connection properties

If you select HashiCorp Vault as your secrets manager, configure connection properties such as the role ID, secret ID, and Vault URI.

Configure the following properties:

Property	Description
Type	Secrets manager type. Choose **HashiCorp Vault**.
Role ID	ID of the AppRole that the Secure Agent should use to authenticate with Vault. The AppRole must have the read and list permissions for secrets.
Secret ID	Secret ID of the AppRole that the Secure Agent should use to authenticate with Vault.
Vault URI	URI of the key vault that stores the connection credentials, for example: `https://my-hashicorp-vault-12343a56.a1b2345c.z1.hashicorp.cloud:8200`

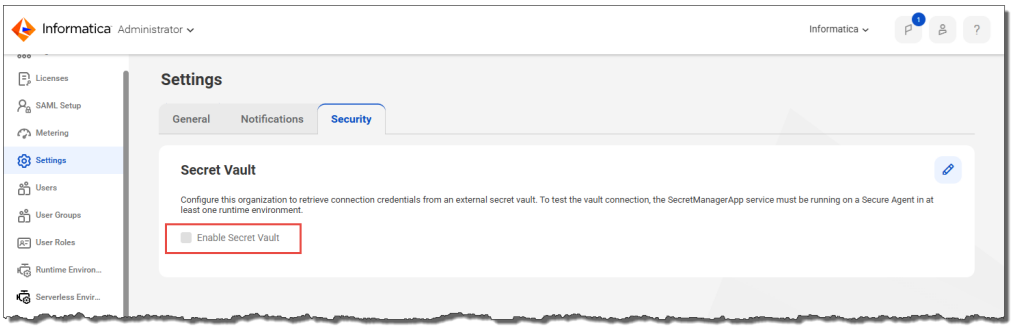
Property	Description
Namespace	Namespace within the key vault, if used.
AppRole Path	Custom path in which the AppRole authentication method was enabed. If not specified, the value is assumed to be the default path, `approle`.

For more information about HashiCorp Vault properties, see the HashiCorp documentation.

Enabling and disabling a secrets manager

Enable and disable the use of a secrets manager on the **Security** tab of the **Settings** page.

1. On the **Settings** page, open the **Security** tab.
2. Click the edit (pencil) icon.
3. Select **Enable Secret Vault**, as shown in the following image:



4. Select the secrets manager that you use, either AWS Secrets Manager, Azure Key Vault, or HashiCorp Vault.
5. Enter the connection details such as the vault URI, authentication type, and region.
6. Test the connection.

   When you test the connection, you need to select a runtime environment. All Secure Agents within the runtime environment must be installed on a local machine or VM, and the SecretManagerApp service must be running on each agent. The Hosted Agent and serverless agents can't connect to an external secrets manager.

When the connection is successful, you can configure connections to use the secrets manager.

To disable the use of a secrets manager, clear the **Enable Secret Vault** option. However, you'll first need to disable the **Use Secret Vault** option in all connections.

Configuring a secrets manager for a sub-organization

You can configure a secrets manager for a sub-organization. To do this, log into the sub-organization and enable the secrets manager.

When you enable a secrets manager for a sub-organization, ensure that the SecretManagerApp service is running on a Secure Agent that is native to the sub-organization. If the SecretManagerApp service runs on a Secure Agent that is shared from the parent organization, it can only connect to the secrets manager that is configured for the parent organization. It can't connect to the secrets manager that is configured for the sub-organization.

For more information about the SecretManagerApp Secure Agent service, see *Secure Agent Services*.

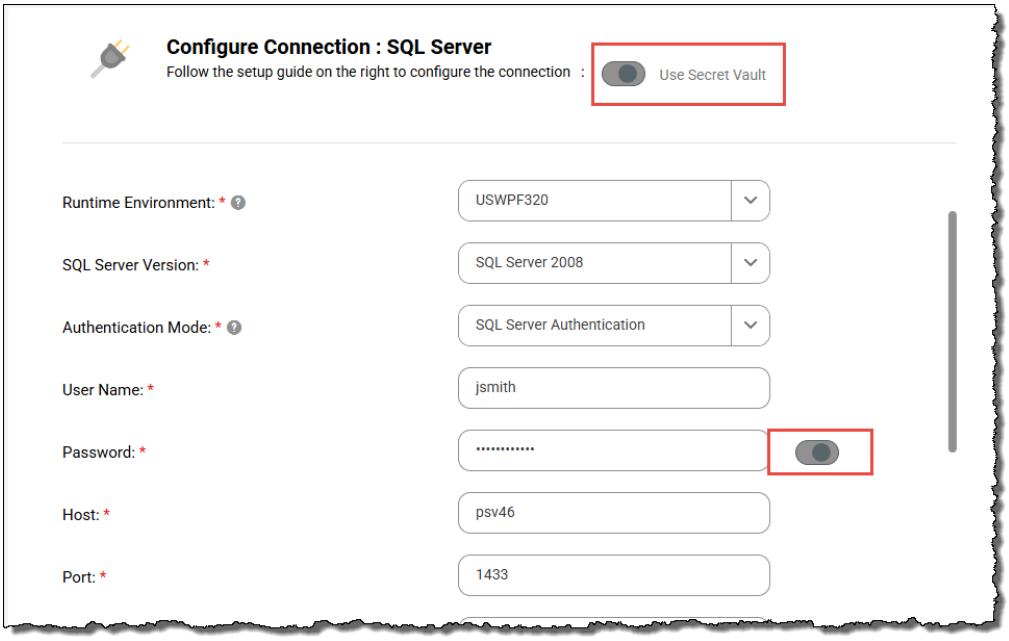
Configuring a connection to use the secrets manager

You can configure any connection that has sensitive credentials to retrieve these credentials from the secrets manager.

1. Open the **Connections** page.
2. Perform either of the following actions:
   - To create a connection, click **New Connection** and enter the connection details.
   - To edit a connection, click the connection name, and then click **Edit**.
3. In the Connection Properties area, select **Use Secret Vault**.
4. Enable the slider next to each property that you store in the secrets manager, and then enter the secret name, including the path, in the corresponding field. If the secret is a JSON object, you'll also need to include the secret key.

   For example, you configure a relational connection and you store the database password in HashiCorp Vault. The secret name is `secret/data/MyCredentials`, and the secret key is `MyPassword`. To retrieve the password from HashiCorp Vault, select **Use Secret Vault**, enable the slider next to the **Password** field, and enter `secret/data/MyCredentials:MyPassword` in the **Password** field.

The following image shows the connection details:



For more information about formatting the connection property, see "Connection property format" on page 85.

5. Select the runtime environment to be used with the connection.

   All Secure Agents within the runtime environment must be installed on a local machine or VM and must be able to access the secrets manager. Additionally, the SecretManagerApp service must be running on each agent.

6. Configure the connection-specific properties.
7. To test the connection, click **Test Connection**.
8. Click **Save**.

For more information about configuring connections, see *Connections*.

Connection property format

To configure a connection property to retrieve its value from a secrets manager, first enable the slider next to the property. Then enter the secret name, including the path, in the corresponding field. If the secret is a JSON object, you'll also need to include the secret key.

The following table shows the value to enter based on the format of the secret:

Format of secret	Format of value to enter

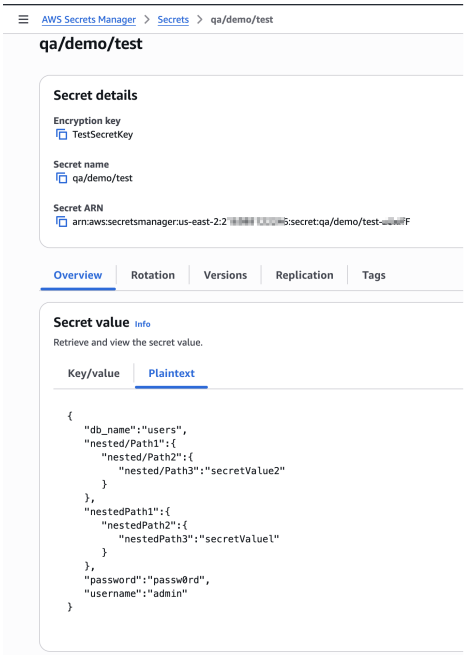
JSON object, for example:
`{`
`"engine": "mysql",`
`"username": "tsmith",`
`"password": "Hello123",`
`"host": "my-database-endpoint.us-west-2.rds.amazonaws.com",`
`"dbname": "myDatabase",`
`"port": "1234"`
`}`
	`<secret name>:<key>`
	If you use AWS Secrets Manager, you can also enter the secret in the following format:*
	`<secret ARN>:<key>`

Simple value, for example: `--name "MyPassword" --value "Hello123"`	`<secret name>` If you use AWS Secrets Manager, you can also enter the full ARN of the secret.*
* If you use AWS Secrets Manager and the account that hosts the Secure Agent differs from the account that hosts the secrets, you must enter the secret using the full ARN.	

Example: Connection property format for AWS Secrets Manager

You want to configure a connection to retrieve a password from AWS Secrets Manager.

The following image shows the secret:



The secret name is `qa/demo/test`, the secret ARN is `arn:aws:secretsmanager:us-east-2:2*********6:secret:qa/demo/test-zUxff*****F`, and the key is `password`.

Enter either of the following values for the **Password** connection property:

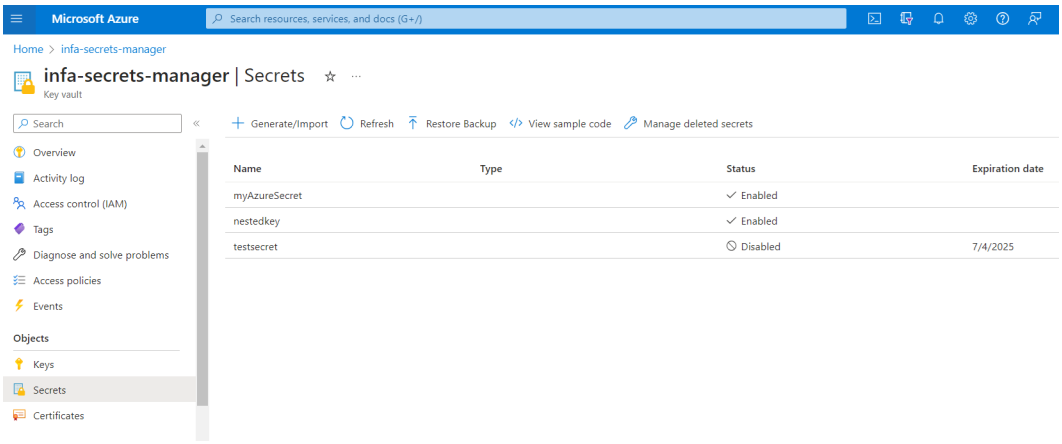
- `qa/demo/test:password`
- `arn:aws:secretsmanager:us-east-2:2*********6:secret:qa/demo/test-zUxff*****F:password`

  Note: If the account that hosts the Secure Agent differs from the account that hosts the secrets, you must enter the password in this format.

Example: Connection property format for Azure Key Vault

You want to configure a connection to retrieve a password from Azure Key Vault. The password is stored in a secret named "nestedkey."

The following image shows the secret:



The secret is stored in the following format:

```
{
"name":"Morpheus",
   "job":"Leader",
   "id":"199",
   "dbdetails":{
      "uname": "root",
      "password": "passw0rd"
   }
}
```

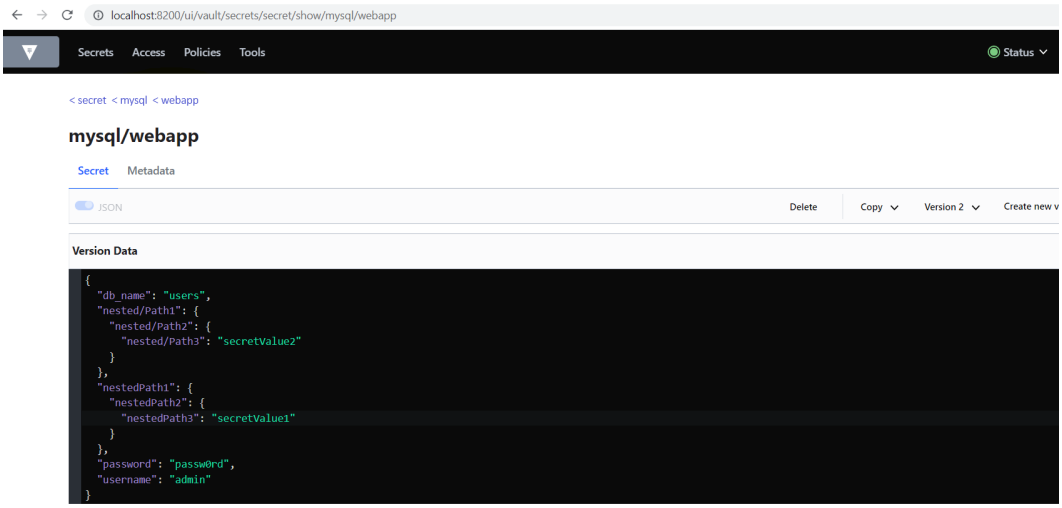
Enter the following value for the **Password** connection property:

`nestedkey:dbdetails:password`

Example: Connection property format for HashiCorp Vault

You want to configure a connection to retrieve a password from HashiCorp Vault.

The following image shows the secret:



The secret name is `secret/mysql/webapp` and the key is `password`.

Enter the following value for the **Password** connection property:

`secret/mysql/webapp:password`

CHAPTER 8

Permissions

Permissions determine the access rights that a user has for a Secure Agent, Secure Agent group, connection, schedule, or asset. Permissions add additional or custom security for an object. Permissions define which users and groups can read, update, delete, execute, and change permissions on the object.

The role assigned to your user account or to a group in which you are a member must have the Set Permission privilege for the object type. For example, to configure permissions on a Secure Agent, you must be assigned a role that has the Set Permission privilege for Secure Agents.

To configure permissions on an object, navigate to the object and set the appropriate permissions. For example, you want only users in the Development Team user group to have access to assets in the Development Data folder. Navigate to the folder, edit the permissions, and grant the Development Team user group permissions on the folder.

Permissions apply to the objects that you configure but not to copies of the object. Therefore, when you copy or export an asset, the permissions are not copied or exported with the asset. For example, you export a mapping task in which only user rjones has execute permission. When you import the mapping task, the imported mapping has no permissions assigned to it. Therefore, any user with privileges to run mapping tasks can run the imported task.

You can configure the following permissions on an object:

Permission	Description
Read	Open and view the object. If the object is source controlled, this permission allows the user or group to pull or check out the object from the source control repository. You must have the read permission to access the integration hub connection to perform any operations. If you select a task, this permission also allows the user or group to use a connection or schedule in the task.
Update	Edit the object. If the object is source controlled, this permission allows the user or group to check in, check out, pull, unlink, or roll back the object. Requires read permission, which is automatically granted.
Delete	Delete the object.
Execute	Run the object. Applies to mappings, tasks, taskflows, and Cloud Integration Hub assets. Monitor, stop, and restart instances of the mapping, task, or taskflow.
Change permissions	Change the permissions that are assigned to the object.

Note: These permissions control permissions within IDMC. They do not control operating system permissions, such as the ability to start, stop, or configure the Secure Agent on Windows or Linux.

Rules and guidelines for permissions

Use the following rules and guidelines for permissions:

- When you configure permissions on an object, verify that the user or group to which you grant permissions is assigned a role with the appropriate privileges for the object type.
- For example, if you grant a user with the Service Consumer role Update privilege on a particular folder, the user cannot update the folder because the Service Consumer role does not have update privileges for folders.
- To edit an asset, the user must have read permission on all assets used within the asset. For example, when you assign a user Read and Update permissions on a synchronization task, verify that the user also has Read permission on the connections, mapplets, schedules, and saved queries that are used in the task.
- To run a subscription or a publication that executes a mapping task, the user must have the Update privilege for the project and folder that contains the mapping task.
- Users with the Admin role or who belong to a group with the Admin role always have "change permission" privileges, regardless of any other setting. You can't revoke the "change permission" privilege for Admin users.
- When a user edits a task, assets without Read permission are not displayed. To avoid unexpected results, the user should cancel all changes and avoid editing the task until the user is granted the appropriate Read permissions.
- When configuring a taskflow, a user needs Execute permission on all tasks to be added to the taskflow.
- To edit a taskflow, a user needs Execute permission on all tasks in the taskflow. Without Execute permission on all tasks, the user cannot save changes to the taskflow.
- To run a taskflow, a user needs Read and Execute permissions on taskflows.
- To view taskflow run details on the Monitor page, a user needs Read permission on the taskflow.
- To run, re-run, suspend, resume, restart, abort, stop, and schedule a taskflow, a user needs Execute permission.
- To monitor jobs or to stop a running job, a user needs Execute permission on the mapping, task, or taskflow.
- If you assign custom permissions to a Data Integration task and invoke the Data Integration task through an Application Integration process or a guide, you must complete either of the following tasks:
  - Give the Application Integration anonymous user permission to run the associated Data Integration asset.
  - Add the Application Integration anonymous user to a user group that has permission to run the associated Data Integration asset.

Configuring permissions

You can configure permissions on an object if you are assigned a role with the Set Permission privilege for the object type. For example, to configure permissions on a folder, you must be assigned a role that has the Set Permission privilege for folders.

1. Navigate to the object for which you want to configure permissions.

   For example:

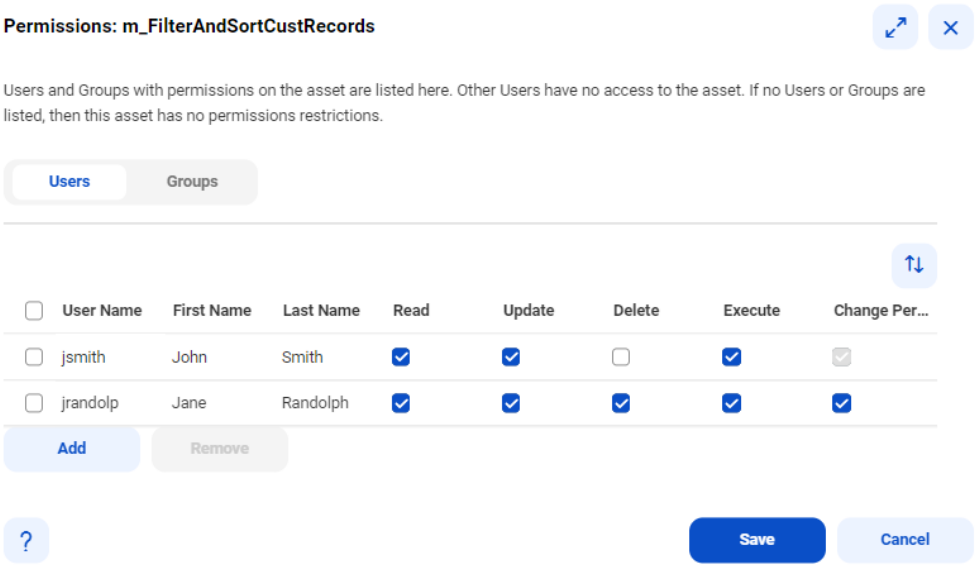
   - To configure permissions on a Secure Agent or Secure Agent group, in Administrator, select **Runtime Environments**.
   - To configure permissions on a connection, in Administrator, select **Connections**.
   - To configure permissions on a mapping, in Data Integration, open the project and folder that contain the mapping.

2. In the row that contains the object, either click **Actions** and select **Permissions**, or click the **Change Permission** icon.

   The **Permissions** dialog box lists the users and groups that have permissions on the object.

   If the **Permissions** dialog box lists no users or groups, then no permissions are configured for the object. Any user with appropriate privileges for the object type can access the object.

   The following image shows the **Permissions** dialog box for a mapping:



3. To configure user permissions on the object:

   a. Select **Users**.

   b. If the user does not appear in the **Users** list, click **Add**, and select a user.

   c. Enable or disable the appropriate permissions on the user.

   Note: When you grant any user permissions on the object, IDMC also adds you as a user with permissions on the object. This prevents you from losing access to the object when you configure permissions.

4. To configure user group permissions on the object:

   a. Select **Groups**.

b. If the group does not appear in the **Groups** list, click **Add**, and select a group.

c. Enable or disable the appropriate permissions on the group.

Note: When you grant any group permissions on the object, IDMC also adds you as a user with permissions on the object. This prevents you from losing access to the object when you configure permissions.

5. To remove all permissions restrictions for the object, remove all users and groups from the **Permissions** dialog box.

   When you remove all users and groups, any user with appropriate privileges for the object type can access the object.

6. Click **Save**.

CHAPTER 9

Schedules

You can create schedules to run tasks or taskflows at specified times or at regular intervals. You can also define a blackout period during which scheduled tasks or jobs don't run.

You must have the Schedule privilege in the Administrator asset privileges to create, read, update, delete, or set permissions on project schedules. Create schedules and configure blackout periods on the **Schedules** page in Administrator. After you create a schedule, you can associate it with tasks and taskflows in another service such as Data Integration.

When you create a schedule, you specify the date and time. You can configure a schedule to run associated assets throughout the day between 12:00 a.m. and 11:55 p.m. IDMC might add a small schedule offset to the start time, end time, and all other time configurations. As a result, scheduled tasks and taskflows might start later than expected. For example, you configure a schedule to run hourly until noon, and the schedule offset for your organization is 10 seconds. IDMC extends the end time for the schedule to 12:00:10 p.m., and the last hourly task or taskflow starts at 12:00:10 p.m. To see the schedule offset for your organization, check the **Schedule Offset** organization property for the Data Integration Service.

You can perform the following tasks with schedules:

Associate a schedule with a task or taskflow

To associate a schedule with a task or taskflow, edit the task or taskflow. For example, to associate a schedule with a mapping task, edit the mapping task in Data Integration, and select the schedule on the **Schedules** page.

When you copy a task or taskflow that includes a schedule, the schedule is not associated with the new asset. To associate a schedule with the new asset, edit the asset.

Monitor scheduled tasks

You can monitor scheduled tasks from the **All Jobs** page in Monitor. Scheduled tasks do not appear on the **My Jobs** page.

Export a schedule

You can export a schedule from your organization and import it into another organization. Export a schedule on the **Schedules** page. If the schedule is associated with a task or taskflow, the task or taskflow is not included in the export file.

Delete a schedule

Delete a schedule on the **Schedules** page in Administrator.

Note: You cannot delete a schedule that is used in a task or taskflow. Remove the schedule from all tasks and taskflows before you delete the schedule.

Configuring a blackout period

A blackout period prevents all scheduled tasks and taskflows in the organization from running during a specified period of time. You can configure a blackout period during which the scheduled Data Integration and Data Ingestion and Replication file publications and file subscriptions don't run.

You can configure one blackout period for an organization or sub-organization. The blackout period you configure for a parent organization does not effect the sub-organizations.

If a task is scheduled to run during a blackout period, the task instance will not be started during the blackout period, and it will not restart automatically when the blackout period ends. After the blackout period, task instances will resume according to the schedule. If a task is already running when a blackout period starts, it will not be stopped.

To configure a blackout period, in Administrator, select **Schedules**, and then click **Blackout Period**. The blackout period is displayed on the **Schedules** page.

Repeat frequency

The repeat frequency determines how often tasks run. You can set the repeat frequency to every N minutes, hourly, daily, weekly, biweekly, or monthly.

The following table describes the repeat frequency options:

Option	Description
Does not repeat	Tasks run as scheduled and do not repeat.
Every N minutes	Tasks run on an interval based on a specified number of minutes. You can configure the following options: - Repeat frequency. Select a frequency in minutes. Options are 5, 10, 15, 20, 30, 45. - Days. Days of the week when you want tasks to run. You can select one or more days of the week. - Time range. Hours of the day when you want tasks to start. Select All Day or configure a time range. You can configure a time range between 00:00-23:55. - Repeat option. The range of days when you want tasks to run. You can select Repeat Indefinitely or configure an end date and time.
Hourly	Tasks run on an hourly interval based on the start time of the schedule. You can configure the following options: - Repeat frequency. Select a frequency in hours. Options are 1, 2, 3, 4, 6, 8, 12. - Days. Days of the week when you want tasks to run. You can select one or more days of the week. - Time range. Hours of the day when you want tasks to start. Select All Day or configure a time range. You can configure a time range between 00:00-23:55. - Repeat option. The range of days when you want tasks to run. You can select Repeat Indefinitely or configure an end date and time.
Daily	Tasks run daily at the start time configured for the schedule. You can configure the following options: - Repeat frequency. The frequency at which you want tasks to run. Select Every Day or Every Weekday. - Repeat option. The range of days when you want tasks to run. You can select Repeat Indefinitely or configure an end date and time.

Option	Description
Weekly	Tasks run on a weekly interval based on the start time of the schedule. You can configure the following options: - Days. Days of the week when you want tasks to run. You can select one or more days of the week. - Repeat option. The range of days when you want tasks to run. You can select Repeat Indefinitely or configure an end date and time. If you do not specify a day, the schedule runs regularly on the same day of the week as the start date.
Biweekly	Tasks run every two weeks based on the start time of the schedule. You can configure the following options: - Days. Days of the week when you want tasks to run. You can select one or more days of the week. You must select at least one day. - Repeat option. The range of days when you want tasks to run. You can select Repeat Indefinitely or configure an end date and time. If you configure a biweekly schedule to start at 5 p.m. on a Tuesday and run tasks every two weeks on Mondays, the schedule begins running tasks on the following Monday.
Monthly	Tasks run on a monthly interval based on the start time of the schedule. You can configure the following options: - Day. Day of the month when you want tasks to run. You can configure one of the following options: - Select the exact date of the month, between 1-28. If you want the task to run on days later in the month, use the <n> <day of the week> option. - Select the <n> <day of the week>. Options for <n> include First, Second, Third, Fourth, and Last. Options for <day of the week> includes Day, and Sunday-Saturday. Tip: With the Day option, you can configure tasks to run on the First Day or the Last Day of the month. - Repeat option. The range of days when you want tasks to run. You can select Repeat Indefinitely or configure an end date and time.

Time zones and schedules

IDMC stores time in Coordinated Universal Time (UTC). When you log in, IDMC converts the time and displays it in the time zone associated with your user profile.

When you create a schedule, you select the time zone for the scheduler to use. You can select a time zone that is different from your time zone or your organization time zone.

Daylight Savings Time changes and schedules

IDMC applies Daylight Savings Time changes to all tasks except biweekly tasks.

When Daylight Savings time goes into effect, tasks scheduled to run between 2:00 a.m. and 2:59 a.m., do not run the day that the time changes from 2:00 a.m. to 3:00 a.m. If a task is scheduled to run biweekly at 2 a.m., it will run at 3 a.m. the day of the time change and at 2 a.m. for the next run.

Daylight Savings Time does not trigger additional runs for tasks that are scheduled to run between 1:00 a.m. - 1:59 a.m. when Standard Time begins. For example, a task is scheduled to run every day at 1:30 a.m. When the time changes from 2 a.m. to 1 a.m., the task does not run again at 1:30 a.m.

Tip: To ensure that IDMC does not skip any scheduled runs near the 2 a.m. time change, do not schedule jobs to run between 12:59 a.m. and 3:01 a.m.

Configuring a schedule

Configure a schedule on the **Schedules** page. For mapping tasks and synchronization tasks, you can also create a new schedule when you configure the task. You can configure a schedule to run once or at a specific interval and to run indefinitely or until a specified end time.

1. In Administrator, select **Schedules**.
2. To create a schedule, click **New Schedule**.

   To edit a schedule, click the edit icon in the row that contains the schedule.
3. Configure the following properties:

Property	Description
Schedule Name	Name of the schedule. Each schedule name must be unique within the organization. Schedule names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Maximum length is 255 characters. Names are not case sensitive.
Description	Description of the schedule. Maximum length is 255 characters.
Starts	Date and time when the schedule starts. The date format is MM/DD/YYYY. Time appears in the 24-hour format. Click the calendar button to select the start date. The start date and time can affect the repeat frequency for tasks and taskflow jobs that repeat at regular intervals. For example, if the start date is November 10 and the repeat frequency is monthly, the schedule runs associated assets on the tenth day of each month. If the start time is 3:10 and the repeat frequency is hourly, the assets run every hour at 10 minutes past the hour. Default is the current date, current time, and time zone of the user who creates the schedule.
Time Zone	Select the time zone for the schedule to use. The time zone can differ from the organization time zone or user time zone.
Repeats	Repeat frequency for the schedule. Select one of the following options: - Does Not Repeat - Every N Minutes - Hourly - Daily - Weekly - Biweekly - Monthly Default is Does Not Repeat.

4. Click **Save**.

Exporting schedules

You can export schedules from your organization and import them into other organizations. Assets that are associated with the schedules are not included in the export file. Export schedules on the **Schedules** page.

1. In Administrator, select **Schedules**.
2. Click **Export**.
3. In the **Export Schedules** dialog box, select the schedules that you want to export.
4. Optionally, update the export job name.

   By default, the job name is `SchedulesExport_<date>`.
5. Click **Export**.

   Administrator creates an export job to export the schedule.
6. To check the status of the export job and download the export file, open the **Import/Export Logs** page in Monitor, and click the **Export** tab.

   You can download the export file in the row that contains the export job or on the job details page.

You can import schedules on the **Explore** page in another service such as Data Integration. For information about importing assets, see the help for that service.

After you import schedules, you can associate them with assets in the target organization.

Troubleshooting scheduled tasks

The task does not run at the scheduled time.

A task does not run at the scheduled time if another instance of it is already running when the schedule tries to start the task. For example, you schedule a task to run every 5 minutes. The first task starts at 12 p.m., but does not complete until 12:06 p.m. The second instance of the task does not run at 12:05 p.m. because the first instance has not completed. Data Integration starts the next task at 12:10 p.m.

To resolve this issue, change the schedule to allow the task to complete before starting the next task run.

CHAPTER 10

CLAIRE preferences

You can configure CLAIRE preferences for your organization to enable or disable CLAIRE generative AI services and CLAIRE recommendations.

You can configure the following settings:

CLAIRE generative AI services

Enables users in your organization to use CLAIRE to discover, analyze, and explore data through natural language prompts.

Default is disabled.

CLAIRE recommendations

Enables in-product recommendations for mapping design based on analysis of metadata from your organization's assets and assets from other IDMC organizations.

Default is enabled for parent organizations and disabled for sub-organizations.

CLAIRE generative AI services

Enable CLAIRE generative AI services to allow the users in your organization to use CLAIRE to discover, analyze, and explore data through natural language prompts.

You can enable CLAIRE generative AI services for the following services:

- CLAIRE GPT
- CLAIRE Copilot for Data Integration, Application Integration, and MDM SaaS services

If your organization uses Data Integration, enable CLAIRE generative AI services to use the following features:

- Generate expressions to define mapping logic with natural language prompts
- Generate transformations in the mapping canvas with natural language prompts

If your organization uses Data Governance and Catalog, enable CLAIRE generative AI services to use the following features:

- Generate name and description for business and technical assets
- Automatically generate data lineage
- Use natural language prompts to analyze descriptions and suggest data quality rules
- Search for assets with natural language prompts. Data Governance and Catalog interprets search intent, displays AI reasoning, and shows generated search queries.

By default, CLAIRE generative AI services are disabled. You can enable or disable them for your organization at any time.

For more information about CLAIRE GPT, see the *CLAIRE GPT* help.

CLAIRE recommendations

Enable CLAIRE recommendations to allow in-product recommendations for mapping design.

When you enable CLAIRE recommendations, CLAIRE makes design recommendations based on the current flow of the mapping and anonymous metadata from prior mappings across IDMC organizations.

By default, CLAIRE recommendations are enabled. When you disable CLAIRE recommendations, recommendations are disabled for all users within your organization. You can enable or disable recommendations for your organization at any time.

By default, CLAIRE recommendations are disabled in sub-organizations. Enable and disable CLAIRE recommendations for sub-organizations from within the sub-organization.

When you enable CLAIRE recommendations, Data Integration users can disable recommendations for individual mappings in the mapping designer.

Enabling CLAIRE recommendations allows CLAIRE to recommend transformation types to include in your mapping flow. If your organization uses additional services, CLAIRE can also recommend additional objects, components, or transformations during mapping design.

For example, if your organization uses Metadata Command Center, CLAIRE can recommend the following objects:

- Additional source objects based on primary key and foreign key relationships
- Mapplets and user-defined functions associated with the same data classifications as the source
- Expression transformations to standardize source data

If your organization uses Advanced Integration, enabling CLAIRE recommendations enables the following features:

- CLAIRE-powered configurations, CLAIRE insights, and CLAIRE recommendations for advanced clusters
- CLAIRE-powered runtime strategies and CLAIRE Tuning for mapping tasks that are based on mappings in advanced mode
- CLAIRE recommendations for jobs that run mappings in advanced mode

CHAPTER 11

Bundle management

A bundle is a set of related assets that you can use to improve productivity and enhance the quality of your IDMC projects. Data Integration users design, create, and publish bundles. Administrators manage bundles.

If you are the administrator for an organization, you can perform the following actions to manage bundles:

Install a bundle.

You can install a public, private, or unlisted bundle that the bundle designer has configured to be used as a reference. When you install a bundle, the bundle is installed into the Add-On Bundles project on the **Explore** page. Users in your organization can use the assets in the bundle, but they cannot edit the assets.

Copy a bundle.

You can copy a public, private, or unlisted bundle that the bundle designer has configured for copying. When you copy a bundle, you select the folder where you want to copy the bundle contents. You can copy a bundle multiple times and save the contents into a different project or folder each time that you copy it. After you copy a bundle, users in your organization can edit the assets.

Upgrade a bundle.

If you installed a bundle and a newer version of the bundle is available, you can upgrade the bundle to get the latest version.

Uninstall a bundle.

If your organization no longer needs an installed bundle, you can uninstall it.

To view the bundles that are installed or are available to your organization, in Administrator, select **Add-On Bundles**. The **Add-on Bundles** page displays information about installed bundles, copied bundles, and bundles that are available for installation or copying.

For information about creating or publishing bundles, see *Asset Management* in the Data Integration service help.

Types of bundles

You specify a bundle type when you create a bundle. The bundle type determines who can use the bundle and how the bundle is published.

You can create the following types of bundles:

Public bundles

A public bundle is available to all organizations and sub-organizations. When you create a public bundle, you submit the bundle to Informatica for approval. When the bundle is approved, Informatica makes the bundle available to all organizations.

Private bundles

A private bundle is available only to a parent organization and its sub-organizations. You create a private bundle in the parent organization and then publish the bundle to the sub-organizations. Published private bundles appear on the **Add-On Bundles** page for sub-organizations. Informatica doesn't review private bundles.

If a pushed bundle contains objects with changes that aren't compatible with existing sub-organization tasks, it might cause the sub-organization tasks to become invalid or be removed.

You can't create private bundles if the organization doesn't have any sub-organizations.

Unlisted bundles

An unlisted bundle is used as a reference to create other bundles. Installing an unlisted bundle requires an access code.

Installing a bundle

You can install a public, private, or unlisted bundle that the bundle designer has configured to be used as a reference. Install a bundle on the Available Bundles tab of the **Add-On Bundles** page.

Before you install an unlisted bundle, get the bundle access code. To get the access code for a bundle that was created in your organization, open the **Bundles** page in Data Integration, click the bundle name, and then click **Copy Access Code**. To get the bundle access code for a bundle that was created outside of your organization, contact the bundle publisher.

1. In Administrator, select **Add-On Bundles**.
2. Click **Available Bundles**.

   The Available Bundles tab lists the public and private bundles that are available for installation or copying.
3. If the bundle that you want to install is an unlisted bundle, enter the bundle access code in the **Find** field.
4. Click the bundle name to open the Bundle Details page.
5. Verify that the **Allow** field is set to **Reference** or to **Reference and Copy**.

   You cannot install a bundle that is configured for copying only.
6. Click **Install**.

In Data Integration, the bundle is added to the Add-On Bundles project, and the assets are ready for use. The bundle is also listed on the **Installed Bundles** tab of the **Add-On Bundles** page in Administrator.

Copying a bundle

You can copy a public, private, or unlisted bundle that the bundle designer has configured for copying. Copy a bundle on the Available Bundles tab of the **Add-On Bundles** page. Each time you copy a bundle, an event is logged on the Copied Bundles tab.

Before you copy an unlisted bundle, get the bundle access code. To get the access code for a bundle that was created in your organization, open the **Bundles** page in Data Integration, click the bundle name, and then click **Copy Access Code**. To get the bundle access code for a bundle that was created outside of your organization, contact the bundle publisher.

1. In Administrator, select **Add-On Bundles**.
2. Click **Available Bundles**.

   The Available Bundles tab lists the public and private bundles that are available for installation or copying.
3. If the bundle that you want to copy is an unlisted bundle, enter the bundle access code in the **Find** field.
4. Click the bundle name to open the Bundle Details page.
5. Verify that the **Allow** field is set to **Copy** or to **Reference and Copy**.

   You cannot copy a bundle that is configured to be used as a reference only.
6. Click **Copy Bundle Content to...**.
7. In the **Browse** dialog box, select the Data Integration project or folder into which you want to copy the bundle contents.
8. Click **Select**.

   The assets in the bundle are copied to the selected project or folder.

Upgrading a bundle

You can upgrade an installed bundle when an updated version becomes available. You can check the bundle status on the Installed Bundles tab of the **Add-On Bundles** page.

1. In Administrator, select **Add-On Bundles**.
2. Click **Installed Bundles**.

   The Bundle Status column indicates whether the bundle is up-to-date or whether an upgrade is available.
3. Click the bundle name to open the Bundle Details page.
4. Click **Upgrade**.

Uninstalling a bundle

Uninstall a bundle if users in your organization no longer need it. Uninstall the bundle on the Installed Bundles tab of the **Add-On Bundles** page.

Note: Uninstalling a bundle removes all of the bundle assets from the organization. If you want to keep tasks that use assets in the bundle, remove the assets from the task before you uninstall the bundle.

1. In Administrator, select **Add-On Bundles**.
2. Click **Installed Bundles**.
3. Click the bundle name to open the Bundle Details page.
4. Click **Uninstall**.

After you uninstall the bundle, it is listed on the Available Bundles tab.

CHAPTER 12

Connection Credentials storage location

You can configure where to store the connection properties for your organization and sub-organizations. To specify where to store the connection properties, configure the **Connection Credentials** on the **Settings** page in Administrator.

You can store connection properties in either of the following locations:

Informatica Cloud

When you store connection properties on the cloud, the connection properties are stored in the IDMC repository and are always available. The connections are encrypted by the IDMC key management service.

IDMC backs up connection properties regularly as part of standard backup procedures.

Local Secure Agent

You might store connection properties with a local Secure Agent if you need the connection properties to reside within your firewall. When you enable this option, the properties for all connections that are listed on the **Connections** page are stored with the local agent.

Note: In organizations subject to FedRAMP, you can't store connection properties with a local Secure Agent.

If you choose this option, you can store connection properties with one Secure Agent. Connection properties are stored in the following directory:

```
<Secure Agent installation directory>/apps/Data_Integration_Server/data
```

When you store properties with a local Secure Agent, the Secure Agent must be running so that tasks can run and users can work with connections. Back up connection properties regularly to prevent loss of data. A best practice is to back up connection properties after you change the location or the encryption key for connection properties.

The connections are encrypted by the IDMC key management service. IDMC uses CBC (Cipher Block Chaining) mode 256 AES encryption to store the connections.

If you use an external secrets manager like AWS Secrets Manager or Azure Key Vault to store sensitive connection credentials, you need to set the connection credential storage to **Informatica Cloud**. When you do this, sensitive credentials are retrieved from the secrets manager and other connection properties are stored in the IDMC repository. You can't use a secrets manager if you store connection credentials on a local Secure Agent. For more information about secrets manager configuration, see Chapter 7, “Secrets managers” on page 70.

You can change where you want to store connection properties. When you do this, IDMC moves the connection properties to the appropriate location. For example, your license expires, so you configure the

organization to store connections on the cloud. IDMC moves the connection properties from the local Secure Agent to IDMC.

CHAPTER 13

Service-specific settings

Configure the settings for your Enterprise Data Catalog and Data Integration services.

Enterprise Data Catalog integration properties

If your organization uses data catalog discovery in Data Integration, you can configure Enterprise Data Catalog integration properties for your organization and sub-organizations. Configure Enterprise Data Catalog integration properties so that users can use catalog assets in mappings, synchronization tasks, and file ingestion and replication tasks.

The Enterprise Data Catalog integration properties that you configure for the organization apply to the data catalog searches that all users in the organization perform. If your organization includes sub-organizations, you can configure different Enterprise Data Catalog integration properties for the parent organization and for each sub-organization.

The following table describes the Enterprise Data Catalog integration properties:

Property	Description
Catalog URL	URL of the Enterprise Data Catalog Service. Use the following format: `http://<fully qualified host name>:<port>` Do not append `/ldmcatalog` at the end of the URL.
Runtime environment	Name of the Secure Agent group that is used to read data from Enterprise Data Catalog. The agents in the group that you select must be able to communicate with Enterprise Data Catalog. Therefore, the Enterprise Data Catalog host must be in the same network as the agent machines or it must have the appropriate ports open for communication.
User name	Enterprise Data Catalog user account that the Secure Agent uses to access Enterprise Data Catalog. This user account must have privileges to view and search for objects in Enterprise Data Catalog and to perform functions using the Enterprise Data Catalog REST API.
Password	Password for the Enterprise Data Catalog user account.
Show the data catalog	Shows and hides the **Data Catalog** page in Data Integration.

Data Integration service properties

Data Integration service properties are used by Data Integration. Configure these properties to set the time zone and default email addresses for job notifications.

You can set the following Data Integration service properties:

Jobs properties

The following table describes the jobs properties:

Property	Description
Schedule Offset	A small amount of time that is added to schedule start times to help prevent server overload at standard schedule start times. An organization has a single schedule offset that is applied to all schedules. The schedule offset does not affect the start time of manually started tasks or taskflows. You cannot change the schedule offset. Even though it is not displayed in the schedule details, the schedule offset for your organization is added to the time range configured for all schedules. This ensures that scheduled tasks run as often as expected. For example, you configure a schedule to run every hour from 8:00 a.m. to 12:00 p.m., and the schedule offset for your organization is 15 seconds. Your schedule runs at 8:00:15, 9:00:15, 10:00:15, 11:00:15, and 12:00:15.
Time Zone	Time zone used to display job execution time stamps in email notifications.

Default email notifications properties

Configure the default email notifications properties to set the default email addresses to use for job failure, warning, and success messages. Enter one or more valid email addresses. Separate email addresses with a comma (,) or semicolon (;).

You can also set email notification properties at the task level. When you set email notifications in a task or taskflow, IDMC sends email to the addresses in the task or taskflow instead of the addresses configured for the organization.

Application Integration properties

For an organization that has a usage-based license and was created before April 30, 2025, you can configure the Application Integration processes to run in the Turbo mode.

You can configure the Turbo mode for all the processes that run on a Secure Agent or serverless runtime environment in the organization. The Turbo mode offers high throughput, improved performance, and low latency. When a process runs in the Turbo mode, the scalar value is **Thousand API Calls**, which is determined by the number of process and subprocess invocations.

In organizations that are created on or after April 30, 2025, all the processes will run only in the Turbo mode.

Note: You can't suspend and restart a process that runs in the Turbo mode.

CHAPTER 14

Custom branding

You can customize the branding configuration to use a custom logo, color theme, and favicon instead of the Informatica defaults. Configure custom branding on the **Settings** page.

Custom branding settings are applied to the following services:

- Data Governance and Catalog
- Data Marketplace
- Metadata Command Center
- MDM - Customer 360 SaaS
- MDM - Product 360 SaaS
- MDM - Reference 360
- MDM - Supplier 360 SaaS
- Multidomain MDM SaaS

To customize branding configuration, you must have the appropriate license.

Logo and favicon guidelines

Use the following guidelines when you upload a logo and favicon for your organization:

- The file size for logo and favicon images must be less than 1 MB.
- The maximum size of logo images must be 80 x 325 pixels. The recommended size is 48 x 325 pixels.
- When you upload a logo image, the image is placed within the outlined region. Use the zoom control option to resize the image so that the image fits in the outlined region.
- The maximum size of favicon images must be 196 x 196 pixels. The recommended size is 32 x 32 pixels.
- Use the PNG, JPG, and GIF format for the custom logo and the PNG format for the favicon.

Configuring custom branding for an organization

You can upload a logo and a favicon, and select a color theme on the **Settings** page.

1. On the **Settings** page in Administrator, click **Edit** in the **Custom Branding** area.
2. Select **Enable Custom Branding**.

3. If you want the sub-organizations to inherit the branding configuration from the parent organization, select **Sub-Organizations inherits custom branding**.
4. To update a logo image, click **Upload** and select the logo file. For more information about the logo guidelines, see “Logo and favicon guidelines” on page 107.
5. To update a favicon image, click **Upload** and select the favicon file. For more information about the favicon guidelines, see “Logo and favicon guidelines” on page 107.
6. Select the color theme to match the logo and favicon. You can also create a theme.
7. Preview and verify the changes in the Preview section before applying the custom branding settings.
8. Click **Save**.

CHAPTER 15

Informatica support access

Administrators can control whether Informatica Global Customer Support is allowed to view organization details on the **Security** tab of the **Settings** page.

When Informatica support access is enabled, Informatica Global Customer Support has read-only visibility into your organization.

Informatica Support can view the following items to help troubleshoot issues:

- Configuration metadata, such as connections, secure agents, users, roles, and schedules
- Design-time objects, such as tasks, mappings, and workflows
- Operational metadata, such as job run history, status, error messages, and logs

Informatica support can't perform the following actions:

- Connect to your internal systems or networks
- Log in to your databases or applications
- View data rows processed between the secure agent and your source or target systems
- Run or modify jobs in your organization

Removing Informatica support access might delay support case resolution. When you remove Informatica support access, note the following results:

- When you file a support ticket, you might need to manually upload the relevant service and task logs or organization assets.
- Informatica support can't monitor your jobs during new releases or other maintenance events to assess any potential impact.
- You might need to share your screen when troubleshooting with Informatica support.
- Informatica can't notify you about how an upcoming release might impact your organization.

Removing Informatica support access

You might want to remove Informatica support access to have more control over your assets and security. When you remove access, you can temporarily enable access to help Informatica resolve your support cases.

1. In Administrator, go to the **Settings** page.
2. In the **Informatica Global Support Access** section, select **Disable Informatica Global Support access**. Only administrators can access this check box.

You can grant temporary access to allow Informatica support to view your organization's data while troubleshooting support cases. To temporarily enable Informatica support access, select **Enable access temporarily** and enter the date and time at which to remove access again.

CHAPTER 16

Event monitoring

You can monitor events for the assets, licenses, users, and Secure Agents in your organization through the asset and security logs. To view the logs, you must be assigned a role that has the Audit Log - View privilege.

You can monitor events through the following logs:

Asset log

Displays the following information:

- Events for assets such as when an asset was created, updated, copied, or deleted and the name of the user who modified the asset.
- Events related to licenses such as when a license was added, removed, or changed.

To open the asset log, in Administrator, select **Logs**, and then select **Asset Logs** at the top of the page.

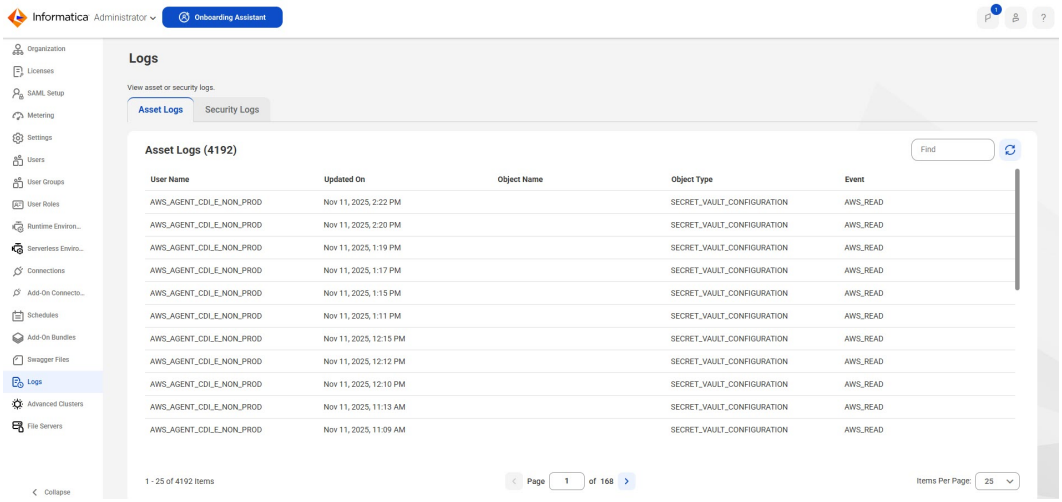
Security log

Displays the following information:

- Authentication events for users such as when a user in the organization logged in to or out of IDMC. Additionally, events for account lockouts or failed login attempts.
- Events for Secure Agents and organizations such as when each agent was created or updated, an agent login failure, when organization information was updated, and the name of the user who modified the agent or organization.
- Changes to allowed IP addresses.
- Email notification changes.
- Password requirement and policy changes.
- Organization name, attribute, or environment type changes.
- CLAIRE preference updates.
- User group or role updates.

To open the security log, in Administrator, select **Logs**, and then select **Security Logs** at the top of the page.

The following image shows the asset log:



Asset logs display events for the past 90 days. Security logs display events for the past 400 days.

You can customize the properties that are displayed in the logs in the following ways:

- To hide a column, right-click the column heading area and uncheck the column that you want to hide.
- To sort the log events, click the column heading for the property that you want to sort by. To reverse the sort order, click the column heading again.
- To search the logs for specific events, enter the search string in the **Find** field. You can search for an object name or event type.

CHAPTER 17

Troubleshooting security

I received the following security violation error:

```
There may have been a security violation while accessing the site. Verify that there are
no malicious scripts running in your browser. This error also appears when you submit
the form multiple times through a browser reload.
```

This error appears when you click an option on a page while the page is still loading from a previous click. Click the Here link to return to Data Integration.

When I try to view the details about an object, such as a connection or replication task, the Object Not Found page displays.

The object was recently deleted. The Object Not Found page appears when an object no longer exists. Refresh the page to display current objects.

When I try to perform a task, the Access Denied page displays.

The Access Denied page displays when you try to perform a task that is not allowed for your user account. You might not have the appropriate role or asset permissions to perform the task. If you need to perform the task, ask your organization administrator to review your user account.

My security scan revealed a "Weak SSL/TLS Key Exchange" vulnerability in my elastic runtime environment.

To resolve this vulnerability, perform the following steps:

1. In Administrator, select **Runtime Environments**.
2. Select your elastic runtime environment from the list.
3. Select the **Service Configuration** tab.
4. In the **Service Configuration** subtab, search for **JRE_OPTS**.
   Verify that the entry found contains Service=Data_Integration_Server, Type=TOMCAT_JRE, Name=JRE_OPTS.
5. Edit the entry and set the value to **-Djdk.tls.ephemeralDHKeySize=2048**
6. Save your changes.

CHAPTER 18

Cipher suites used by IDMC

IDMC uses specific cipher suites for TLS 1.2 and 1.3. It's critical that your organization's infrastructure supports at least one of these suites to prevent disruption in connecting to IDMC, including Secure Agent connectivity and task failures.

IDMC uses the following cipher suites:

TLS 1.3 cipher suites

TLS_AES_128_GCM_SHA256

TLS_AES_256_GCM_SHA384

TLS 1.2 cipher suites

TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256

TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384

Deprecated cipher suites

The following cipher suites are no longer used by IDMC:

- TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
- TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256

Verifying your organization's cipher support

To verify whether your organization is using a valid cipher suite for IDMC, run the following curl command:

```
curl -vhttps://global-package.dm.informaticacloud.com/package-manager/mgmtapi/version
```

If a proxy is in use, run the following command instead:

```
curl -x://:@: -v https://global-package.dm.informaticacloud.com/package-manager/mgmtapi/
version
```

For example:

```
curl -x http://proxy.example.com:8080 https://global-package.dm.informaticacloud.com/
package-manager/mgmtapi/version
```

A response of '200' indicates that you are using a valid cipher suite.

Note: If your network or system administrator is testing cipher availability using the `nmap` utility ensure that `nmap` is version 7.95 or later. Earlier versions can have issues identifying TLS1.3 ciphers.

INDEX

A

add-on bundles
  See bundles. 99
Administrator service
  overview 9
advanced clusters
  metering usage reports 55
Application Ingestion and Replication
  metering usage reports 55
Application Integration
  metering usage reports 55
asset logs
  maximum log entries 18, 106
  viewing 111
authentication
  organizations 16
AWS Secrets Manager
  configuration 71
AWS Secrets Managers
  secret names 78
Azure DevOps user credentials 62
Azure Key Vault
  configuration 80
  secret names 80

B

Bitbucket user credentials 62
blackout period
  configuring for the organization 93
bundles
  copying 101
  installing 100
  managing 99
  uninstalling 102
  upgrading 101
  viewing 99

C

CLAIRE
  preferences 97
Cloud Application Integration community
  URL 7
Cloud Developer community
  URL 7
connections
  storing properties 16, 103
custom branding
  configuring for an organization 107
  settings 11
customer-managed keys
  configuring 65
customer-managed keys *(continued)*
  settings 11

D

Data Ingestion and Replication service
  metering usage reports 55
Data Integration community
  URL 7
Data Integration Data Catalog page
  showing and hiding 17, 105
Database ingestion and Replication
    File Ingestion and Replication
      metering usage reports 55
  metering usage reports 55
Daylight Savings Time
  schedules 94

E

encryption key password
  for connection properties 16, 103
Enterprise Data Catalog
  integration with Informatica Intelligent Data Management Cloud 17, 105
events
  monitoring 111

F

fingerprint authentication
  organizations 16

G

GitHub user credentials 62
guidelines
  logo and favicon 107

H

HashiCorp Vault
  authentication methods 81
  configuration 80
  secrets format 81

I

Informatica Global Customer Support
  contact information 8

Informatica Intelligent Data Management Cloud
web site 7
IP address filtering
configuring 14
IPU meters 34, 35
IPU usage
billing periods 41
disabled and deleted sub-organizations 39
monitoring 40
reports 43

J

job limits
monitoring 49
job usage
monitoring 49

K

key vaults *See secrets managers*

L

license meters 49
license metrics
viewing 49
licenses
configuring the sub-organization type 32
editing sub-organization licenses 32
expiration 33
management 30
sub-organizations 31
types 30
login denied
troubleshooting 113

M

maintenance outages 8
metering
IPU meters 34, 35
IPU scalars 34
IPU usage reports 43
meter definitions 50
organizations and sub-organizations 34
serverless compute units 53
usage reports 55
viewing all meters 49
viewing IPU metrics 40
viewing IPU usage 40
viewing license metrics 49
viewing usage details 54
viewing usage graphs 54
metering usage reports
information 55
monitoring
events 111

O

organization hierarchies
creating a sub-organization 23
organization hierarchies *(continued)*
unlinking a sub-organization 24
organizations
synchronizing sub-organization licenses 32
adding sub-organizations 22
additional production 27
authentication properties 14
changing source control repository 61
CLAIRE preferences 97
configuring custom branding 107
creating a sub-organization 23
creating additional production organizations 28
creating sandbox organizations 28
custom branding configuration 107
customer-managed key settings 11
customer-managed keys 65
Data Integration Service properties 18, 106
deleting a sub-organization 25
disabling and enabling sub-organizations 25
disabling source control 62
enabling source control 59
Enterprise Data Catalog integration properties 17, 105
fingerprint authentication 16
license expiration 33
linking an organization as a sub-organization 23
metering 34
overview 13
properties 18
removing sub-organizations 24
sandbox 27
schedule offset 18, 106
secrets manager configuration 11, 70
session idle timeout 14
setting up 14
source control best practices 63
source control configuration 57
source control settings 11
storing connection properties 16, 103
switching to another organization 25
types 13
unlinking a sub-organization 24

P

passwords
expiration 14
minimum character mix 14
minimum length 14
reuse 14
permissions
best practices 89
configuring for objects 90
for copied assets 88
for imported assets 88
overview 88
permission descriptions 88
rules and guidelines 89

R

repeat frequency
description 95
schedules 93

S

scalars
  IPU meters 34
schedules
  associating with tasks or taskflows 92
  configuring 95
  configuring a blackout period 93
  Daylight Savings Time 94
  deleting 92
  description 92
  exporting 96
  importing 96
  monitoring scheduled tasks 92
  repeat frequency 93
  schedule offset 18, 106
  time zones 94
secret vaults *See secrets managers*
secrets managers
  AWS Secrets Manager connection properties 78
  Azure Key Vault connection properties 80
  configuring for an organization 70
  connection configuration 83
  connection property format 85
  cross-account access configuration for AWS Secrets Manager 75
  disabling for an organization 82
  enabling for an organization 82
  examples 85
  HashiCorp Vault connection properties 81
  IAM role configuration for AWS Secrets Manager 72
  instance profile configuration for AWS Secrets Manager 74
  restrictions for sub-organizations 83
Secure Agent
  storing connection properties 16, 103
Secure Agent services
  upgrade settings 11
security
  troubleshooting 113
security logs
  maximum log entries 18, 106
  viewing 111
serverless runtime environments
  metering usage reports 55
  serverless compute units 53
session idle timeout
  configuring 14
source control
  best practices 63
  changing the repository URL 61
  configuring access to the repository 62
  configuring access using OAuth 58
  configuring for a sub-organization 58
  configuring for an organization 57
  configuring read-only access to the repository 57
  configuring read/write access to the repository 57
  development guidelines 63
  disabling for an organization 62
  enabling for an organization 59
  on-premises repositories 59
  settings 11
  setup guidelines 63
status
  IDMC 8
Streaming Ingestion and Replication
  metering usage reports 55
sub-organizations
  synchronizing licenses 32
  add-on connectors 26
  adding 22
  authentication properties 14
  changing source control repository 61
  CLAIRE preferences 97
  configuring custom branding 107
  creating 23
  customer-managed key settings 11
  Data Integration Service properties 18, 106
  deleting an existing sub-organization 25
  denying parent organization access 26
  disabling and enabling 25
  disabling and IPU usage 39
  disabling source control 62
  editing licenses 32
  enabling source control 59
  Enterprise Data Catalog integration properties 17, 105
  example 21
  exporting and importing assets 26
  license expiration 33
  licenses 31
  linking an existing organization 23
  metering 34
  organization type 32
  properties 18
  reasons to create 21
  removing 24
  schedule offset 18, 106
  secrets manager configuration 11
  source control configuration 58
  source control settings 11
  storing connection properties 16, 103
  switching to another organization 25
  unlinking from a parent organization 24
system status 8

T

time zones
  description 94
troubleshooting
  security 113
trust site
  description 8
trusted IP ranges
  configuring 14

U

upgrade notifications 8
usage-based licenses
  meters 35
  IPU metrics 34

W

web site 7